

TYPE II REGULARITY FOR NAVIER–STOKES

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ABSTRACT. We establish a local Type II exclusion criterion for the three-dimensional Navier–Stokes equations near a possible singular point. From a positive Caffarelli–Kohn–Nirenberg concentration sequence we select renormalized local windows around a retained core and analyze the resulting alternatives by compactness, pressure reconstruction, and local energy estimates. The argument combines a compact single-core exclusion criterion, a repaired-gauge representation for nondegenerate concentration cores, local Calderon–Zygmund pressure control, a Caccioppoli estimate on compact cylinders, multibubble and cascade reductions, and scale-collapse cost estimates. The conclusion is conditional on the analytic inputs and rigidity hypotheses stated in the paper; no additional stationary or self-similar classification theorem is used implicitly.

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1. LOCAL SINGLE-CORE CRITERION

1.1. Introduction. We first isolate a local compactness mechanism for Type II concentration in the three-dimensional Navier–Stokes equations, in the standard class of suitable weak solutions introduced by Leray and the partial-regularity theory of Scheffer and Caffarelli–Kohn–Nirenberg [Ler34, Sch76, CKN82, Lin98]. The setting is a sequence of renormalized solutions on fixed parabolic cylinders, centered at a selected concentration core and written in scale-translation variables. The result excludes the case in which one compact core carries a fixed positive Caffarelli–Kohn–Nirenberg density while the localized dissipation along selected windows vanishes. The theorem is conditional on the local representation and pressure decomposition/stability inputs stated below; the compact-cylinder A, E, C, D bounds are part of the branch hypothesis.

The notation and estimates follow the usual local energy, critical-space, and monograph conventions for Navier–Stokes [Lad69, Tem77, CF88, Kat84, ESS03, LR02]. The argument is local. Uniform Caffarelli–Kohn–Nirenberg bounds and bounded modulation coefficients give compactness on smaller cylinders. Vanishing localized dissipation forces the compactness limit to be spatially constant. If the constant is zero, strong velocity convergence and pressure stability contradict the positive core density. If the constant is nonzero, the sequence has a nonzero bounded selected-window limit, which is incompatible with the local Type II condition.

1.2. Setup and definitions. For $R > 0$, write

$$Q_R = B_R \times (-R^2, 0).$$

For a suitable pair (v, q) on Q_R , define

$$\begin{aligned} A_v(R) &= R^{-1} \operatorname{ess\,sup}_{-R^2 < s < 0} \int_{B_R} |v(y, s)|^2 dy, \\ E_v(R) &= R^{-1} \int_{Q_R} |\nabla v|^2 dy ds, \\ C_v(R) &= R^{-2} \int_{Q_R} |v|^3 dy ds, \\ D_q(R) &= R^{-2} \int_{-R^2}^0 \int_{B_R} |q(y, s) - (q)_{B_R}(s)|^{3/2} dy ds. \end{aligned}$$

All pressure quantities are understood modulo functions of time.

Definition 1.1 (Bounded selected-window limit). Let (V_n, P_n) be a sequence of represented local rescalings on a compact cylinder Q_r . A selected-window subsequence has a bounded selected-window limit on Q_r if, after passing to a subsequence,

$$V_n \rightarrow V_* \quad \text{strongly in } L^3(Q_r),$$

where $V_* \in L^\infty(Q_r)$. The limit is nonzero if $V_* \not\equiv 0$ on Q_r .

Definition 1.2 (Local Type II sequence). A represented sequence is called a local Type II sequence if it has positive local Caffarelli–Kohn–Nirenberg concentration on some fixed compact cylinder and no selected-window subsequence has a nonzero bounded selected-window limit on any compact cylinder in the sense of Theorem 1.1.

Definition 1.3 (Compact single-core vanishing-dissipation branch). Fix radii $0 < r_* < R_*$, constants $M < \infty$, $\eta_0 > 0$, and a sequence of represented local rescalings (V_n, P_n, a_n, b_n) on Q_{R_*} . The sequence is a compact single-core vanishing-dissipation branch if:

- (i) each (V_n, P_n) is suitable on Q_{R_*} and solves the represented Navier–Stokes equation
- (1) $\partial_s V_n + (V_n \cdot \nabla) V_n + \nabla P_n = \nu \Delta V_n + a_n(s)(V_n + y \cdot \nabla V_n) + b_n(s) \cdot \nabla V_n, \quad \nabla \cdot V_n = 0;$

(ii) the compact-cylinder suitable estimates are uniformly bounded:

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \leq M;$$

(iii) the selected core has positive CKN concentration on Q_{r_*} :

$$C_{V_n}(r_*) + D_{P_n}(r_*) \geq \eta_0 \quad \text{for all } n;$$

(iv) the localized scale and center are nondegenerate for the selected core, so the representation theorem gives the variables in (1);

(v) the modulation coefficients are uniformly bounded on the selected cylinder:

$$\|a_n\|_{L^\infty(-R_*^2, 0)} + \|b_n\|_{L^\infty(-R_*^2, 0)} \leq M;$$

(vi) the branch has vanishing localized dissipation in the sense of Theorem 1.4 below.

Definition 1.4 (Vanishing localized dissipation). Let $r_* < \rho < R_*$. Selected windows are first translated and rescaled to the fixed cylinders $Q_{r_*} \Subset Q_\rho \Subset Q_{R_*}$. The localized dissipation on Q_ρ is

$$\mathcal{D}_n(\rho) = \int_{Q_\rho} |\nabla V_n|^2 dy ds.$$

The branch has vanishing localized dissipation if, after passing to a selected-window subsequence, there is $\rho \in (r_*, R_*)$ such that

$$\mathcal{D}_n(\rho) \rightarrow 0.$$

1.3. Local analytic inputs. The compact single-core argument is based on the following local analytic statements.

Theorem 1.5 (Local concentration dichotomy). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$, and suppose $\Sigma(T) \neq \emptyset$. Then, using the pressure-normalized quantities C, D , there exist $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

Moreover, at the concentration point either a local Type I scale-invariant bound holds, or the point lies in the local Type II alternative, meaning that no such local Type I bound holds.

Proof. The statement is the standing local concentration dichotomy assumed in the sequel. $\square$

Proposition 1.6 (Bridge to local Type II sequences). *Let (x_0, T) be a concentration point in the local Type II alternative of Theorem 1.5. After choosing the positive concentration scales given there and applying the local scale-translation selection used in the present section to a nondegenerate compact single core, the represented selected windows form a sequence with positive local CKN concentration on a fixed compact cylinder. In the selected-window Type II branch, no subsequence has a nonzero bounded selected-window limit on any compact cylinder. Hence the represented selected windows are a local Type II sequence in the sense of Theorem 1.2.*

Proof. The local concentration dichotomy supplies the positive CKN concentration sequence and the exclusion of the local Type I alternative at the chosen concentration point. The scale-translation selection rewrites those shrinking cylinders as fixed selected windows. The last condition is the selected-window formulation of the Type II branch considered here: any branch with a nonzero bounded selected-window limit belongs to the bounded-window alternative, not to Theorem 1.2. $\square$

Proposition 1.7 (Local representation input). *For a nondegenerate selected single core, the local scale-translation construction gives represented variables (V_n, P_n, a_n, b_n) satisfying (1) on compact cylinders. Failure of the localized nondegeneracy condition is a multibubble, cascade, or gauge-degenerate alternative, not a compact single-core branch.*

Proposition 1.8 (Local pressure decomposition and stability). *For every $B_r \Subset B_R$, the represented pressure decomposes as*

$$P_n = P_{\text{loc},n} + H_n, \quad -\Delta P_{\text{loc},n} = \partial_i \partial_j (\zeta V_{n,i} V_{n,j}),$$

where $\zeta \in C_c^\infty(B_R)$ equals one on B_r , and H_n is harmonic in B_r . The local part satisfies the Calderon–Zygmund bound [CZ52, Ste70]

$$\|P_{\text{loc},n}\|_{L^{3/2}(B_R)} \leq C \|V_n\|_{L^3(B_R)}^2$$

for a.e. time, and the harmonic part is controlled modulo spatial means by the local pressure norm.

Moreover, if $V_n \rightarrow 0$ strongly in $L^3(Q_R)$, then for every $r < R$,

$$P_n - (P_n)_{B_r} \rightarrow 0 \quad \text{strongly in } L^{3/2}(Q_r).$$

Lemma 1.9 (Localization of pressure oscillation bounds). *Let $0 < r < R$, and let $q \in L^{3/2}(Q_R)$. Then*

$$D_q(r) \leq C \left(\frac{R}{r} \right)^2 D_q(R),$$

where C is universal.

Proof. For a.e. $s \in (-r^2, 0)$, Jensen's inequality gives

$$|(q)_{B_r}(s) - (q)_{B_R}(s)|^{3/2} \leq \frac{1}{|B_r|} \int_{B_r} |q(y, s) - (q)_{B_R}(s)|^{3/2} dy.$$

Hence

$$\int_{B_r} |q - (q)_{B_r}(s)|^{3/2} dy \leq C \int_{B_r} |q - (q)_{B_R}(s)|^{3/2} dy.$$

Integrating over $(-r^2, 0)$ and using $Q_r \subset Q_R$,

$$\begin{aligned} D_q(r) &\leq C r^{-2} \int_{-r^2}^0 \int_{B_r} |q - (q)_{B_R}(s)|^{3/2} dy ds \\ &\leq C r^{-2} \int_{-R^2}^0 \int_{B_R} |q - (q)_{B_R}(s)|^{3/2} dy ds = C \left(\frac{R}{r} \right)^2 D_q(R). \end{aligned}$$

□

1.4. Compactness and pressure stability.

Proposition 1.10 (Compactness on selected windows). *Let (V_n, P_n) be suitable on Q_{R_*} , solve (1), and satisfy*

$$A_{V_n}(R_*) + E_{V_n}(R_*) + C_{V_n}(R_*) + D_{P_n}(R_*) \leq M,$$

with

$$\|a_n\|_{L^\infty(-R_*^2, 0)} + \|b_n\|_{L^\infty(-R_*^2, 0)} \leq M.$$

Then for every $r < R_*$, after passing to a subsequence there are (V, P) such that

$$\begin{aligned} V_n &\overset{*}{\rightharpoonup} V \quad \text{in } L^\infty(-r^2, 0; L^2(B_r)), \\ V_n &\rightharpoonup V \quad \text{in } L^2(-r^2, 0; H^1(B_r)), \\ V_n &\rightarrow V \quad \text{strongly in } L^3(Q_r), \end{aligned}$$

and

$$P_n - (P_n)_{B_r} \rightharpoonup P \quad \text{in } L^{3/2}(Q_r).$$

After passing to a further subsequence, a_n and b_n converge weak- $*$ in L^∞ to bounded coefficients, and the limit satisfies the corresponding represented equation distributionally on smaller cylinders.

Proof. The bounds on A and E give weak-* compactness in $L^\infty L^2$ and weak compactness in $L^2 H^1$ on smaller cylinders. Fix $r < r_1 < R_*$, and choose m so large that

$$H_0^m(B_{r_1}) \hookrightarrow W_0^{1,\infty}(B_{r_1}).$$

We claim that $\partial_s V_n$ is uniformly bounded in

$$L^1(-r_1^2, 0; H^{-m}(B_{r_1})).$$

Let $\varphi \in H_0^m(B_{r_1})$. Pairing (1) with φ , integrating by parts in space where needed, and using the embedding above gives

$$\begin{aligned} |\langle (V_n \cdot \nabla) V_n, \varphi \rangle| &= \left| \int_{B_{r_1}} V_{n,i} V_{n,j} \partial_j \varphi_i dy \right| \leq C \|V_n\|_{L^3(B_{r_1})}^2 \|\varphi\|_{H^m}, \\ |\langle \nu \Delta V_n, \varphi \rangle| &\leq C \|\nabla V_n\|_{L^2(B_{r_1})} \|\varphi\|_{H^m}, \\ |\langle \nabla P_n, \varphi \rangle| &= \left| \int_{B_{r_1}} (P_n - (P_n)_{B_{r_1}}) \nabla \cdot \varphi dy \right| \leq C \|P_n - (P_n)_{B_{r_1}}\|_{L^{3/2}(B_{r_1})} \|\varphi\|_{H^m}. \end{aligned}$$

The modulation terms obey

$$\begin{aligned} |\langle a_n V_n, \varphi \rangle| &\leq C \|V_n\|_{L^2(B_{r_1})} \|\varphi\|_{H^m}, \\ |\langle a_n y \cdot \nabla V_n, \varphi \rangle| &= \left| \int_{B_{r_1}} a_n V_{n,i} \partial_j (y_j \varphi_i) dy \right| \leq C \|V_n\|_{L^2(B_{r_1})} \|\varphi\|_{H^m}, \end{aligned}$$

and

$$|\langle b_n \cdot \nabla V_n, \varphi \rangle| = \left| \int_{B_{r_1}} V_{n,i} b_{n,j} \partial_j \varphi_i dy \right| \leq C \|V_n\|_{L^2(B_{r_1})} \|\varphi\|_{H^m},$$

where the constant uses r_1 and the uniform L^∞ bounds for a_n, b_n . Integrating these estimates in time gives a uniform bound because

$$\begin{aligned} \int_{-r_1^2}^0 \|V_n(s)\|_{L^3(B_{r_1})}^2 ds &\leq r_1^{2/3} \|V_n\|_{L^3(Q_{r_1})}^2, \\ \int_{-r_1^2}^0 \|\nabla V_n(s)\|_{L^2(B_{r_1})} ds &\leq r_1 \|\nabla V_n\|_{L^2(Q_{r_1})}, \\ \int_{-r_1^2}^0 \|P_n(s) - (P_n)_{B_{r_1}}(s)\|_{L^{3/2}(B_{r_1})} ds &\leq r_1^{2/3} \|P_n - (P_n)_{B_{r_1}}\|_{L^{3/2}(Q_{r_1})}, \end{aligned}$$

and

$$\int_{-r_1^2}^0 \|V_n(s)\|_{L^2(B_{r_1})} ds \leq r_1^2 \|V_n\|_{L^\infty(-r_1^2, 0; L^2(B_{r_1}))}.$$

The uniform C , E , D , and A bounds therefore prove the claimed time-derivative bound; the D -bound on Q_{r_1} follows from the bound on Q_{R_*} by [Theorem 1.9](#).

The compact embedding $H^1(B_{r_1}) \hookrightarrow L^2(B_r)$, together with Aubin–Lions compactness [[Aub63](#), [LM72](#), [Sim87](#)], gives

$$V_n \rightarrow V \quad \text{strongly in } L^2(Q_r)$$

after passing to a subsequence. The same $L^\infty L^2 \cap L^2 H^1$ bounds give the parabolic interpolation estimate

$$\|V_n\|_{L^{10/3}(Q_r)} \leq C(r, M).$$

Indeed, for a.e. s , the Sobolev and Gagliardo–Nirenberg inequalities give

$$\|V_n(s)\|_{L^{10/3}(B_r)} \leq C \|V_n(s)\|_{L^2(B_r)}^{2/5} \|\nabla V_n(s)\|_{L^2(B_r)}^{3/5}.$$

Raising to the power $10/3$, integrating in time, and using the A and E bounds yields

$$\int_{-r^2}^0 \|V_n(s)\|_{L^{10/3}(B_r)}^{10/3} ds \leq C \left(\operatorname{ess\,sup}_{-r^2 < s < 0} \|V_n(s)\|_{L^2(B_r)}^2 \right)^{2/3} \int_{Q_r} |\nabla V_n|^2 dy ds \leq C(r, M).$$

The same estimate applies to V by weak lower semicontinuity, and therefore to $V_n - V$ by the triangle inequality. Interpolating between L^2 and $L^{10/3}$, one obtains

$$\|V_n - V\|_{L^3(Q_r)} \leq \|V_n - V\|_{L^2(Q_r)}^{1/6} \|V_n - V\|_{L^{10/3}(Q_r)}^{5/6}.$$

The second factor is uniformly bounded and the first tends to zero, so

$$V_n \rightarrow V \quad \text{strongly in } L^3(Q_r).$$

The pressure compactness follows from the uniform D -bound: after subtracting spatial means,

$$P_n - (P_n)_{B_r} \rightharpoonup P \quad \text{in } L^{3/2}(Q_r).$$

We pass to the limit distributionally using the preceding convergences. It remains to justify the modulation terms. For every compactly supported test field φ ,

$$\int a_n(s) y \cdot \nabla V_n \cdot \varphi = - \int a_n(s) V_{n,i} \partial_j (y_j \varphi_i),$$

where summation over repeated spatial indices is understood. Equivalently,

$$\partial_j (y_j \varphi_i) = 3\varphi_i + y_j \partial_j \varphi_i.$$

Hence this term passes to the limit by strong local L^2 convergence of V_n and weak-* convergence of a_n : if $F_n \rightarrow F$ in L^1 and $a_n \rightharpoonup^* a$ in L^∞ , then

$$\int a_n F_n - \int a F = \int a_n (F_n - F) + \int (a_n - a) F \rightarrow 0.$$

The term $a_n V_n$ is immediate from the same argument, while

$$\int b_n(s) \cdot \nabla V_n \cdot \varphi = - \int V_{n,i} b_{n,j}(s) \partial_j \varphi_i$$

passes to the limit by strong local L^2 convergence of V_n and weak-* convergence of b_n . These passages give the limiting represented equation in the sense of distributions. $\square$

Remark 1.11 (Use of bounded modulation). The uniform L^∞ bounds on a_n and b_n enter the compactness argument in exactly two places. First, they give the negative Sobolev time-derivative bound for the three lower-order modulation terms in (1). Second, after subsequential weak-* convergence of a_n, b_n , they allow the modulation terms to pass to the distributional limit once $V_n \rightarrow V$ strongly in L^2_{loc} .

Lemma 1.12 (Stability of local pressure decomposition). *Let $V_n \rightarrow V$ strongly in $L^3(Q_R)$, and use the pressure decomposition of Theorem 1.8. Then for every $r < R$,*

$$P_{\text{loc},n} \rightarrow P_{\text{loc}} \quad \text{strongly in } L^{3/2}(Q_r),$$

where $-\Delta P_{\text{loc}} = \partial_i \partial_j (\zeta V_i V_j)$. If $V \equiv 0$ on Q_R , then

$$P_n - (P_n)_{B_r} \rightarrow 0 \quad \text{strongly in } L^{3/2}(Q_r).$$

More locally in time, if $V_n \rightarrow 0$ strongly in

$$L^3(B_R \times (-r^2, 0)),$$

then

$$P_n - (P_n)_{B_r} \rightarrow 0 \quad \text{strongly in } L^{3/2}(Q_r).$$

Proof. Strong convergence in L^3 gives

$$V_{n,i}V_{n,j} \rightarrow V_iV_j \quad \text{strongly in } L^{3/2}(Q_R).$$

Indeed,

$$\|V_{n,i}V_{n,j} - V_iV_j\|_{L^{3/2}(Q_R)} \leq (\|V_n\|_{L^3(Q_R)} + \|V\|_{L^3(Q_R)})\|V_n - V\|_{L^3(Q_R)}.$$

Calderon–Zygmund estimates for the compactly supported localized source yield strong convergence of the localized pressures in $L^{3/2}$ on smaller balls:

$$\|P_{\text{loc},n} - P_{\text{loc}}\|_{L^{3/2}(Q_r)} \leq C_{r,R}\|V_{n,i}V_{n,j} - V_iV_j\|_{L^{3/2}(Q_R)}.$$

Assume now that $V \equiv 0$ on Q_R . Then $P_{\text{loc}} = 0$. By the zero-limit stability part of [Theorem 1.8](#),

$$P_n - (P_n)_{B_r} \rightarrow 0 \quad \text{strongly in } L^{3/2}(Q_r).$$

The same proof is local in time. If $V_n \rightarrow 0$ strongly in $L^3(B_R \times (-r^2, 0))$, the Calderon–Zygmund estimate and harmonic pressure control are applied only for times $s \in (-r^2, 0)$, giving the strong convergence on Q_r . $\square$

Lemma 1.13 (Zero-dissipation limit). *If*

$$V_n \rightharpoonup V \quad \text{in } L^2(I; H^1(B_\rho))$$

and

$$\int_I \int_{B_\rho} |\nabla V_n|^2 dy ds \rightarrow 0,$$

then $\nabla V = 0$ a.e. on $B_\rho \times I$. Hence for a.e. time, $V(\cdot, s)$ is spatially constant on B_ρ .

Proof. Weak lower semicontinuity gives

$$\int_I \int_{B_\rho} |\nabla V|^2 dy ds \leq \liminf_{n \rightarrow \infty} \int_I \int_{B_\rho} |\nabla V_n|^2 dy ds = 0.$$

Thus $\nabla V = 0$ a.e. on $B_\rho \times I$. By Fubini, for a.e. $s \in I$ one has $\nabla V(\cdot, s) = 0$ in $L^2(B_\rho)$. Since B_ρ is connected, every $H^1(B_\rho)$ function with zero weak gradient is a.e. equal to a constant. Hence $V(\cdot, s)$ is spatially constant for a.e. s . $\square$

Lemma 1.14 (Spatially constant limits). *Let V be a limit on Q_ρ and suppose $\nabla V = 0$ a.e. there. Then $V(y, s) = c(s)$ on Q_ρ , with $c \in L^\infty(-\rho^2, 0)$. If $c \not\equiv 0$ on Q_ρ , then the approximating selected-window subsequence has a nonzero bounded selected-window limit.*

Proof. The spatial constancy follows from [Theorem 1.13](#). Since $V(\cdot, s) = c(s)$ on B_ρ , the local $L_s^\infty L_y^2$ bound gives

$$|c(s)|^2 |B_\rho| \leq \text{ess sup}_{-\rho^2 < s < 0} \int_{B_\rho} |V(y, s)|^2 dy,$$

so $c \in L^\infty(-\rho^2, 0)$. If $c \not\equiv 0$ on Q_ρ , then the strong L^3 convergence of [Theorem 1.10](#) gives a subsequential limit to a nonzero bounded function on Q_ρ , which is precisely a nonzero bounded selected-window limit in the sense of [Theorem 1.1](#). $\square$

1.5. Contradiction lemmas.

Lemma 1.15 (Persistence of positive core concentration). *For a compact single-core branch in the sense of Theorem 1.3, there are fixed $r_* > 0$ and $\eta_0 > 0$ such that along the selected-window subsequence,*

$$C_{V_n}(r_*) + D_{P_n}(r_*) \geq \eta_0 \quad \text{for all } n.$$

Proof. The lower bound is part of the definition of the selected compact core and is preserved by passing to subsequences of the selected windows. $\square$

Lemma 1.16 (Zero constant contradicts positive CKN concentration). *Let $r_* < \rho$. Suppose the compactness limit in Theorem 1.10 is spatially constant on Q_ρ and is identically zero on Q_{r_*} . Then*

$$C_{V_n}(r_*) + D_{P_n}(r_*) \rightarrow 0.$$

Consequently this case contradicts Theorem 1.15.

Proof. Strong $L^3(Q_{r_*})$ convergence to zero gives

$$C_{V_n}(r_*) = r_*^{-2} \int_{Q_{r_*}} |V_n|^3 dy ds = r_*^{-2} \|V_n\|_{L^3(Q_{r_*})}^3 \rightarrow 0.$$

It remains to control the pressure term. Since V is spatially constant on Q_ρ , the identity $V = 0$ on Q_{r_*} implies

$$V = 0 \quad \text{on } B_\rho \times (-r_*^2, 0).$$

Indeed, for a.e. $s \in (-r_*^2, 0)$, the spatially constant value on B_ρ vanishes on the nonempty subball B_{r_*} , hence vanishes on all of B_ρ . The strong L^3_{loc} convergence therefore gives

$$V_n \rightarrow 0 \quad \text{strongly in } L^3(B_\rho \times (-r_*^2, 0)).$$

Applying Theorem 1.12 with $R = \rho$ and $r = r_*$, in its local-in-time form, gives

$$P_n - (P_n)_{B_{r_*}} \rightarrow 0 \quad \text{strongly in } L^{3/2}(Q_{r_*}).$$

Therefore

$$D_{P_n}(r_*) = r_*^{-2} \|P_n - (P_n)_{B_{r_*}}\|_{L^{3/2}(Q_{r_*})}^{3/2} \rightarrow 0.$$

Thus

$$C_{V_n}(r_*) + D_{P_n}(r_*) \rightarrow 0,$$

contradicting the uniform lower bound

$$C_{V_n}(r_*) + D_{P_n}(r_*) \geq \eta_0.$$

$\square$

Lemma 1.17 (Nonzero constant contradicts the Type II regime). *If the compactness limit is spatially constant and nonzero on a compact subcylinder, then the original sequence is not a local Type II sequence.*

Proof. By Theorem 1.14, the selected-window subsequence has a nonzero bounded selected-window limit. This contradicts Theorem 1.2. $\square$

1.6. Main theorem.

Theorem 1.18 (Local single-core Type II criterion). *Under the analytic inputs [Theorems 1.6 to 1.8](#), a compact single-core vanishing-dissipation branch in the sense of [Theorem 1.3](#) cannot be a local Type II sequence.*

Proof. By the definition of vanishing localized dissipation, after passing to a selected-window subsequence there is a fixed radius $\rho \in (r_*, R_*)$ such that

$$\int_{Q_\rho} |\nabla V_n|^2 dy ds \rightarrow 0$$

and the positive core bound on Q_{r_*} remains valid along this subsequence. Apply compactness on the fixed compact cylinder Q_ρ and pass to a subsequence. The compactness proposition gives

$$V_n \rightharpoonup V \quad \text{in } L^2(-\rho^2, 0; H^1(B_\rho))$$

and strong convergence in L^3 on every smaller cylinder. Since the localized dissipation tends to zero, the zero-dissipation lemma gives

$$\nabla V = 0 \quad \text{a.e. on } Q_\rho.$$

Thus $V(y, s) = c(s)$ on Q_ρ , and in particular on the selected core cylinder Q_{r_*} .

If this spatially constant limit is zero on the core cylinder, the zero-limit contradiction lemma gives

$$C_{V_n}(r_*) + D_{P_n}(r_*) \rightarrow 0,$$

contradicting persistence of positive core concentration.

If the spatially constant limit is not identically zero on Q_{r_*} , the nonzero-limit lemma produces a nonzero bounded selected-window limit, contradicting the definition of a local Type II sequence. The two cases exhaust all spatially constant limits, so the branch cannot occur. $\square$

Remark 1.19 (Minimal local inputs). The proof uses two local analytic inputs from the Type II argument: the localized representation for a nondegenerate selected core, the compact-ball pressure reconstruction with stability on smaller balls. The compact-cylinder A, E, C, D bounds and vanishing localized dissipation are part of the branch hypothesis, not separate analytic inputs. Multibubble, cascade, gauge-degenerate, and compactness-failure alternatives are outside this single-core theorem; they are treated by the local multibubble and cascade exclusion theorem.

2. REPAIRED-GAUGE CONSTRUCTION

2.1. Assumptions and notation. Let $u : \mathbb{R}^3 \times (T - \delta, T) \rightarrow \mathbb{R}^3$, $p : \mathbb{R}^3 \times (T - \delta, T) \rightarrow \mathbb{R}$ be a suitable weak Navier–Stokes solution with viscosity $\nu > 0$. For $z = (x_0, t_0)$ and $R > 0$,

$$Q_R(z) := B_R(x_0) \times (t_0 - R^2, t_0), \quad Q_R^* := B_R \times I_*.$$

Fix a reference physical center-time $(x_\bullet, t_\star)$ around which $B_{R_*}(x_\bullet)$ is taken. For a fixed spatial center x_0 we also use

$$Q_R(t) := Q_R(x_0, t).$$

Fix a compact window $I_* := [\tau_0 - \ell, \tau_0 + \ell]$, $\ell > 0$, and write $I_{\text{ph}} := [t_\star - \Lambda_{\max}^2 \ell, t_\star + \Lambda_{\max}^2 \ell]$.

For a cylinder $Q_R(x_0, t_0) = B_R(x_0) \times (t_0 - R^2, t_0)$, set

$$\begin{aligned} A(u; Q_R(x_0, t_0)) &:= R^{-1} \sup_{s \in (t_0 - R^2, t_0)} \int_{B_R(x_0)} |u(x, s)|^2 dx, \\ E(u; Q_R(x_0, t_0)) &:= R^{-1} \int_{Q_R(x_0, t_0)} |\nabla u|^2 dx ds, \\ C(u; Q_R(x_0, t_0)) &:= R^{-2} \int_{Q_R(x_0, t_0)} |u|^3 dx ds, \\ D(p; Q_R(x_0, t_0)) &:= R^{-2} \int_{t_0 - R^2}^{t_0} \int_{B_R(x_0)} |p(x, s) - \langle p(s) \rangle_{B_R(x_0)}|^{3/2} dx ds. \end{aligned}$$

Here $\langle p(s) \rangle_{B_R(x_0)} := |B_R|^{-1} \int_{B_R(x_0)} p(x, s) dx$; pressure is always understood modulo time-dependent constants.

Definition 2.1 (Single-core hypotheses). Let $(R_*, r_*, M, \eta, \kappa, \Lambda_{\min}, \Lambda_{\max})$ satisfy $\Lambda_{\min} \leq \Lambda_{\max} < \infty$. We say (u, p) satisfies SC($R_*, r_*, M, \eta, \kappa$) on $Q_{R_*}(x_\bullet, t_\star)$ if the following hold:

- (SC1) $A(u; Q_{R_*}(x_\bullet, t)) + E(u; Q_{R_*}(x_\bullet, t)) + C(u; Q_{R_*}(x_\bullet, t)) + D(p; Q_{R_*}(x_\bullet, t)) \leq M$ for every $t \in [t_\star - R_*^2, t_\star]$.
- (SC2) There exists a physically selected core map $(x_c(t), \rho_c(t))$ on I_{ph} such that $\rho_c(t) \in (0, R_*]$, $x_c(t) \in B_{R_*/2}(x_\bullet)$, and for some $\Theta_0 > 0$,

$$\int_{B_{r_*}(0)} |\rho_c(t)^{-1} u(x_c(t) + \rho_c(t)y, t)|^3 dy \geq \eta \quad \text{for a.e. } t \in I_{\text{ph}}.$$

- (SC3) There exists $0 < r_{\text{sep}} \leq r_*/2$ such that for every $0 < r \leq r_{\text{sep}}$, every competitor pair

$$\mathcal{C}_{\text{core}}(t) := \left\{ (\bar{x}, \bar{\rho}) : |\bar{x} - x_\bullet| \leq \frac{R_*}{2}, 0 < \bar{\rho} \leq R_* \right\},$$

$(\bar{x}, \bar{\rho}) \in \mathcal{C}_{\text{core}}(t) \setminus \{(x_c(t), \rho_c(t))\}$ with

$$\rho_c(t)^{-1} |\bar{x} - x_c(t)| + \left| \frac{\bar{\rho}}{\rho_c(t)} - 1 \right| \geq r_{\text{sep}}$$

has separated local weighted mass in the same core-scale variables:

$$\left| \int_{B_r} |\rho_c(t)^{-1} u(x_c(t) + \rho_c(t)y, t)|^3 dy - \int_{B_r} |\bar{\rho}^{-1} u(\bar{x} + \bar{\rho}y, t)|^3 dy \right| \geq \eta.$$

This quantitative non-competing core condition is imposed on the selected physical core directly; it is not a consequence of the repaired-gauge construction.

(SC4) For the selected core,

$$\sup_{t \in I_{\text{ph}}} \int_{B_{2R_*}} |y|^{-p} |\rho_c(t)^{-1} u(x_c(t) + \rho_c(t)y, t)|^3 dy \leq M,$$

for some $0 < p < 3$.

(SC5) The preliminary frame associated with the selected core is the absolutely continuous representative

$$0 < \Lambda_{\min} \leq \Lambda_{\text{pre}}(\tau) \leq \Lambda_{\max}, \quad \Lambda_{\text{pre}} \in AC(I_*), \quad X_{\text{pre}} \in AC(I_*; \mathbb{R}^3).$$

It is tied to the selected physical core by the time change $dt_c/d\tau = \Lambda_{\text{pre}}(\tau)^2$, $t_c(\tau_0) = t_*$: one has

$$X_{\text{pre}}(\tau) = x_c(t_c(\tau)), \quad \Lambda_{\text{pre}}(\tau) = \rho_c(t_c(\tau))$$

for a.e. $\tau \in I_*$.

The parameter κ denotes the quantitative constants obtained in the local gauge estimates and is not used in items (SC1)–(SC5).

Remark 2.2. The selected core (x_c, ρ_c) and its separation/nondegeneracy conditions are specified in physical variables; no repaired-gauge constraint is used in their definition. This avoids the circularity of defining the core via the chart to be constructed.

2.2. Preliminary frame map from the selected core. The transport map is constructed from the selected core; the core selection itself is part of the hypotheses.

Define

$$\frac{dt_c}{d\tau} = \Lambda_{\text{pre}}(\tau)^2, \quad t_c(\tau_0) = t_*,$$

and use the compatibility in (SC5) to identify $\Lambda_{\text{pre}}(\tau) = \rho_c(t_c(\tau))$ and $X_{\text{pre}}(\tau) = x_c(t_c(\tau))$ a.e.

Proposition 2.3 (Preliminary frame map regularity). *Under $\text{SC}(R_*, r_*, M, \eta, \kappa)$, the map*

$$\Phi_{\text{pre}}(\tau, Y) := (x, t) = (X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)Y, t_c(\tau))$$

is well-defined and absolutely continuous in τ . Moreover, $1/\Lambda_{\text{pre}}, \Lambda_{\text{pre}} \in L^\infty(I_)$, $X_{\text{pre}} \in W_{\text{loc}}^{1,1}(I_*; \mathbb{R}^3)$, and for every $R \leq R_*$,*

$$\Phi_{\text{pre}} : Q_R^* \rightarrow \Phi_{\text{pre}}(Q_R^*) \subset Q_{2R_*}(x_\bullet, t_\bullet)$$

is bi-Lipschitz with $\det \nabla_Y \Phi_{\text{pre}} = \Lambda_{\text{pre}}^3$, hence Jacobian and inverse-Jacobian bounds depend only on $\Lambda_{\min}, \Lambda_{\max}$.

Proof. From (SC5), $\Lambda_{\text{pre}} \in AC(I_*)$, $0 < \Lambda_{\min} \leq \Lambda_{\text{pre}} \leq \Lambda_{\max} < \infty$, and $X_{\text{pre}} \in AC(I_*; \mathbb{R}^3)$. The scalar ODE

$$\frac{dt_c}{d\tau} = \Lambda_{\text{pre}}(\tau)^2, \quad t_c(\tau_0) = t_*$$

therefore has an absolutely continuous strictly increasing solution on I_* , and

$$\frac{dt_c}{d\tau} = \Lambda_{\text{pre}}(\tau)^2 \quad \text{a.e. on } I_*.$$

The compatibility condition (SC5) identifies this absolutely continuous frame with the selected physical core a.e.; no differentiability of x_c or ρ_c beyond the frame regularity in (SC5) is used. For fixed τ , $\nabla_Y \Phi_{\text{pre}}(\tau, \cdot) = \Lambda_{\text{pre}}(\tau)I$, so

$$\det \nabla_Y \Phi_{\text{pre}}(\tau, \cdot) = \Lambda_{\text{pre}}(\tau)^3, \quad \Lambda_{\min}^3 \leq \det \nabla_Y \Phi_{\text{pre}} \leq \Lambda_{\max}^3.$$

Hence, if $Y_1, Y_2 \in B_R$,

$$\Lambda_{\min}|Y_1 - Y_2| \leq |\Phi_{\text{pre}}(\tau, Y_1) - \Phi_{\text{pre}}(\tau, Y_2)| \leq \Lambda_{\max}|Y_1 - Y_2|,$$

and therefore $\Phi_{\text{pre}}(\tau, \cdot)$ is bi-Lipschitz on B_R . Its inverse has Lipschitz constant $\Lambda_{\min}^{-1}$, so Jacobian and inverse-Jacobian bounds are controlled by $\Lambda_{\min}, \Lambda_{\max}$.

For image inclusions, if $x = X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)Y$ with $|Y| \leq R$, then

$$|x - X_{\text{pre}}(\tau)| \leq \Lambda_{\max} R, \quad X_{\text{pre}}(\tau) \in B_{R_*/2}(x_\bullet),$$

so $x \in B_{R_*/2 + \Lambda_{\max} R}(x_\bullet)$. As $\tau \in I_*$, we also have $t = t_c(\tau) \in I_{\text{ph}}$. Therefore

$$\Phi_{\text{pre}}(Q_R^*) \subset B_{R_*/2 + \Lambda_{\max} R}(x_\bullet) \times I_{\text{ph}}.$$

□

Definition 2.4 (Preliminary profile). For $(x, t) = \Phi_{\text{pre}}(\tau, Y)$, define

$$V_{\text{pre}}(Y, \tau) := \Lambda_{\text{pre}}(\tau) u(x, t), \quad P_{\text{pre}}(Y, \tau) := \Lambda_{\text{pre}}(\tau)^2 p(x, t).$$

Hence $u(x, t) = \Lambda_{\text{pre}}(\tau)^{-1} V_{\text{pre}}(Y, \tau)$ and $p(x, t) = \Lambda_{\text{pre}}(\tau)^{-2} P_{\text{pre}}(Y, \tau)$ under Φ_{pre} .

2.3. Exact renormalized change of variables. Define renormalized fields on Q_{2R}^* by

$$Y = \frac{x - X(\tau)}{\Lambda(\tau)}, \quad u(x, t) = \Lambda(\tau)^{-1} V(Y, \tau), \quad p(x, t) = \Lambda(\tau)^{-2} P(Y, \tau),$$

where $dt/d\tau = \Lambda(\tau)^2$, $X : I_* \rightarrow \mathbb{R}^3$, $\Lambda : I_* \rightarrow (0, \infty)$, and $\Lambda, X \in AC(I_*)$. Write

$$a(\tau) := -\Lambda'(\tau), \quad b(\tau) := -X'(\tau).$$

Lemma 2.5 (Exact differential identities). *For smooth data and fixed (x, t) , the chain rule gives*

$$\partial_t u = \Lambda^{-3} (\partial_\tau V - (X' + \Lambda' Y) \cdot \nabla_Y V - \Lambda' V),$$

$$\nabla_x u = \Lambda^{-2} \nabla_Y V, \quad \Delta_x u = \Lambda^{-3} \Delta_Y V, \quad \nabla_x p = \Lambda^{-3} \nabla_Y P.$$

In particular

$$\nabla_x \cdot u = \Lambda^{-2} \nabla_Y \cdot V.$$

Proof. For fixed (x, t) , $\tau = \tau(t)$ and $Y = (x - X(\tau))/\Lambda(\tau)$. Differentiate

$$u(x, t) = \Lambda(\tau)^{-1} V(Y, \tau)$$

in t :

$$\partial_t u = -\frac{\Lambda'}{\Lambda^2} V + \frac{1}{\Lambda} \partial_\tau V \frac{d\tau}{dt} + \frac{1}{\Lambda} \nabla_Y V \cdot \frac{dY}{dt}.$$

Since $d\tau/dt = \Lambda^{-2}$ and

$$Y_\tau = -\Lambda^{-1} X' - \frac{\Lambda'}{\Lambda} Y, \quad \frac{dY}{dt} = Y_\tau \frac{d\tau}{dt},$$

we have

$$\partial_t u = \Lambda^{-3} (\partial_\tau V - (X' + \Lambda' Y) \cdot \nabla_Y V - \Lambda' V).$$

For spatial derivatives, $Y = (x - X)/\Lambda$ gives $\nabla_x Y = \Lambda^{-1} I$, hence

$$\nabla_x u = \Lambda^{-2} \nabla_Y V, \quad \Delta_x u = \Lambda^{-3} \Delta_Y V.$$

Similarly $p(x, t) = \Lambda^{-2} P(Y, \tau)$ yields

$$\nabla_x p = \Lambda^{-3} \nabla_Y P.$$

Taking divergence of $u = \Lambda^{-1} V$ gives

$$\nabla_x \cdot u = \Lambda^{-2} \nabla_Y \cdot V.$$

□

Theorem 2.6 (Renormalized Navier–Stokes equation). *Assume u, p solve*

$$\partial_t u + (u \cdot \nabla_x)u + \nabla_x p - \nu \Delta_x u = 0, \quad \nabla_x \cdot u = 0$$

in distributions. Under the transformation above, V, P satisfy, on the image renormalized cylinder, the exact equation

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu \Delta V + a(\tau)(V + Y \cdot \nabla V) + b(\tau) \cdot \nabla V = 0, \quad \nabla \cdot V = 0,$$

with the sign convention fixed by $a = -\Lambda'$, $b = -X'$. Conversely, any V, P satisfying this equation pull back to a distributional solution u, p of Navier–Stokes on the physical image.

Proof. To avoid hidden regularity loss, first write the physical equation tested against $\Phi \in C_c^\infty$ in the physical cylinder and then use the smooth pull-back formula (an approximation argument gives distributions). By Lemma 2.5,

$$\partial_t u + (u \cdot \nabla_x)u + \nabla_x p - \nu \Delta_x u = 0$$

becomes

$$\Lambda^{-3}(\partial_\tau V - (X' + \Lambda'Y) \cdot \nabla V - \Lambda'V) + \Lambda^{-3}(V \cdot \nabla)V + \Lambda^{-3}\nabla P - \Lambda^{-3}\nu \Delta V = 0.$$

Multiplying by Λ^3 and using $a = -\Lambda'$, $b = -X'$ gives

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu \Delta V + a(V + Y \cdot \nabla V) + b \cdot \nabla V = 0.$$

The divergence condition is

$$\nabla_x \cdot u = \Lambda^{-2}\nabla \cdot V = 0 \implies \nabla \cdot V = 0.$$

Conversely, assume (V, P) solve this equation on the renormalized cylinder. Set u, p by the same change-of-variables map. Then the preceding identities run in reverse and give

$$\partial_t u + (u \cdot \nabla_x)u + \nabla_x p - \nu \Delta_x u = 0, \quad \nabla_x \cdot u = 0$$

in $\mathcal{D}'$ on the physical image. $\square$

2.4. Domain localization and support. Let $R > 0$ be fixed and $0 < \ell \leq \ell_0$. Set $Q_R^* = B_R \times I_*$. With Φ as above define physical support maps

$$\mathcal{Q}_R^{\text{ph}} := \Phi(Q_R^*), \quad \mathcal{Q}_{2R}^{\text{ph}} := \Phi(Q_{2R}^*).$$

Proposition 2.7 (Compact support transport). *Assume $\Lambda_{\min} \leq \Lambda \leq \Lambda_{\max}$ on I_* . Then*

$$X(\tau) \in B_{R_*/2}(x_\bullet), \quad |x - X(\tau)| \leq \Lambda_{\max}R,$$

$$\mathcal{Q}_R^{\text{ph}} \subset B_{R_*/2+\Lambda_{\max}R}(x_\bullet) \times I_{\text{ph}},$$

$$\mathcal{Q}_{2R}^{\text{ph}} \subset B_{R_*/2+2\Lambda_{\max}R}(x_\bullet) \times I_{\text{ph}},$$

for every $\tau \in I_$ and $x = X(\tau) + \Lambda(\tau)Y$ with $Y \in B_{2R}$. In particular, if $|c(\tau)| \leq \delta$ from Proposition 2.20, then $X(\tau) \in B_{R_*/2+\Lambda_{\max}\delta}(x_\bullet)$ on I_* . Hence, $\chi_R \in C_c^\infty(B_{2R})$ pulls back to $\chi_R^\tau := \chi_R \circ \Phi^{-1} \in C_c^\infty(\mathcal{Q}_{2R}^{\text{ph}})$ with support depending only on $R, \Lambda_{\min}, \Lambda_{\max}$.*

Proof. Fix $\tau \in I_*$ and $Y \in B_R$. Since $x = X(\tau) + \Lambda(\tau)Y$,

$$|x - X(\tau)| \leq |Y| \Lambda_{\max} \leq \Lambda_{\max}R, \quad \Lambda_{\max}^{-1}|x - X(\tau)| \leq |Y|.$$

Because $X(\tau) \in B_{R_*/2}(x_\bullet)$, this gives

$$x \in B_{R_*/2+\Lambda_{\max}R}(x_\bullet).$$

Similarly, if $Y \in B_{2R}$, then $x \in B_{R_*/2+2\Lambda_{\max}R}(x_\bullet)$. Together with $t = t_c(\tau) \in I_{\text{ph}}$, these give the claimed inclusions for $\mathcal{Q}_R^{\text{ph}}$ and $\mathcal{Q}_{2R}^{\text{ph}}$.

For pull-backs of cutoffs, Φ is bi-Lipschitz at every τ , so composition

$$\chi_R^\tau(x, t) := \chi_R\left(\frac{x - X(\tau)}{\Lambda(\tau)}\right)$$

is smooth on $\mathcal{Q}_{2R}^{\text{ph}}$, supported where $Y \in B_{2R}$, hence supported where $(x, t) \in \mathcal{Q}_{2R}^{\text{ph}}$. The support size depends only on $R, \Lambda_{\min}, \Lambda_{\max}$ and the fixed C_c^∞ norm of χ_R , not on the particular branch. $\square$

2.5. Transport of local quantities under the chart. Let $t = t(\tau) = t_c(\tau)$ and, for $R > 0$, define

$$J_\tau^-(R) := (t(\tau) - \Lambda_{\max}^2 R^2, t(\tau)].$$

Define renormalized local quantities on $Q_R^*(\tau)$ by

$$\begin{aligned} A_V(\tau, R) &:= \sup_{s \in (\tau - R^2, \tau)} \int_{B_R} |V(\cdot, s)|^2 dY, & E_V(\tau, R) &:= \int_{\tau - R^2}^\tau \int_{B_R} |\nabla V|^2 dY ds, \\ C_V(\tau, R) &:= \int_{\tau - R^2}^\tau \int_{B_R} |V|^3 dY ds, & D_V(\tau, R) &:= \int_{\tau - R^2}^\tau \int_{B_R} |P - \langle P \rangle_{B_R}|^{3/2} dY ds. \end{aligned}$$

For interval notation and a center X , define

$$\begin{aligned} A_u(J, R; X) &:= \sup_{s \in J} \int_{B_R(X)} |u(\cdot, s)|^2 dx, & E_u(J, R; X) &:= \int_J \int_{B_R(X)} |\nabla u|^2 dx ds, \\ C_u(J, R; X) &:= \int_J \int_{B_R(X)} |u|^3 dx ds, & D_u(J, R; X) &:= \int_J \int_{B_R(X)} |p - \langle p \rangle_{B_R(X)}|^{3/2} dx ds. \end{aligned}$$

In the inequalities below we abbreviate

$$\begin{aligned} A_u(J, R) &:= A_u(J, R; X(\tau)), & E_u(J, R) &:= E_u(J, R; X(\tau)), \\ C_u(J, R) &:= C_u(J, R; X(\tau)), & D_u(J, R) &:= D_u(J, R; X(\tau)). \end{aligned}$$

Proposition 2.8 (Cylinder comparison on compact windows). *Assume $\Lambda_{\min} \leq \Lambda \leq \Lambda_{\max}$ and formulas of Lemma 2.5. Then for every $\tau \in I_*$, $R > 0$,*

$$\begin{aligned} A_V(\tau, R) &\leq \Lambda_{\min}^{-1} A_u(J_\tau^-(R), \Lambda_{\max} R), \\ E_V(\tau, R) &\leq \Lambda_{\min}^{-3} E_u(J_\tau^-(R), \Lambda_{\max} R), \\ C_V(\tau, R) &\leq \Lambda_{\min}^{-2} C_u(J_\tau^-(R), \Lambda_{\max} R), \\ D_V(\tau, R) &\leq \Lambda_{\min}^{-2} D_u(J_\tau^-(R), \Lambda_{\max} R). \end{aligned}$$

In particular, all four renormalized quantities are bounded whenever the physical quantities are bounded on the corresponding compact physical cylinders.

Proof. These are direct changes of variables under Φ_{pre} :

$$\begin{aligned} V(\cdot, \tau) &= \Lambda(\tau) u(\cdot, t(\tau)), & P(\cdot, \tau) &= \Lambda(\tau)^2 p(\cdot, t(\tau)), \\ \nabla_Y V &= \Lambda \nabla_x u, & dY d\tau &= \Lambda^{-5} dx dt, & d\tau &= \Lambda^2 dt. \end{aligned}$$

Insert these formulas into each definition and use $\Lambda_{\min} \leq \Lambda \leq \Lambda_{\max}$, plus

$$\tau - s \in (0, R^2) \implies t(\tau) - \Lambda_{\max}^2 R^2 \leq t(s) \leq t(\tau) - \Lambda_{\min}^2 R^2.$$

$\square$

2.6. Local repaired gauge map. Fix $R > 0$, $0 < p < 3$, and $\chi_R \in C_c^\infty(B_{2R})$, $\chi_R \equiv 1$ on B_R . Let the local energy profile space be

$$\mathcal{V}_R := \left\{ V \in L^3(B_{2R}) \cap W^{1,3/2}(B_{2R}; \mathbb{R}^3) : |Y|^{-p/3} V \in L^3(B_{2R}) \right\},$$

with norm

$$\|V\|_{\mathcal{V}_R} := \|V\|_{L^3(B_{2R})} + \||Y|^{-p/3} V\|_{L^3(B_{2R})} + \|\nabla V\|_{L^{3/2}(B_{2R})}.$$

For differentiating the repaired scale and centering gauges we use the admissible gauge class

$$\mathcal{A}_{p,R} := \left\{ V : \begin{array}{l} V \in L^3(B_{4R}) \cap W_{\text{loc}}^{1,3}(B_{4R}; \mathbb{R}^3), \\ \int_{B_{4R}} |Y|^{-p} |V|^3 dY < \infty, \quad \int_{B_{4R}} |Y|^{-p} |V| |Y \cdot \nabla V| dY < \infty, \\ \int_{B_{4R}} |Y|^{-p-1} |V|^3 dY < \infty \end{array} \right\}.$$

The larger ball is only a domain buffer: all gauge integrals are supported in B_{2R} , while (λ, c) is restricted near $(1, 0)$ so that $\lambda^{-1}(B_{2R} - c) \subset B_{4R}$.

Definition 2.9 (Gauge actions). For $(\lambda, c) \in (0, \infty) \times \mathbb{R}^3$, define

$$V_{\lambda,c}(Y) := \lambda^{-1} V(\lambda^{-1}(Y - c)).$$

For pressure profiles use

$$P_{\lambda,c}(Y) := \lambda^{-2} P(\lambda^{-1}(Y - c)).$$

We also write $V^{\lambda,c} := V_{\lambda,c}$ and $P^{\lambda,c} := P_{\lambda,c}$.

Definition 2.10 (Finite-dimensional gauge map). For $V \in \mathcal{V}_R$ and $(\lambda, c) \in (0, \infty) \times \mathbb{R}^3$, set

$$\mathcal{F}(\lambda, c; V) := (\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3),$$

with

$$\mathcal{F}_0(\lambda, c; V) := \int_{\mathbb{R}^3} \chi_R(Y) |Y|^{-p} |V_{\lambda,c}(Y)|^3 dY - \Theta_0, \quad \mathcal{F}_j(\lambda, c; V) := \int_{\mathbb{R}^3} Y_j \chi_R(Y) |V_{\lambda,c}(Y)|^2 dY$$

for $j = 1, 2, 3$. The gauge conditions are $\mathcal{F}(\lambda, c; V) = 0$. Codomain is $\mathbb{R}^4$, domain $(0, \infty) \times \mathbb{R}^3 \times \mathcal{V}_R$.

Definition 2.11 (Localized modulation matrix). For $J \subset I_*$ and a profile path $W : J \rightarrow \mathcal{V}_R$, define

$$J_W(\tau) := D_{(\lambda,c)} \mathcal{F}(1, 0; W(\tau)) \in \mathbb{R}^{4 \times 4}.$$

The rows are ordered as the scale constraint followed by the three centering constraints.

Lemma 2.12 (Weighted integrability and differentiation under integrals). Assume $V, W \in \mathcal{A}_{p,R}$. Then $G_0(V) := \int \chi_R |Y|^{-p} |V|^3$ is finite and

$$\left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \int \chi_R |Y|^{-p} |V + \epsilon W|^3 dY = \int 3\chi_R |Y|^{-p} |V| V \cdot W dY.$$

Proof. For $W \in \mathcal{A}_{p,R}$, define

$$F_\epsilon(Y) := \chi_R(Y) |Y|^{-p} |V(Y) + \epsilon W(Y)|^3.$$

For each Y , $\epsilon \mapsto F_\epsilon(Y)$ is C^1 and

$$\partial_\epsilon F_\epsilon = 3\chi_R |Y|^{-p} |V + \epsilon W| (V + \epsilon W) \cdot W.$$

For $|\epsilon| \leq 1$,

$$|F_\epsilon(Y)| + |\partial_\epsilon F_\epsilon| \leq C\chi_R |Y|^{-p} (|V|^3 + |W|^3) \in L^1(B_{2R})$$

by the defining weighted L^3 condition of $\mathcal{A}_{p,R}$. Hence dominated convergence allows differentiation under the integral, and

$$\left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \int_{B_{2R}} \chi_R |Y|^{-p} |V + \varepsilon W|^3 dY = \int_{B_{2R}} 3\chi_R |Y|^{-p} |V| V \cdot W dY.$$

□

Lemma 2.13 (Gauge covariance). *For $V \in \mathcal{V}_R$, $(\lambda, c) \in (0, \infty) \times \mathbb{R}^3$, and $\alpha = 0, \dots, 3$,*

$$\mathcal{F}_\alpha(1, 0; V_{\lambda,c}) = \mathcal{F}_\alpha(\lambda, c; V).$$

Proof. For each component this is a direct substitution:

$$\begin{aligned} \mathcal{F}_0(1, 0; V_{\lambda,c}) &= \int \chi_R(Y) |Y|^{-p} |V_{\lambda,c}(Y)|^3 - \Theta_0 dY = \mathcal{F}_0(\lambda, c; V), \\ \mathcal{F}_j(1, 0; V_{\lambda,c}) &= \int Y_j \chi_R(Y) |V_{\lambda,c}(Y)|^2 dY = \mathcal{F}_j(\lambda, c; V), \quad j = 1, 2, 3. \end{aligned}$$

□

Proposition 2.14 (Differentiability of $\mathcal{F}$). *For each fixed $V \in \mathcal{A}_{p,R}$, the map $(\lambda, c) \mapsto \mathcal{F}(\lambda, c; V)$ is C^1 in a neighborhood of $(1, 0)$. In each component $\mathcal{F}_\alpha$ all derivatives under the integral sign are legitimate.*

Proof. Fix (λ, c) near $(1, 0)$, and abbreviate

$$V_{\lambda,c}(Y) := \lambda^{-1} V(\lambda^{-1}(Y - c)).$$

By direct differentiation,

$$\partial_\lambda V_{\lambda,c} = -\lambda^{-1} V_{\lambda,c} - \lambda^{-1}(Y - c) \cdot \nabla V_{\lambda,c}, \quad \partial_{c_k} V_{\lambda,c} = -\lambda^{-2} \partial_k V(\lambda^{-1}(Y - c)) = -\partial_k V_{\lambda,c}.$$

The neighborhood of $(1, 0)$ is chosen so that the arguments of V remain inside B_{4R} for $Y \in \text{supp} \chi_R$. The scale derivative is controlled by the two weighted integrability conditions

$$\int |Y|^{-p} |V|^3 < \infty, \quad \int |Y|^{-p} |V| |Y \cdot \nabla V| < \infty,$$

transported by the smooth change of variables. The translation derivative in the scale row is treated by the source identity

$$3 \int |Y|^{-p} |V| V \cdot \partial_k V = p \int Y_k |Y|^{-p-2} |V|^3,$$

obtained first for smooth compactly supported profiles and then by approximation in $\mathcal{A}_{p,R}$; the right-hand side is finite because $\int |Y|^{-p-1} |V|^3 < \infty$. The centering rows are supported in B_{2R} and are controlled by $V \in L^2_{\text{loc}}$ and $\nabla V \in L^3_{\text{loc}}$. These bounds give an integrable majorant for the difference quotients, so dominated convergence applies.

$$\mathcal{F}_0(\lambda, c; V) = \int_{\mathbb{R}^3} \chi_R(Y) |Y|^{-p} |V_{\lambda,c}|^3 dY - \Theta_0,$$

for which dominated convergence gives

$$\begin{aligned} \partial_\lambda \mathcal{F}_0 &= \int_{\mathbb{R}^3} \chi_R |Y|^{-p} 3 |V_{\lambda,c}| V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} dY, \\ \partial_{c_k} \mathcal{F}_0 &= \int_{\mathbb{R}^3} \chi_R |Y|^{-p} 3 |V_{\lambda,c}| V_{\lambda,c} \cdot \partial_{c_k} V_{\lambda,c} dY. \end{aligned}$$

Similarly

$$\mathcal{F}_j(\lambda, c; V) = \int_{\mathbb{R}^3} Y_j \chi_R(Y) |V_{\lambda,c}|^2 dY, \quad j = 1, 2, 3,$$

hence

$$\partial_\lambda \mathcal{F}_j = \int_{\mathbb{R}^3} 2Y_j \chi_R V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} dY,$$

$$\partial_{c_k} \mathcal{F}_j = \int_{\mathbb{R}^3} 2Y_j \chi_R V_{\lambda,c} \cdot \partial_{c_k} V_{\lambda,c} dY.$$

Thus each component admits pointwise C^1 formulas and these are continuous in (λ, c) near $(1, 0)$. Therefore $(\lambda, c) \mapsto \mathcal{F}(\lambda, c; V) \in C^1$ near $(1, 0)$. $\square$

Proposition 2.15 (Full local Jacobian at $(1, 0)$). *Let $V \in \mathcal{A}_{p,R}$. Set $V_0(Y) = V(Y)$. At $(\lambda, c) = (1, 0)$, the matrix*

$$J(V) := D_{(\lambda,c)} \mathcal{F}(1, 0; V) \in \mathbb{R}^{4 \times 4}$$

has rows

$$J_\alpha = (\partial_\lambda \mathcal{F}_\alpha, \partial_{c_1} \mathcal{F}_\alpha, \partial_{c_2} \mathcal{F}_\alpha, \partial_{c_3} \mathcal{F}_\alpha), \quad \alpha = 0, \dots, 3.$$

Writing $W_{\text{scl}} = V_0 + Y \cdot \nabla V_0$, one has

$$(2) \quad \partial_\lambda \mathcal{F}_0(1, 0; V) = -3 \int_{\mathbb{R}^3} \chi_R |Y|^{-p} |V_0| V_0 \cdot W_{\text{scl}} dY,$$

$$(3) \quad \partial_{c_k} \mathcal{F}_0(1, 0; V) = -3 \int_{\mathbb{R}^3} \chi_R |Y|^{-p} |V_0| V_0 \cdot \partial_k V_0 dY,$$

$$(4) \quad \partial_\lambda \mathcal{F}_j(1, 0; V) = -2 \int_{\mathbb{R}^3} Y_j \chi_R V_0 \cdot W_{\text{scl}} dY,$$

$$(5) \quad \partial_{c_k} \mathcal{F}_j(1, 0; V) = -2 \int_{\mathbb{R}^3} \chi_R Y_j V_0 \cdot \partial_k V_0 dY,$$

for $j, k = 1, 2, 3$.

Proof. Differentiate the formulas in Definition 2.10 at $(\lambda, c) = (1, 0)$.

Since $V_{\lambda,c}(Y) = \lambda^{-1} V(\lambda^{-1}(Y - c))$,

$$\partial_\lambda V_{\lambda,c} = -\lambda^{-1} V_{\lambda,c} - \lambda^{-1} Y \cdot \nabla V_{\lambda,c}, \quad \partial_{c_k} V_{\lambda,c} = -\lambda^{-2} \partial_k V(\lambda^{-1}(Y - c)).$$

At $(1, 0)$, $\partial_\lambda V_{\lambda,c}|_{(1,0)} = -(V + Y \cdot \nabla V)$, $\partial_{c_k} V_{\lambda,c}|_{(1,0)} = -\partial_k V$.

For $\mathcal{F}_0$, write

$$\mathcal{F}_0(\lambda, c; V) = \int \chi_R(Y) |Y|^{-p} |V_{\lambda,c}(Y)|^3 dY - \Theta_0.$$

Derivative in λ :

$$\partial_\lambda \mathcal{F}_0 = \int \chi_R |Y|^{-p} 3 |V_{\lambda,c}| V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} dY$$

Evaluating at $(1, 0)$ yields (2).

For $k = 1, 2, 3$, c_k -derivative of $\mathcal{F}_0$ has no cutoff term:

$$\partial_{c_k} \mathcal{F}_0(1, 0; V) = \int \chi_R |Y|^{-p} 3 |V| V \cdot (-\partial_k V) dY,$$

which is (3).

For $\mathcal{F}_j$,

$$\mathcal{F}_j(\lambda, c; V) = \int Y_j \chi_R(Y) |V_{\lambda,c}|^2 dY.$$

Hence

$$\partial_\lambda \mathcal{F}_j = \int 2Y_j \chi_R V_{\lambda,c} \cdot \partial_\lambda V_{\lambda,c} dY,$$

and at $(1, 0)$:

$$\partial_\lambda \mathcal{F}_j(1, 0; V) = -2 \int Y_j \chi_R V \cdot W_{\text{scl}} dY,$$

which is (4). Similarly,

$$\partial_{c_k} \mathcal{F}_j(1, 0; V) = \int 2Y_j \chi_R V \cdot \partial_{c_k} V_{\lambda, c}|_{(1,0)} dY = -2 \int \chi_R Y_j V \cdot \partial_k V dY,$$

giving (5). $\square$

Proposition 2.16 (Translation block and Schur-complement invertibility). *Let $V : J \rightarrow \mathcal{A}_{p,R}$ satisfy the repaired gauge constraints*

$$\mathcal{F}(1, 0; V(\tau)) = 0 \quad \text{for a.e. } \tau \in J.$$

Write the modulation matrix in block form

$$J(V(\tau)) = \begin{pmatrix} \alpha(\tau) & u(\tau)^T \\ v(\tau) & A(\tau) \end{pmatrix},$$

where $\alpha = \partial_\lambda \mathcal{F}_0$, $u_\ell = \partial_{c_\ell} \mathcal{F}_0$, $v_j = \partial_\lambda \mathcal{F}_j$, and $A_{j\ell} = \partial_{c_\ell} \mathcal{F}_j$. Assume that for a.e. $\tau \in J$

$$m_\chi(\tau) := \int |V(Y, \tau)|^2 \chi_R(Y) dY \geq m_0 > 0,$$

$$\int_{A_R} |V(Y, \tau)|^2 dY \leq \varepsilon_0, \quad 3C_\chi \varepsilon_0 \leq \frac{m_0}{2},$$

where $A_R = \{R \leq |Y| \leq 2R\}$ and

$$C_\chi := \max_{j,\ell} \|Y_j \partial_\ell \chi_R\|_{L^\infty}.$$

Assume also

$$C_u := \text{ess sup}_{\tau \in J} \int |Y|^{-p-1} |V(Y, \tau)|^3 dY < \infty$$

and, with $C_\nabla := \| |Y| \nabla \chi_R \|_{L^\infty}$,

$$6p \left(\frac{2}{m_0} \right) C_u R C_\nabla \varepsilon_0 \leq \frac{p\Theta_0}{2}.$$

Then $J(V(\tau))$ is invertible for a.e. $\tau \in J$, and

$$\text{ess sup}_{\tau \in J} \|J(V(\tau))^{-1}\|_{\infty \rightarrow \infty} \leq K(R, p, \Theta_0, m_0, C_u, \varepsilon_0, \chi_R).$$

Proof. The entries are finite by $V(\tau) \in \mathcal{A}_{p,R}$ and the compact support of χ_R . With the present parameter convention, $\partial_\lambda V_{\lambda, c}|_{(1,0)} = -(V + Y \cdot \nabla V)$. Hence the scale entry satisfies the signed transversality identity

$$\alpha(\tau) = -p \int |Y|^{-p} |V(Y, \tau)|^3 dY = -p\Theta_0$$

on the gauge surface. Thus $|\alpha| \geq p\Theta_0$.

For the centering block, integration by parts gives

$$A_{j\ell} = -\delta_{j\ell} \int |V|^2 \chi_R - \int |V|^2 Y_j \partial_\ell \chi_R.$$

Writing $A = -m_\chi I_3 + E$, the annular support of $\nabla \chi_R$ gives

$$|E_{j\ell}| \leq C_\chi \int_{A_R} |V|^2 \leq C_\chi \varepsilon_0, \quad \|E\|_{\infty \rightarrow \infty} \leq 3C_\chi \varepsilon_0 \leq m_0/2.$$

Thus $A = -m_\chi(I_3 - m_\chi^{-1}E)$ and $\|m_\chi^{-1}E\|_{\infty \rightarrow \infty} \leq 1/2$, so A^{-1} exists and $\|A^{-1}\|_{\infty \rightarrow \infty} \leq 2/m_0$.

For the scale-to-translation block, the repaired scale-row formula gives

$$|u_\ell| \leq p \int |Y|^{-p-1} |V|^3 dY \leq p \int |Y|^{-p-1} |V|^3 dY,$$

so $\|u\|_\infty \leq pC_u$. For the translation-to-scale block,

$$v_j = - \int |V|^2 Y_j (Y \cdot \nabla \chi_R) dY$$

because the centering constraint cancels the interior term. Therefore

$$\|v\|_\infty \leq 2R C_\nabla \varepsilon_0.$$

The Schur complement $S = \alpha - u^T A^{-1} v$ satisfies

$$|u^T A^{-1} v| \leq 3\|u\|_\infty \|A^{-1}\|_{\infty \rightarrow \infty} \|v\|_\infty \leq 6p \left(\frac{2}{m_0} \right) C_u R C_\nabla \varepsilon_0 \leq \frac{p\Theta_0}{2}.$$

Since $|\alpha| = p\Theta_0$, this gives $|S| \geq p\Theta_0/2$. The block inverse formula

$$J^{-1} = \begin{pmatrix} S^{-1} & -S^{-1} u^T A^{-1} \\ -A^{-1} v S^{-1} & A^{-1} + A^{-1} v S^{-1} u^T A^{-1} \end{pmatrix}$$

then gives the displayed bound. $\square$

2.7. Repaired gauge by implicit inversion in time.

Lemma 2.17 (Time chain rule for preliminary profile). *For each $\alpha = 0, \dots, 3$, set $G_\alpha(\tau) := \mathcal{F}_\alpha(1, 0; V_{\text{pre}}(\tau))$. Then $G_\alpha \in W_{\text{loc}}^{1,1}(I_*)$ and for a.e. τ ,*

$$G'_\alpha(\tau) = D_V \mathcal{F}_\alpha(1, 0; V_{\text{pre}}(\tau)) [\partial_\tau V_{\text{pre}}(\tau)].$$

The right-hand side is obtained from the profile formulas of Proposition 2.14 and is in $L_{\text{loc}}^1(I_)$.*

Proof. Let $V_{\text{pre}}^\varepsilon$ be a spatial and temporal Steklov mollification inside a slightly larger compact cylinder. For the smoothed profiles the ordinary chain rule gives

$$\frac{d}{d\tau} \mathcal{F}_\alpha(1, 0; V_{\text{pre}}^\varepsilon(\tau)) = D_V \mathcal{F}_\alpha(1, 0; V_{\text{pre}}^\varepsilon(\tau)) [\partial_\tau V_{\text{pre}}^\varepsilon(\tau)].$$

The transformed equation expresses $\partial_\tau V_{\text{pre}}^\varepsilon$ as the mollification of $\nu \Delta V_{\text{pre}} - (V_{\text{pre}} \cdot \nabla) V_{\text{pre}} - \nabla P_{\text{pre}}$ plus the chart transport terms. Pairing this distribution with the compactly supported Gateaux gradients below is legitimate by the local suitable bounds, the pressure bound, and the weighted L^3 control in (SC4). Passing $\varepsilon \downarrow 0$ gives the asserted $W_{\text{loc}}^{1,1}$ identity. For $W \in \mathcal{V}_R$, these derivative formulas hold:

$$\begin{aligned} D_V \mathcal{F}_0(1, 0; V)[W] &= \int_{\mathbb{R}^3} 3\chi_R |Y|^{-p} |V| V \cdot W dY, \\ D_V \mathcal{F}_j(1, 0; V)[W] &= \int_{\mathbb{R}^3} 2Y_j \chi_R V \cdot W dY, \quad j = 1, 2, 3, \end{aligned}$$

where both right-hand sides are finite on $\mathcal{V}_R$ by compact support, $p < 3$, and $V, W \in L^3 \cap W_{\text{loc}}^{1,3/2}$. The right-hand side is finite by the explicit formulas above and by the distributional equation paired with compactly supported test gradients; the local energy, pressure, and weighted L^3 bounds give an $L_{\text{loc}}^1(I_*)$ majorant. $\square$

Lemma 2.18 (AC implicit inversion with parameters). *Let $U \subset \mathbb{R}^4$ be open and $G : I_* \times U \rightarrow \mathbb{R}^4$. Assume:*

- (i) *for each $s \in U$, $\tau \mapsto G(\tau, s) \in W_{\text{loc}}^{1,1}(I_*; \mathbb{R}^4)$, and $D_\tau G \in L_{\text{loc}}^1(I_*; L^\infty(U; \mathbb{R}^4))$;*
- (ii) *for a.e. $\tau \in I_*$, $s \mapsto G(\tau, s)$ is C^1 on U , with $D_s G$ continuous in (τ, s) ;*

(iii) for some $(\tau_0, s_0) \in I_* \times U$, $G(\tau_0, s_0) = 0$, and there is $\rho, C > 0$ such that

$$\sup_{(\tau, s) \in I_* \times B_\rho(s_0)} \|D_s G(\tau, s)^{-1}\|_{\mathcal{L}(\mathbb{R}^4, \mathbb{R}^4)} \leq C.$$

Then after possibly shrinking I_* to a subinterval $J \ni \tau_0$, there is a unique map $s \in AC(J; U)$, $s(\tau_0) = s_0$, with

$$G(\tau, s(\tau)) = 0 \quad \text{a.e. in } J.$$

Moreover

$$s'(\tau) = -D_s G(\tau, s(\tau))^{-1} D_\tau G(\tau, s(\tau)) \quad \text{a.e. on } J.$$

Proof. By (ii), the matrix field $D_s G$ is continuous in s , so there is $r_0 \in (0, \rho)$ such that $\inf_{(\tau, s) \in I_* \times B_{r_0}(s_0)} |\det D_s G(\tau, s)| > 0$ after possibly shrinking I_* ; this keeps $D_s G^{-1}$ bounded and continuous on $B_{r_0}(s_0)$.

Set

$$F(\tau, s) := -D_s G(\tau, s)^{-1} D_\tau G(\tau, s).$$

Then $F(\tau, \cdot)$ is continuous on $B_{r_0}(s_0)$ for a.e. τ and $F(\cdot, s) \in L^1_{\text{loc}}(I_*; \mathbb{R}^4)$ by (i), hence F is a Carathéodory right-hand side. Since the gauge map is C^1 in the finite variables, after shrinking r_0 there is $L \in L^1(J)$ such that

$$|F(\tau, s_1) - F(\tau, s_2)| \leq L(\tau) |s_1 - s_2| \quad \text{for } s_1, s_2 \in B_{r_0}(s_0).$$

Choose $\delta > 0$ so small that

$$\int_{\tau_0 - \delta}^{\tau_0 + \delta} \sup_{s \in B_{r_0}(s_0)} |F(\sigma, s)| d\sigma < r_0.$$

On $J = [\tau_0 - \delta, \tau_0 + \delta] \cap I_*$ define the Picard operator

$$(\mathcal{T}s)(\tau) := s_0 + \int_{\tau_0}^{\tau} F(\sigma, s(\sigma)) d\sigma$$

on $C(J; B_{r_0}(s_0))$. The preceding bound keeps $\mathcal{T}$ in the same closed ball, and the Lipschitz estimate gives

$$\|\mathcal{T}s_1 - \mathcal{T}s_2\|_{C(J)} \leq \left(\int_J L(\sigma) d\sigma \right) \|s_1 - s_2\|_{C(J)}.$$

After decreasing δ once more, the coefficient is < 1 , so Banach's fixed-point theorem gives a unique solution

$$s(\tau) = s_0 + \int_{\tau_0}^{\tau} F(\sigma, s(\sigma)) d\sigma, \quad \tau \in J,$$

which stays in $B_{r_0}(s_0) \subset U$. This solution is absolutely continuous.

Define $H(\tau) := G(\tau, s(\tau))$. Since $G(\cdot, s(\cdot)) \in W^{1,1}_{\text{loc}}$ from (i), the chain rule gives for a.e. $\tau \in J$

$$H'(\tau) = D_\tau G(\tau, s(\tau)) + D_s G(\tau, s(\tau)) s'(\tau) = D_\tau G(\tau, s(\tau)) + D_s G(\tau, s(\tau)) F(\tau, s(\tau)) = 0.$$

Hence $H \equiv G(\tau_0, s_0) = 0$ on J , which is the claimed constraint. Uniqueness follows from Grönwall on the difference of two AC solutions on $B_{r_0}(s_0)$, using Lipschitz continuity of $F(\tau, \cdot)$ on that ball. $\square$

Definition 2.19 (Retained nondegenerate repaired-gauge regime). Let $J \subset I_*$ be compact. The retained nondegenerate repaired-gauge regime over the preliminary profile V_{pre} consists of branches for which there are

$$\lambda \in AC(J; (0, \infty)), \quad c \in AC(J; \mathbb{R}^3),$$

such that, for a.e. $\tau \in J$,

$$V(\tau) := V_{\text{pre}}^{\lambda(\tau), c(\tau)} \in \mathcal{A}_{p,R}, \quad \mathcal{F}(1, 0; V(\tau)) = 0.$$

The path is quantitatively nondegenerate if the translation-block, annular, weighted first-moment, and Schur-complement bounds in Proposition 2.16 hold on J . If no such local AC selection exists, or if the quantitative nondegeneracy hypotheses fail persistently, the branch lies in the gauge-degenerate, multibubble, or cascade alternatives rather than in the retained regime.

Proposition 2.20 (Gauge regularity on the retained regime). *Let (λ, c) be the repaired-gauge selection of the retained nondegenerate regime on J . Then*

$$\mathcal{F}(\lambda(\tau), c(\tau); V_{\text{pre}}(\tau)) = 0 \quad \text{for a.e. } \tau \in J,$$

and $(\lambda, c) \in AC(J; \mathbb{R}^4)$ by definition. If the path is quantitatively nondegenerate, then

$$\text{ess sup}_{\tau \in J} \|J(V(\tau))^{-1}\| < \infty.$$

Proof. The constraint follows from Lemma 2.13 applied to $V(\tau) = V_{\text{pre}}^{\lambda(\tau), c(\tau)}$. The absolute continuity is part of Definition 2.19. The uniform inverse bound is Proposition 2.16. $\square$

Definition 2.21 (Repaired chart). Define

$$\Lambda(\tau) := \Lambda_{\text{pre}}(\tau)\lambda(\tau), \quad X(\tau) := X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)c(\tau),$$

$$V(\tau) := V_{\text{pre}}^{\lambda(\tau), c(\tau)}, \quad P(\tau) := P_{\text{pre}}^{\lambda(\tau), c(\tau)} \text{ (same physical pressure pull-back under } \Phi).$$

Theorem 2.22 (AC repaired parameters). *Assume $SC(R_*, r_*, M, \eta, \kappa)$, and suppose the branch lies in the retained nondegenerate repaired-gauge regime on J , with selection (λ, c) . Then the corrected chart (Λ, X) is absolutely continuous on J , satisfies*

$$a(\tau) = -\Lambda'(\tau) \in L_{\text{loc}}^1(J), \quad b(\tau) = -X'(\tau) \in L_{\text{loc}}^1(J),$$

and the gauge constraints $\mathcal{F}(1, 0; V(\tau)) = 0$ hold a.e. Equivalently,

$$\mathcal{F}(\lambda(\tau), c(\tau); V_{\text{pre}}(\tau)) = 0$$

for the repaired-gauge selection (λ, c) .

Proof. By Definition 2.19, $(\lambda, c) \in AC(J)$ and the repaired gauge constraints hold for $V(\tau)$ a.e. By definition $V(\tau) = V_{\text{pre}}^{\lambda(\tau), c(\tau)}$, so Lemma 2.13 applied at $(\lambda, c) = (\lambda(\tau), c(\tau))$ gives

$$\mathcal{F}(1, 0; V_{\text{pre}}^{\lambda, c}) = \mathcal{F}(\lambda, c; V_{\text{pre}})$$

implies $\mathcal{F}(1, 0; V(\tau)) = 0$ a.e.

Since $\Lambda = \Lambda_{\text{pre}}\lambda$ and $X = X_{\text{pre}} + \Lambda_{\text{pre}}c$, and $\Lambda_{\text{pre}}, X_{\text{pre}}$ are AC by (SC5), the products and sums preserve AC, hence $(\Lambda, X) \in AC(I_*)$. Differentiating,

$$a = -\Lambda' = -(\Lambda'_{\text{pre}}\lambda + \Lambda_{\text{pre}}\lambda') \in L_{\text{loc}}^1, \quad b = -X' = -(X'_{\text{pre}} + \Lambda'_{\text{pre}}c + \Lambda_{\text{pre}}c') \in L_{\text{loc}}^1.$$

This gives the claimed regularity. $\square$

Remark 2.23 (Gauge degeneracy). Failure of invertibility of J is the gauge-degenerate alternative to the retained single-core regime for this fixed renormalization mechanism. In the Type II decomposition of alternatives this case belongs to the multibubble/cascade and scale-rigidity alternatives, rather than to the representation theorem.

2.8. Pressure transport and decomposition.

Lemma 2.24 (Renormalized pressure equation). *If V, P satisfy the renormalized Navier–Stokes equation in Theorem 2.6, then*

$$-\Delta P = \partial_i \partial_j (V_i V_j) \quad \text{in } \mathcal{D}'(Q_{2R}^*).$$

Proof. Take test $\varphi \in C_c^\infty(Q_{2R}^*)$. Since $V, P \in L_{\text{loc}}^2 \times L_{\text{loc}}^{3/2}$ on Q_{2R}^* , all pairings below are legitimate. Expand

$$\nabla \cdot ((V \cdot \nabla) V) = \sum_{i,j} \partial_i \partial_j (V_i V_j) - \sum_{i,j} \partial_i V_i \partial_j V_j.$$

Because $\nabla \cdot V = 0$, this reduces to $\partial_i \partial_j (V_i V_j)$. Taking divergence of

$$\partial_\tau V + (V \cdot \nabla) V + \nabla P - \nu \Delta V + a(V + Y \cdot \nabla V) + b \cdot \nabla V = 0$$

and using $\nabla \cdot V = 0$, $\nabla \cdot (V + Y \cdot \nabla V) = 0$, $\nabla \cdot (b \cdot \nabla V) = b \cdot \nabla (\nabla \cdot V) = 0$, and $\nabla \cdot \Delta V = \Delta (\nabla \cdot V) = 0$, we get in $\mathcal{D}'$:

$$\Delta P + \partial_i \partial_j (V_i V_j) = 0.$$

Hence $-\Delta P = \partial_i \partial_j (V_i V_j)$ on Q_{2R}^* . $\square$

Theorem 2.25 (Localized pressure decomposition). *Let $\zeta \in C_c^\infty(B_{2R})$, $\zeta \equiv 1$ on B_R , and assume $V \in L_{\text{loc}}^1(I_*; L^3(B_{2R}))$, $P \in L_{\text{loc}}^1(I_*; L^{3/2}(B_{2R}))$. For a.e. $\tau \in I_*$ there exists*

$$P(\cdot, \tau) = P_{\text{loc}}(\cdot, \tau) + H(\cdot, \tau)$$

on B_R such that

$$\begin{cases} -\Delta P_{\text{loc}}(\cdot, \tau) = \partial_i \partial_j (\zeta V_i V_j), \\ -\Delta H(\cdot, \tau) = 0 \text{ on } B_R, \quad \int_{B_R} H(\cdot, \tau) dY = 0. \end{cases}$$

Moreover

$$\|P_{\text{loc}}(\tau)\|_{L^{3/2}(B_R)} \lesssim \|V(\tau)\|_{L^3(B_{2R})}^2, \quad \|H(\tau)\|_{L^q(B_r)} \lesssim_{q,R,r} \|P(\tau)\|_{L^{3/2}(B_{2R})} + \|V(\tau)\|_{L^3(B_{2R})}^2$$

for each $1 \leq q < \infty$ and each $0 < r < R$, with constants depending only on R, r , and q .

Proof. For fixed τ , set

$$P_{\text{loc}} = \mathcal{R}_i \mathcal{R}_j (\zeta V_i V_j), \quad H := P - P_{\text{loc}}.$$

$\zeta V_i V_j \in L_{\text{loc}}^1(I_*; L^{3/2}(B_{2R}))$, so the Calderón–Zygmund operator $f \mapsto \mathcal{R}_i \mathcal{R}_j f$ gives

$$P_{\text{loc}}(\cdot, \tau) = \mathcal{R}_i \mathcal{R}_j (\zeta V_i V_j)(\cdot, \tau), \quad \|P_{\text{loc}}(\tau)\|_{L^{3/2}(B_R)} \lesssim \|V(\tau)\|_{L^3(B_{2R})}^2.$$

Because $\mathcal{R}_i \mathcal{R}_j$ is linear bounded on $L^{3/2}$,

$$P_{\text{loc}} \in L_{\text{loc}}^1(I_*; L^{3/2}(B_R)).$$

Since $-\Delta P_{\text{loc}} = \partial_i \partial_j (\zeta V_i V_j)$ and $\zeta \equiv 1$ on B_R ,

$$-\Delta H(\cdot, \tau) = \partial_i \partial_j ((1 - \zeta) V_i V_j) = 0 \quad \text{in } B_R,$$

so $H(\cdot, \tau)$ is harmonic on B_R . Imposing $\int_{B_R} H = 0$ fixes the time-dependent additive constant uniquely in each time slice. For $q \in [1, \infty)$ and $0 < r < R$, harmonicity and local L^p -to- L^q interior estimates imply

$$\|H(\tau)\|_{L^q(B_r)} \lesssim_{q,R,r} \|H(\tau)\|_{L^{3/2}(B_R)}.$$

Finally, $\|H(\tau)\|_{L^{3/2}(B_R)} \leq \|P(\tau)\|_{L^{3/2}(B_R)} + \|P_{\text{loc}}(\tau)\|_{L^{3/2}(B_R)}$, giving the announced bound in terms of P and V . $\square$

Remark 2.26 (Measurability and normalization). If V, P are measurable in τ , then $\tau \mapsto P_{\text{loc}}(\cdot, \tau)$ is measurable in $L^{3/2}(B_R)$ by boundedness of $\mathcal{R}_i \mathcal{R}_j$, hence $\tau \mapsto H(\cdot, \tau)$ is measurable in each $L^q(B_r)$, $0 < r < R$. The normalization $\oint_{B_R} H(\cdot, \tau) dY = 0$ fixes the time-dependent additive constant in H uniquely for each τ , and therefore the decomposition is a legitimate renormalized pressure splitting.

2.9. Modulation from differentiated constraints.

Lemma 2.27 (Differentiation of constraints). *For the repaired profile V constructed above, with $|Y|^{-p}|V|^3 \in L^1_{\text{loc}}(I_*; L^1)$, Then $G_\alpha(\tau) := \mathcal{F}_\alpha(1, 0; V(\tau))$ are absolutely continuous and*

$$\frac{d}{d\tau} G_\alpha(\tau) = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \partial_\tau V dY$$

for a.e. τ . Here $\mathbf{D}_\alpha$ is the L^1 -integrable Gateaux gradient associated to $\mathcal{F}_\alpha$.

Proof.

$$G_\alpha(\tau) = \mathcal{F}_\alpha(1, 0; V(\tau))$$

is first evaluated on spatial-temporal mollifications V^ε . For those smooth profiles the ordinary chain rule gives

$$\frac{d}{d\tau} G_\alpha^\varepsilon(\tau) = D_V \mathcal{F}_\alpha(1, 0; V^\varepsilon(\tau)) [\partial_\tau V^\varepsilon(\tau)].$$

The local suitable bounds, pressure estimates, and compact support of the Gateaux gradients below allow passage to the limit $\varepsilon \downarrow 0$, which gives the displayed derivative formula for V . For $\mathcal{F}_0$,

$$\mathbf{D}_0(V, \nabla V, \nabla \chi_R) = 3\chi_R |Y|^{-p}|V| V;$$

for $\alpha = 1, 2, 3$,

$$\mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) = 2\chi_R Y_\alpha V.$$

Each coefficient is $L^1(B_{2R})$: $3\chi_R |Y|^{-p}|V| V \in L^{3/2}(B_{2R}) \subset L^1(B_{2R})$ from $p < 3$ and $|Y|^{-p}|V|^3 \in L^1$, and $2\chi_R Y_\alpha V \in L^2(B_{2R}) \subset L^1(B_{2R})$ by compact support.

$$\frac{d}{d\tau} G_\alpha(\tau) = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \cdot \partial_\tau V dY,$$

Since $\mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \in L^1(B_{2R})$ and $\partial_\tau V \in L^1_{\text{loc}}(I_*; L^2(B_{2R}))$, the right-hand side is in $L^1_{\text{loc}}(I_*)$, so $G_\alpha \in W^{1,1}_{\text{loc}}(I_*) \subset AC_{\text{loc}}(I_*)$. $\square$

Lemma 2.28 (Directional derivatives in the profile variable). *For $\mathcal{A}_\tau = V + Y \cdot \nabla V$ and $\alpha = 0, 1, 2, 3$,*

$$D_V \mathcal{F}_\alpha(1, 0; V)[\mathcal{A}_\tau] = -\partial_\lambda \mathcal{F}_\alpha(1, 0; V), \quad D_V \mathcal{F}_\alpha(1, 0; V)[\partial_k V] = -\partial_{c_k} \mathcal{F}_\alpha(1, 0; V).$$

Indeed, from $V_{\lambda,c}(Y) = \lambda^{-1}V(\lambda^{-1}(Y - c))$,

$$\partial_\lambda V_{\lambda,c}|_{(1,0)} = -(V + Y \cdot \nabla V) = -\mathcal{A}_\tau, \quad \partial_{c_k} V_{\lambda,c}|_{(1,0)} = -\partial_k V.$$

Substituting these into the Gateaux formulas from Proposition 2.14 gives the two identities.

Theorem 2.29 (Modulation system). *Let (V, P) be the repaired pair on the compact window under consideration. Writing $\theta := (a, b)$, define*

$$\mathcal{A}_\tau(Y) := V(\tau, Y) + Y \cdot \nabla V(\tau, Y).$$

For $\alpha = 0, 1, 2, 3$, define

$$\mathcal{R}_\alpha(\tau) := - \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \cdot ((V \cdot \nabla)V + \nabla P - \nu \Delta V) dY.$$

Then for a.e. $\tau \in I_*$,

$$\partial_\lambda \mathcal{F}_\alpha(1, 0; V) a(\tau) + \sum_{k=1}^3 \partial_{c_k} \mathcal{F}_\alpha(1, 0; V) b_k(\tau) = \mathcal{R}_\alpha(\tau),$$

Equivalently,

$$J(\tau)\theta(\tau) = \mathcal{R}(\tau).$$

Hence

$$\theta(\tau) = J(\tau)^{-1} \mathcal{R}(\tau) \in L_{\text{loc}}^1(I_*)^4, \quad |\theta(\tau)| \leq \kappa |\mathcal{R}(\tau)|.$$

Proof. Set $G_\alpha(\tau) := \mathcal{F}_\alpha(1, 0; V(\tau))$. By Theorem 2.22, $G_\alpha(\tau) = 0$ a.e.; by Lemma 2.27, $G_\alpha \in W_{\text{loc}}^{1,1}$. Hence for a.e. τ ,

$$0 = \frac{d}{d\tau} G_\alpha(\tau) = D_V \mathcal{F}_\alpha(1, 0; V(\tau)) [\partial_\tau V(\tau)].$$

Insert the equation from Theorem 2.6,

$$\partial_\tau V = -(V \cdot \nabla) V - \nabla P + \nu \Delta V - a \mathcal{A}_\tau - b \cdot \nabla V.$$

This gives

$$0 = -D_V \mathcal{F}_\alpha[(V \cdot \nabla) V + \nabla P - \nu \Delta V] - a D_V \mathcal{F}_\alpha[\mathcal{A}_\tau] - \sum_{k=1}^3 b_k D_V \mathcal{F}_\alpha[\partial_k V].$$

By Lemma 2.27, for any profile direction W ,

$$D_V \mathcal{F}_\alpha(1, 0; V)[W] = \int_{\mathbb{R}^3} \mathbf{D}_\alpha(V, \nabla V, \nabla \chi_R) \cdot W \, dY.$$

By Lemma 2.28,

$$D_V \mathcal{F}_\alpha[\mathcal{A}_\tau] = -\partial_\lambda \mathcal{F}_\alpha(1, 0; V), \quad D_V \mathcal{F}_\alpha[\partial_k V] = -\partial_{c_k} \mathcal{F}_\alpha(1, 0; V).$$

Substituting into the previous line yields the linear system

$$\partial_\lambda \mathcal{F}_\alpha(1, 0; V) a(\tau) + \sum_{k=1}^3 \partial_{c_k} \mathcal{F}_\alpha(1, 0; V) b_k(\tau) = \mathcal{R}_\alpha(\tau),$$

which is equivalent to $J(\tau)\theta(\tau) = \mathcal{R}(\tau)$. By definition of $\mathbf{D}_\alpha$, each $\mathcal{R}_\alpha \in L_{\text{loc}}^1(I_*)$, and Proposition 2.16 gives $|J^{-1}(\tau)| \leq K$ a.e., hence

$$\theta(\tau) = J^{-1}(\tau) \mathcal{R}(\tau), \quad \theta \in L_{\text{loc}}^1(I_*), \quad |\theta| \leq K |\mathcal{R}| \text{ a.e..}$$

□

Corollary 2.30 (Regularity of a, b). *For the repaired pair of Theorem 2.29, $a, b \in L_{\text{loc}}^1(I_*)$. If in addition $\mathcal{R} \in L_{\text{loc}}^\infty(I_*)$, then $a, b \in L_{\text{loc}}^\infty(I_*)$.*

Proof. By Theorem 2.29, $\theta = (a, b) = J^{-1} \mathcal{R}$ a.e., and $J^{-1} \in L_{\text{loc}}^\infty(I_*)$ by Theorem 2.16. The first claim is therefore immediate from $\mathcal{R} \in L_{\text{loc}}^\infty$, while the second follows if $\mathcal{R} \in L_{\text{loc}}^\infty$. □

2.10. Repaired suitable structure in renormalized variables.

Proposition 2.31 (Transport of suitable local energy inequality). *Suppose (u, p) is suitable on $\mathcal{Q}_{2R}^{\text{ph}}$, and (V, P) is obtained by the repaired map Φ . Set*

$$\tilde{a}(\tau) := -\Lambda(\tau)^{-1}\Lambda'(\tau), \quad \tilde{b}(\tau) := -\Lambda(\tau)^{-1}X'(\tau).$$

Then for every nonnegative $\phi \in C_c^\infty(Q_R^*)$ and $\tau_1 < \tau_2$,

$$\begin{aligned} \mathcal{E}_\phi(V, P) &:= \frac{1}{2}|V|^2\partial_\tau(\phi^2) + \frac{\nu}{2}|V|^2\Delta_Y\phi^2 + \frac{1}{2}|V|^2V \cdot \nabla(\phi^2) \\ &\quad + (P_{\text{loc}} + H)V \cdot \nabla(\phi^2) + \frac{\tilde{a}}{2}|V|^2\phi^2 + \frac{\tilde{a}}{2}|V|^2Y \cdot \nabla(\phi^2) + \frac{1}{2}|V|^2\tilde{b} \cdot \nabla(\phi^2). \end{aligned}$$

$$\begin{aligned} \frac{1}{2} \int_{B_R} |V(\tau_2)|^2 \phi(\cdot, \tau_2)^2 + \nu \int_{\tau_1}^{\tau_2} \int_{B_R} |\nabla V|^2 \phi^2 dY d\tau &\leq \frac{1}{2} \int_{B_R} |V(\tau_1)|^2 \phi(\cdot, \tau_1)^2 \\ &\quad + \int_{\tau_1}^{\tau_2} \int_{B_R} \mathcal{E}_\phi(V, P) dY d\tau. \end{aligned}$$

The right-hand side is controlled on compact windows by $\|\tilde{a}\|_{L^1}$, $\|\tilde{b}\|_{L^1}$, M , and the local bounds from Theorem 2.25, and support bounds from Section 2.4. Hence (V, P) is locally suitable on Q_R^* .

Proof. Apply the physical local energy inequality on $\mathcal{Q}_{2R}^{\text{ph}}$: for nonnegative $\psi \in C_c^\infty(\mathcal{Q}_{2R}^{\text{ph}})$,

$$\frac{1}{2} \int |u|^2 \psi|_{t_1}^{t_2} + \nu \int_{t_1}^{t_2} \int |\nabla_x u|^2 \psi dx dt \leq \int_{t_1}^{t_2} \int \left(\frac{1}{2} |u|^2 (\partial_t \psi + \nu \Delta_x \psi) + \left(\frac{|u|^2}{2} u + p u \right) \cdot \nabla_x \psi \right) dx dt.$$

Choose

$$\psi(x, t) := \Lambda(t)^{-1} \phi \left(\frac{x - X(t)}{\Lambda(t)}, \tau(t) \right)^2, \quad \tau = \tau(t), \quad d\tau/dt = \Lambda^{-2},$$

let $t_i = t(\tau_i)$, $i = 1, 2$. By Proposition 2.3 and Section 2.4, $\psi \in C_c^\infty(\mathcal{Q}_{2R}^{\text{ph}})$.

$$\nabla_x \psi = \Lambda^{-2} \nabla_Y \phi^2, \quad \Delta_x \psi = \Lambda^{-3} \Delta_Y \phi^2, \quad \partial_t \psi = -\Lambda^{-4} \Lambda' \phi^2 + \Lambda^{-3} \partial_\tau(\phi^2) - \Lambda^{-4} (X' + \Lambda' Y) \cdot \nabla_Y \phi^2.$$

Using $u = \Lambda^{-1}V$, $p = \Lambda^{-2}P$, $\nabla_x u = \Lambda^{-2} \nabla_Y V$, $dx dt = \Lambda^5 dY d\tau$, each term in the physical inequality becomes

$$\begin{aligned} \frac{1}{2} \int |u|^2 \psi|_{t_1}^{t_2} &= \frac{1}{2} \int |V|^2 \phi^2|_{\tau_1}^{\tau_2}, \\ \nu \int |\nabla_x u|^2 \psi dx dt &= \nu \int |\nabla V|^2 \phi^2 dY d\tau, \\ \frac{1}{2} \int |u|^2 \partial_t \psi dx dt &= \frac{1}{2} \int |V|^2 \left(\partial_\tau(\phi^2) + \tilde{a} \phi^2 + \tilde{a} Y \cdot \nabla_Y \phi^2 + \tilde{b} \cdot \nabla_Y \phi^2 \right) dY d\tau, \\ \frac{\nu}{2} \int |u|^2 \Delta_x \psi dx dt &= \frac{\nu}{2} \int |V|^2 \Delta_Y \phi^2 dY d\tau, \\ \int \left(\frac{|u|^2}{2} u + p u \right) \cdot \nabla_x \psi dx dt &= \int \left(\frac{1}{2} |V|^2 V + (P_{\text{loc}} + H)V \right) \cdot \nabla_Y \phi^2 dY d\tau. \end{aligned}$$

Substituting these identities and grouping terms gives the proposition's right-hand side and coefficients $\tilde{a}, \tilde{b}$. $\square$

2.11. Stability under subsequences.

Proposition 2.32. *Let $(u^{(n)}, p^{(n)})$ satisfy $\text{SC}(R_*, r_*, M, \eta, \kappa)$ with the same data parameters and let $(\Lambda^{(n)}, X^{(n)}, V^{(n)}, P^{(n)}, a^{(n)}, b^{(n)})$ be obtained from retained quantitatively nondegenerate repaired-gauge selections on the same I_* , with the constants in Proposition 2.16 uniform in n . After extraction $n_j \rightarrow \infty$, there exist subsequences and limits (Λ, X, V, P, a, b) such that*

$$\begin{aligned} \Lambda^{(n_j)} &\rightarrow \Lambda \text{ in } L_{\text{loc}}^1, & X^{(n_j)} &\rightarrow X \text{ in } L_{\text{loc}}^1, \\ \Lambda^{(n_j)'} &\rightharpoonup \Lambda' \text{ in } L_{\text{loc}}^1, & X^{(n_j)'} &\rightharpoonup X' \text{ in } L_{\text{loc}}^1, \\ V^{(n_j)} &\rightarrow V \text{ in } L_{\text{loc}}^2(I_*; L_{\text{loc}}^2), & V^{(n_j)} &\rightharpoonup V \text{ in } L_{\text{loc}}^2(I_*; H_{\text{loc}}^1), \\ P^{(n_j)} &\rightharpoonup P \text{ in } L_{\text{loc}}^{3/2}(Q_R^*), & (a^{(n_j)}, b^{(n_j)}) &\rightharpoonup (a, b) \text{ in } L_{\text{loc}}^1(I_*)^4. \end{aligned}$$

Moreover $a = -\Lambda'$, $b = -X'$ a.e. on I_* , and the limit obeys all conclusions below.

Proof. From SC and the retained repaired-gauge selections, $(V^{(n)}, P^{(n)})$ are uniformly bounded in $L_\tau^\infty L^2(B_{2R}) \cap L_\tau^2 H^1(B_{2R})$ and $L_{\text{loc}}^{3/2}(Q_R^*)$. Theorem 2.16 gives uniform invertibility bounds for $J^{(n)}$, and Theorem 2.29 gives $(a^{(n)}, b^{(n)})$ uniformly in L_{loc}^1 (since $\mathcal{R}^{(n)} \in L_{\text{loc}}^1$). Hence $\Lambda^{(n)}$ and $X^{(n)}$ are uniformly bounded in $W_{\text{loc}}^{1,1}$, so by BV compactness and 1D subsequences,

$$\Lambda^{(n_j)} \rightarrow \Lambda \text{ in } L_{\text{loc}}^1, \quad X^{(n_j)} \rightarrow X \text{ in } L_{\text{loc}}^1,$$

and $a^{(n_j)} \rightharpoonup -\Lambda'$, $b^{(n_j)} \rightharpoonup -X'$ in L_{loc}^1 . From the renormalized equation,

$$\partial_\tau V^{(n)} \in L_\tau^1(H_{\text{loc}}^{-1}) + L_\tau^{4/3}(L_{\text{loc}}^{12/11}),$$

uniformly on compact windows. Aubin–Lions compactness on bounded balls gives

$$V^{(n_j)} \rightarrow V \text{ in } L_{\text{loc}}^2(I_*; L_{\text{loc}}^2), \quad V^{(n_j)} \rightharpoonup V \text{ in } L_{\text{loc}}^2(I_*; H_{\text{loc}}^1).$$

By Calderón–Zygmund and Theorem 2.25, $P^{(n_j)} \rightharpoonup P$ in $L_{\text{loc}}^{3/2}(Q_R^*)$, and with normalized pressure split $P = P_{\text{loc}} + H$, the same for P_{loc} and H . Passing to limits in $\mathcal{F}(1, 0; V^{(n)}(\tau)) = 0$, Theorem 2.6, and the linear modulation system yields the corresponding limit constraints and equations. $\square$

2.12. Final representation theorem.

Theorem 2.33 (Chart and repaired-gauge representation). *Assume $\text{SC}(R_*, r_*, M, \eta, \kappa)$, and suppose the branch lies in the retained repaired-gauge regime with selection (λ, c) on $J = [\tau_0 - \ell, \tau_0 + \ell] \subset I_*$. Then there are fields (V, P) on Q_{2R}^J , with $(\Lambda, X) \in AC(J)$, $a = -\Lambda'$, $b = -X' \in L_{\text{loc}}^1(J)$, such that*

$$u(x, t) = \Lambda(\tau)^{-1} V\left(\frac{x - X(\tau)}{\Lambda(\tau)}, \tau\right), \quad \frac{dt}{d\tau} = \Lambda(\tau)^2,$$

and $\mathcal{F}(1, 0; V(\tau)) = 0$ for $\tau \in J$.

Proof. From (SC5) and Proposition 2.3, $(\Lambda_{\text{pre}}, X_{\text{pre}})$ are AC on I_* , so V_{pre} is defined on Q_{2R}^* . The retained repaired-gauge regime supplies $(\lambda, c) \in AC(J; (0, \infty) \times \mathbb{R}^3)$ and $\mathcal{F}(1, 0; V_{\text{pre}}^{\lambda, c}) = 0$ a.e. By Definition 2.21 and Theorem 2.22, setting

$$\Lambda(\tau) = \Lambda_{\text{pre}}(\tau)\lambda(\tau), \quad X(\tau) = X_{\text{pre}}(\tau) + \Lambda_{\text{pre}}(\tau)c(\tau), \quad V(\tau) = V_{\text{pre}}^{\lambda(\tau), c(\tau)},$$

and $P(\tau) = P_{\text{pre}}^{\lambda(\tau), c(\tau)}$, gives $\Lambda, X \in AC(J)$ and $a = -\Lambda'$, $b = -X' \in L_{\text{loc}}^1(J)$, and

$$\mathcal{F}(1, 0; V(\tau)) = 0 \text{ for a.e. } \tau \in J.$$

Finally $u = \Lambda^{-1}V(\cdot, \tau) \circ \Phi^{-1}$ by definition of the repaired map, hence the physical identity with $\frac{dt}{d\tau} = \Lambda^2$ follows. $\square$

Theorem 2.34 (Pressure and modulation). *Assume, in addition to Theorem 2.33, that the retained repaired-gauge selection is quantitatively nondegenerate in the sense of Proposition 2.16. Then (V, P) satisfies the renormalized Navier–Stokes equation with the sign convention of Theorem 2.6; there is a decomposition $P = P_{\text{loc}} + H$ on Q_R^* as in Theorem 2.25; and (a, b) is the unique solution of*

$$J(\tau) \begin{pmatrix} a(\tau) \\ b(\tau) \end{pmatrix} = \mathcal{R}(\tau), \quad \begin{pmatrix} a \\ b \end{pmatrix} \in L_{\text{loc}}^1(J)^4,$$

with bounds $|(a, b)| \leq K|\mathcal{R}|$, where K depends on $R, p, \Theta_0, m_0, C_u, \varepsilon_0, \chi_R$.

Proof. Equation and sign convention are inherited directly from Theorem 2.6 applied to (Λ, X) from Theorem 2.33 and the definition of V . By Theorem 2.25 applied at each time slice, the decomposition on Q_R^J is defined and unique once $\int_{B_R} H = 0$ is imposed. The modulation identity is Theorem 2.29; uniqueness follows from invertibility of J (Theorem 2.16) and the pointwise identity $(a, b) = J^{-1}\mathcal{R}$. Uniform bounds on (a, b) are obtained by the bound on J^{-1} there and the stated bound on $\mathcal{R}$. $\square$

Theorem 2.35 (Compact-window representation theorem). *Under the assumptions of Theorems 2.33 and 2.34, for every compact window $J \subset I_*$ and each fixed $R > 0$, setting $Q_R^J := B_R \times J$, one has:*

- (1) the exact renormalized equation on Q_{2R}^J ,
- (2) the four repaired-gauge constraints $\mathcal{F}(1, 0; V(\tau)) = 0$,
- (3) bounds on $a, b \in L_{\text{loc}}^1(J)$ (and L_{loc}^∞ under $\mathcal{R} \in L_{\text{loc}}^\infty$),
- (4) local pressure decomposition $P = P_{\text{loc}} + H$ with $L^{3/2}$ local bounds and harmonic remainder controls,
- (5) renormalized local suitable energy inequality on Q_R^J ,
- (6) stability under subsequences in the topologies above.

The later reductions use only these conclusions.

Proof. From Theorem 2.33 and the choice of $J \subset I_*$, all quantities are defined on Q_{2R}^J . Items (1) and (2) are Theorems 2.6 and 2.22; item (3) is Theorem 2.29 together with the inverse bound from Proposition 2.16; item (4) is Theorem 2.25; item (5) is Proposition 2.31; item (6) is Proposition 2.32. Combining these results gives the stated compact-window representation. $\square$

Remark 2.36. The preceding results separate two alternatives:

- (1) the nondegenerate single-core repaired-gauge regime, where the representation theorems apply;
- (2) gauge degeneracy ($\det J = 0$) or compact-structure failure, which belongs to the multi-bubble, cascade, or scale-rigidity alternatives.

3. MULTIBUBBLE AND CASCADE EXCLUSION

Analytic inputs.

The local reduction theorem uses the following analytic inputs from the preceding sections, with profile decompositions and decoupling estimates in their standard critical-space sense [Gal01, BL83]. The remaining work is the multibubble, cascade, and terminal-frame decoupling.

- (i) From Section 1, the compact single-core criterion Theorem 1.18: a compact single-core finite-cost Type II branch with positive local CKN concentration, nondegenerate scale-center selection, pressure reconstruction, bounded compact-cylinder suitable estimates, bounded modulation, and vanishing localized dissipation is excluded.
- (ii) From Section 2, the renormalized equation Theorem 2.6, pressure decomposition Theorem 2.25, modulation system Theorem 2.29, and repaired-gauge representation Theorems 2.33 to 2.35: the renormalized equation, pressure decomposition modulo functions of time, bounded modulation coefficients, renormalized suitable structure, and stability of these structures under subsequences.
- (iii) From Section 4, the compact-cylinder Caccioppoli estimate Theorem 4.4 and rough-core exclusion Theorem 4.7.
- (iv) From Section 5, the collapse cost exclusion Theorems 5.6 and 5.7 and autonomous reduced-limit theorem Theorem 5.19 for scale-rigid or scale-collapsing branches.

3.1. Local setting and active frames. *Normalized local Navier–Stokes setting.* All quantities are written in rescaled local variables $(x, t) \in Q_R = B_R \times (-R^2, 0)$ with viscosity $\nu = 1$ (critical scaling has absorbed ν). A local pair (u_n, p_n) on such a cylinder satisfies

$$\partial_t u_n - \Delta u_n + (u_n \cdot \nabla) u_n + \nabla p_n = 0, \quad \nabla \cdot u_n = 0,$$

and pressure is normalised through spatial averaging

$$(p_n)_{B_\rho}(t) := |B_\rho|^{-1} \int_{B_\rho} p_n(x, t) dx$$

to factor out time-dependent constants.

For $r > 0$, $x_0 \in \mathbb{R}^3$, and $t_0 \in \mathbb{R}$, let

$$Q_r(x_0, t_0) := B_r(x_0) \times (t_0 - r^2, t_0), \quad Q_r := Q_r(0, 0) = B_r \times (-r^2, 0).$$

Let (u_n, p_n) be parabolic rescalings around a potential singular point with positive local Caffarelli–Kohn–Nirenberg concentration on compact cylinders. Concretely, for a base triple (x_n, t_n, λ_n) ,

$$u_n(y, s) = \lambda_n u(x_n + \lambda_n y, t_n + \lambda_n^2 s), \quad p_n(y, s) = \lambda_n^2 p(x_n + \lambda_n y, t_n + \lambda_n^2 s).$$

For each n and $R > 0$, define the scaling-invariant local quantities

$$A_n(R) := R^{-1} \operatorname{ess\,sup}_{-R^2 < t < 0} \int_{B_R} |u_n(x, t)|^2 dx,$$

$$E_n(R) := R^{-1} \int_{Q_R} |\nabla u_n(x, t)|^2 dx dt,$$

$$C_n(R) := R^{-2} \int_{Q_R} |u_n(x, t)|^3 dx dt, \quad D_n(R) := R^{-2} \int_{-R^2}^0 \int_{B_R} |p_n - (p_n)_{B_R}(t)|^{3/2} dx dt,$$

where

$$(p_n)_{B_R}(t) := |B_R|^{-1} \int_{B_R} p_n(x, t) dx.$$

For an active frame (x_n, λ_n) , $\lambda_n > 0$, define the critical rescaling

$$\mathcal{T}_{x_n, \lambda_n} f(y) := \lambda_n f(x_n + \lambda_n y).$$

If $\phi \in L^3_\sigma(\mathbb{R}^3)$ is observed in this frame, its contribution is

$$\phi_n(y) = \mathcal{T}_{x_n, \lambda_n} \phi(y) = \lambda_n \phi(x_n + \lambda_n y).$$

The critical Jacobian identity is exact:

$$\|\mathcal{T}_{x_n, \lambda_n} f\|_{L^3(E)}^3 = \int_{x_n + \lambda_n E} |f(x)|^3 dx$$

for every measurable $E \subset \mathbb{R}^3$. In particular

$$\|\mathcal{T}_{x_n, \lambda_n} f\|_{L^3(\mathbb{R}^3)}^3 = \|f\|_{L^3(\mathbb{R}^3)}^3,$$

so $\mathcal{T}_{x_n, \lambda_n}$ is an isometry on $L^3(\mathbb{R}^3)$.

Definition 3.1 (Bounded selected-window limit). A selected-window subsequence has a bounded selected-window limit on a compact cylinder Q_R if there exists $u_\infty \in L^\infty(Q_R)$ such that, after extracting a subsequence,

$$\forall \varepsilon > 0 \exists N : \forall n \geq N, \|u_n - u_\infty\|_{L^3(Q_R)} < \varepsilon.$$

The limit is nonzero if $u_\infty \not\equiv 0$ on Q_R .

Definition 3.2 (Local Type II sequence). A positive local concentration sequence is called a local Type II sequence if every selected-window rescaling has zero bounded selected-window limit on every compact cylinder Q_R :

$$\forall R > 0, \quad \lim_{n \rightarrow \infty} \|u_n\|_{L^3(Q_R)} = 0.$$

Definition 3.3 (Active frame). Fix a concentrating time sequence $t_n \uparrow T$ and a compact time interval $I \Subset (-\infty, 0]$. An active frame is a tuple

$$\mathfrak{F}^j = (x_n^j, \lambda_n^j, I, U_n^j, P_n^j), \quad \lambda_n^j \downarrow 0,$$

where $x_n^j \in \mathbb{R}^3$ and

$$U_n^j(y, s) := \lambda_n^j u(x_n^j + \lambda_n^j y, t_n + (\lambda_n^j)^2 s), \quad P_n^j(y, s) := (\lambda_n^j)^2 p(x_n^j + \lambda_n^j y, t_n + (\lambda_n^j)^2 s)$$

on $B_{2R} \times I$ for every fixed compact cylinder under consideration. The frame is active at threshold $\eta_* > 0$ if for some fixed $R_0 < \infty$

$$\liminf_{n \rightarrow \infty} \int_{Q_{R_0}} |U_n^j|^3 dy ds \geq \eta_*^3.$$

Definition 3.4 (Relations between active frames). Let $\mathfrak{F}^i$ and $\mathfrak{F}^j$ be two active frames. After passing to a subsequence exactly one of the following parameter regimes holds.

(i) They are *same-point comparable-scale* if

$$0 < c \leq \frac{\lambda_n^i}{\lambda_n^j} \leq C < \infty, \quad \frac{|x_n^i - x_n^j|}{\max(\lambda_n^i, \lambda_n^j)} \leq C$$

for all large n .

(ii) They form a *same-point cascade* if, after relabeling,

$$\frac{\lambda_n^i}{\lambda_n^j} \rightarrow 0, \quad \frac{|x_n^i - x_n^j|}{\lambda_n^j} = O(1).$$

The frame i is then the inner frame and j is an outer frame.

(iii) They are *separated-point* if

$$\frac{|x_n^i - x_n^j|}{\lambda_n^i} \rightarrow \infty \quad \text{and} \quad \frac{|x_n^i - x_n^j|}{\lambda_n^j} \rightarrow \infty.$$

Definition 3.5 (Terminal frame and perturbative remainder). Given a finite active family, a terminal frame is chosen by selecting an active scale with minimal order:

$$\lambda_n^{j*} \leq C_j \lambda_n^j \quad \text{for every active } j$$

after passing to a subsequence. Comparable frames at the same physical point are grouped before this choice; if several grouped frames remain tied, one is chosen by a fixed lexicographic ordering of their labels. A remainder r_n in a terminal frame is perturbative on $B_{2R} \times I$ if

$$\|r_n\|_{L_T^\infty L^3(B_{2R})} \rightarrow 0$$

and its equation residual and pressure satisfy

$$\|f_n\|_{L_t^1 H^{-1}(B_R)} + \|\nabla \pi_n^r\|_{L_t^1 H^{-1}(B_R)} \rightarrow 0$$

for every compact $B_R \times I$.

Definition 3.6 (Local multibubble/cascade branch). A local multibubble/cascade branch is a positive-concentration local Type II branch for which at least one of the following alternatives occurs:

- (i) the compact-cylinder suitable estimates $A_n + E_n + C_n + D_n$ are unbounded on some fixed rescaled cylinder;
- (ii) positive CKN mass splits into two or more comparable active cores;
- (iii) a strict same-point scale cascade appears inside bounded rescaled cylinders;
- (iv) separated physical points remain active in different local rescaled frames;
- (v) the localized scale or centering selection degenerates (no unique core selected).

Definition 3.7 (Active profile family). An active profile family is a family

$$\{(\phi^j, x_n^j, \lambda_n^j)\}_{j \in J}, \quad \phi^j \in L_\sigma^3(\mathbb{R}^3),$$

with finite or countable index set J , extracted along a positive local concentration sequence. In physical variables write

$$\Lambda_{\lambda, x_0} \phi(x) := \lambda^{-1} \phi\left(\frac{x - x_0}{\lambda}\right).$$

For every finite $J_0 \subset J$ the family gives an expansion

$$u(\cdot, t_n) = \sum_{j \in J_0} \Lambda_{\lambda_n^j, x_n^j} \phi^j + r_n^{J_0}$$

on the compact terminal windows under consideration. After the finite active subfamily has been selected, the corresponding remainder satisfies the perturbative bounds of [Theorem 3.5](#). The parameters are considered only after grouping same-point comparable-scale profiles into compound profiles as in [Theorem 3.8](#). A profile is active if

$$\|\phi^j\|_{L^3(\mathbb{R}^3)} \geq \eta_* > 0.$$

The family has finite critical mass when

$$\sum_{j \in J} \|\phi^j\|_{L^3(\mathbb{R}^3)}^3 \leq M_* < \infty.$$

It captures positive local concentration if every compact terminal-window sequence with

$$\liminf_{n \rightarrow \infty} \int_{Q_R} |u_n|^3 > 0$$

either contains an active profile observed in that window or is absorbed into the selected terminal frame; equivalently, no positive terminal-window concentration remains in the perturbative remainder.

Definition 3.8 (Compound profile). Let $J_k \subset J$ be a same-point comparable-scale class. Choose one representative $(x_n^{j_k}, \lambda_n^{j_k})$. After passing to a subsequence,

$$\rho_j := \lim_{n \rightarrow \infty} \frac{\lambda_n^j}{\lambda_n^{j_k}} \in (0, \infty), \quad z_j := \lim_{n \rightarrow \infty} \frac{x_n^j - x_n^{j_k}}{\lambda_n^{j_k}} \in \mathbb{R}^3.$$

The compound profile in the representative frame is the L_σ^3 -sum

$$\Phi^k(y) := \sum_{j \in J_k} \rho_j^{-1} \phi^j \left(\frac{y - z_j}{\rho_j} \right),$$

whenever J_k is finite; finiteness follows for active classes from [Theorem 3.9](#). If $\|\Phi^k\|_{L^3} < \varepsilon_*$, where ε_* is the critical small-data threshold, the class is perturbative. Otherwise Φ^k is treated as one active object in its representative frame.

Lemma 3.9 (Finite active profile count). *Every active profile family with finite critical mass contains only finitely many active profiles. More precisely,*

$$\#J \leq M_* \eta_*^{-3}.$$

Proof. For each active profile $j \in J$,

$$\|\phi^j\|_{L^3(\mathbb{R}^3)}^3 \geq \eta_*^3.$$

Let $J_0 \subset J$ be finite. Summing over J_0 gives

$$\#J_0 \eta_*^3 \leq \sum_{j \in J_0} \|\phi^j\|_{L^3(\mathbb{R}^3)}^3 \leq \sum_{j \in J} \|\phi^j\|_{L^3(\mathbb{R}^3)}^3 \leq M_*.$$

Hence $\#J_0 \leq M_* \eta_*^{-3}$. Taking the supremum over all finite subsets J_0 yields the same bound for the cardinality of J , so J is finite and

$$\#J \leq M_* \eta_*^{-3}.$$

□

Lemma 3.10 (Exhaustive parameter partition). *Let $\mathfrak{F}^i, \mathfrak{F}^j$ be two active frames. After passing to a subsequence, either the pair is same-point comparable-scale, same-point cascade, or separated-point in the sense of [Theorem 3.4](#); if none of these relations holds in a selected compact frame, then one of the two contributions is locally invisible there in L^3 .*

Proof. Pass to a subsequence so that

$$\frac{\lambda_n^i}{\lambda_n^j} \rightarrow \ell \in [0, \infty]$$

in the extended half-line. If $0 < \ell < \infty$, then either

$$\frac{|x_n^i - x_n^j|}{\max(\lambda_n^i, \lambda_n^j)}$$

is bounded along a further subsequence, giving the same-point comparable-scale case, or it tends to infinity, giving separated points because the two scales are comparable.

If $\ell = 0$, then either

$$\frac{|x_n^i - x_n^j|}{\lambda_n^j}$$

is bounded, giving a same-point cascade with i inner and j outer, or it tends to infinity. In the latter case the j -profile is locally invisible in the i -frame by [Theorem 3.13](#). The case $\ell = \infty$ is the same after interchanging i and j . These alternatives exhaust all subsequential limits of the scale ratio and normalized center distance. $\square$

Theorem 3.11 (Minimal bad-configuration extraction). *Let a positive local Type II branch admit an active profile family with finite critical mass, perturbative remainder in the sense of [Theorem 3.5](#), and concentration capture in the sense of [Theorem 3.7](#). If the branch is not already a single active terminal frame satisfying the single-core hypotheses, then after passing to a subsequence one of the following occurs:*

- (i) *compact-cylinder suitable control fails, and [Theorem 3.15](#) produces a bounded-cylinder concentration core;*
- (ii) *there are at least two active frames in a same-point comparable-scale class, hence a compound profile in the sense of [Theorem 3.8](#);*
- (iii) *there is a same-point cascade;*
- (iv) *there are separated active physical points;*
- (v) *the scale-center selection of the selected compound or terminal frame is degenerate.*

In every case the positive local concentration is carried by an active frame or compound profile, not by the perturbative remainder.

Proof. If compact-cylinder suitable control fails, item (i) follows from [Theorem 3.15](#). Assume therefore that the compact-cylinder family is bounded on the windows under consideration.

By concentration capture, positive local concentration cannot remain solely in the perturbative remainder. Hence either one selected active frame carries the terminal-window concentration, or another active frame remains non-negligible. If there is one nondegenerate selected active frame and no competing active object, the branch is in the single-core regime. Since this case is excluded by hypothesis, either the selected scale-center map is degenerate, giving item (v), or at least two active frames remain.

For two active frames, apply [Theorem 3.10](#). Same-point comparable-scale frames give item (ii), same-point scale separation gives item (iii), and separated points give item (iv). The remaining alternative in [Theorem 3.10](#) is local invisibility of one contribution in the selected compact frame; such a contribution is absorbed into the perturbative remainder and cannot carry the retained positive local concentration by the capture property. This gives the extraction statement. $\square$

3.2. Elementary local decoupling estimates.

Lemma 3.12 (Critical rescalings that escape are locally invisible). *Let $\phi \in L^3(\mathbb{R}^3)$, let $R < \infty$, and define*

$$W_n(y) = \rho_n \phi(z_n + \rho_n y),$$

where $\rho_n > 0$. Assume that the sets

$$z_n + \rho_n B_R$$

either have measure tending to zero, or escape to spatial infinity in the sense that for every compact set $K \subset \mathbb{R}^3$,

$$(z_n + \rho_n B_R) \cap K = \emptyset$$

for all sufficiently large n . Then

$$\|W_n\|_{L^3(B_R)} \rightarrow 0.$$

Proof. By the critical change of variables,

$$\|W_n\|_{L^3(B_R)}^3 = \int_{B_R} \rho_n^3 |\phi(z_n + \rho_n y)|^3 dy = \int_{z_n + \rho_n B_R} |\phi(x)|^3 dx.$$

Set

$$m_n := \int_{z_n + \rho_n B_R} |\phi(x)|^3 dx.$$

Since $\|W_n\|_{L^3(B_R)}^3 = m_n$, it is enough to show $m_n \rightarrow 0$.

Fix $\varepsilon > 0$. We split into the two structural alternatives.

Case 1: shrinking measure. If $|z_n + \rho_n B_R| \rightarrow 0$, then by absolute continuity of the integral $|\phi|^3 \in L^1(\mathbb{R}^3)$ there exists $\delta > 0$ such that

$$|E| < \delta \implies \int_E |\phi|^3 dx < \varepsilon.$$

Choose N with $|z_n + \rho_n B_R| < \delta$ for all $n \geq N$. Then

$$m_n = \int_{z_n + \rho_n B_R} |\phi|^3 dx < \varepsilon, \quad n \geq N.$$

Case 2: escape to infinity. Assume now that for every compact set $K \subset \mathbb{R}^3$, $(z_n + \rho_n B_R) \cap K = \emptyset$ for all sufficiently large n . By absolute continuity at infinity, choose compact $K \subset \mathbb{R}^3$ such that

$$\int_{\mathbb{R}^3 \setminus K} |\phi|^3 < \varepsilon.$$

For every such compact K , there is $N(K)$ with $(z_n + \rho_n B_R) \cap K = \emptyset$ for all $n \geq N(K)$. For these indices,

$$z_n + \rho_n B_R \subset \mathbb{R}^3 \setminus K,$$

and therefore, for $n \geq N(K)$,

$$m_n = \int_{z_n + \rho_n B_R} |\phi|^3 dx \leq \int_{\mathbb{R}^3 \setminus K} |\phi|^3 < \varepsilon.$$

In either case, for every $\varepsilon > 0$, $m_n < \varepsilon$ for n sufficiently large. Hence $m_n \rightarrow 0$, and therefore

$$\|W_n\|_{L^3(B_R)} \rightarrow 0.$$

□

Lemma 3.13 (Separated active profiles vanish in the selected frame). *Let (x_n^p, λ_n^p) and (x_n^q, λ_n^q) be two active frames with separated physical centers in the sense that*

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} \rightarrow \infty.$$

If $\phi^q \in L^3(\mathbb{R}^3)$, then the q -profile observed in the p -frame converges to zero locally in L^3 : for every $R < \infty$,

$$\left\| \frac{\lambda_n^p}{\lambda_n^q} \phi^q \left(\frac{x_n^p - x_n^q}{\lambda_n^q} + \frac{\lambda_n^p}{\lambda_n^q} y \right) \right\|_{L^3(B_R)} \rightarrow 0.$$

Proof. The displayed function has the form in [Theorem 3.12](#) with

$$z_n = \frac{x_n^p - x_n^q}{\lambda_n^q}, \quad \rho_n = \frac{\lambda_n^p}{\lambda_n^q}.$$

The corresponding physical set in the q -profile variables is

$$z_n + \rho_n B_R = \frac{x_n^p - x_n^q}{\lambda_n^q} + \frac{\lambda_n^p}{\lambda_n^q} B_R.$$

For every $y \in B_R$,

$$|z_n + \rho_n y| \geq |z_n| - \rho_n R.$$

Set $E_n := z_n + \rho_n B_R$. Then

$$\text{dist}(E_n, 0) = |z_n| - \rho_n R = \frac{\lambda_n^p}{\lambda_n^q} \left(\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \right).$$

Since $\frac{|x_n^q - x_n^p|}{\lambda_n^p} \rightarrow \infty$, there exists $N_0(R)$ such that

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \geq 1, \quad n \geq N_0(R),$$

and hence for such n , $\text{dist}(E_n, 0) \geq \rho_n$.

There are two cases.

(a) If $\rho_n \rightarrow 0$, then

$$|z_n + \rho_n B_R| = \rho_n^3 |B_R| \rightarrow 0.$$

Then the first alternative in [Theorem 3.12](#) applies.

(b) Otherwise, after passing to a subsequence, $\rho_n \geq \rho_* > 0$. Then

$$\text{dist}(z_n + \rho_n B_R, 0) \geq \rho_* \left(\frac{|x_n^q - x_n^p|}{\lambda_n^p} - R \right) \rightarrow \infty,$$

so the sets escape every fixed compact subset of $\mathbb{R}^3$. The second alternative in [Theorem 3.12](#) applies.

In both subcases the $L^3(B_R)$ -norm of the displayed expression converges to zero. $\square$

Lemma 3.14 (Strict outer scales vanish in the innermost frame). *Let several active profiles have the same physical center and scales $\lambda_n^1, \dots, \lambda_n^J$. Suppose λ_n^1 is an innermost scale and*

$$\frac{\lambda_n^1}{\lambda_n^j} \rightarrow 0 \quad (j \neq 1).$$

Then every outer profile $\phi^j \in L^3(\mathbb{R}^3)$, viewed in the λ_n^1 -frame, converges to zero in $L^3(B_R)$ for each fixed $R < \infty$. Any constant ambient velocity arising from the regular part of the outer flow is absorbed into the translation modulation.

Proof. In the innermost variables, the j -th outer contribution has the form

$$W_n^j(y) = \frac{\lambda_n^1}{\lambda_n^j} \phi^j \left(\frac{x_n^1 - x_n^j}{\lambda_n^j} + \frac{\lambda_n^1}{\lambda_n^j} y \right).$$

Set

$$\rho_n := \frac{\lambda_n^1}{\lambda_n^j} \rightarrow 0, \quad z_n^j := \frac{x_n^1 - x_n^j}{\lambda_n^j}.$$

Then

$$W_n^j(y) = \rho_n \phi^j(z_n^j + \rho_n y),$$

and by the critical change of variables,

$$\|W_n^j\|_{L^3(B_R)}^3 = \int_{z_n^j + \rho_n B_R} |\phi^j(x)|^3 dx.$$

Since $\rho_n \rightarrow 0$, the translated domains have

$$|z_n^j + \rho_n B_R| = \rho_n^3 |B_R| \rightarrow 0.$$

By absolute continuity of the integral for $\phi^j \in L^3(\mathbb{R}^3)$,

$$\|W_n^j\|_{L^3(B_R)} \rightarrow 0.$$

To treat a general ϕ^j , let $\phi_\varepsilon^j \in C_c^\infty(\mathbb{R}^3)$ with $\|\phi^j - \phi_\varepsilon^j\|_{L^3} < \varepsilon$. By critical isometry,

$$\|W_n^j - (W_n^j)_\varepsilon\|_{L^3(B_R)} \leq \|\phi^j - \phi_\varepsilon^j\|_{L^3} < \varepsilon,$$

where $(W_n^j)_\varepsilon$ is the same scaling of ϕ_ε^j . The compactly supported term has vanishing local norm by the shrinking-domain argument above, and then letting first $n \rightarrow \infty$, then $\varepsilon \downarrow 0$ implies the claimed convergence. A constant vector field from the regular part of an outer flow contributes only a Galilean drift in the selected frame; it is therefore incorporated into the translation modulation and does not remain as an exterior perturbation. $\square$

Lemma 3.15 (Compactness failure produces another core). *If the compact-cylinder suitable estimates are unbounded on a fixed rescaled cylinder, then after passing to a subsequence the local $C_n + D_n$ quantity is unbounded on a comparable cylinder. Consequently the branch contains an additional local CKN concentration core.*

Proof. Fix $r > 0$. Apply the local Caccioppoli inequality with outer radius $2r$. For each index n , suitable weak solutions satisfy, with C_r independent of n ,

$$A_n(r) + E_n(r) \leq C_r(1 + C_n(2r) + C_n(2r)^{2/3} + D_n(2r)),$$

where the pressure is normalized by subtracting spatial means $(p_n)_{B_{2r}}(t)$.

Suppose that $A_n(r) + E_n(r) + C_n(r) + D_n(r)$ is unbounded along a subsequence. If $C_n(2r) + D_n(2r)$ were bounded on this subsequence, the preceding estimate would give uniform boundedness of $A_n(r) + E_n(r)$. Monotonicity of the local integrals also gives $C_n(r) + D_n(r) \leq C_n(2r) + D_n(2r)$, contradicting unboundedness of the full family on Q_r . Therefore, after extracting a subsequence,

$$C_n(2r) + D_n(2r) \rightarrow \infty.$$

Define the localized CKN concentration at variable center and radius

$$G_n(x, t, \rho) := \rho^{-2} \int_{Q_\rho(x, t)} |u_n|^3 dx ds + \rho^{-2} \int_{Q_\rho(x, t)} |p_n - (p_n)_{B_\rho}(s)|^{3/2} dx ds.$$

Set

$$\mathbf{M}_n := \sup\{G_n(x, t, \rho) : 0 < \rho \leq 2r, Q_\rho(x, t) \subset Q_{4r}\}.$$

The choice $(x, t, \rho) = (0, 0, 2r)$ is admissible because $Q_{2r}(0, 0) \subset Q_{4r}$, so

$$\mathbf{M}_n \geq C_n(2r) + D_n(2r) \rightarrow \infty.$$

After extracting a further subsequence, we may assume $\mathbf{M}_n \geq 1$ for all n . Choose (x_n, t_n, ρ_n) with $0 < \rho_n \leq 2r$, $Q_{\rho_n}(x_n, t_n) \subset Q_{4r}$, such that

$$G_n(x_n, t_n, \rho_n) \geq \frac{1}{2} \mathbf{M}_n \geq \frac{1}{2}.$$

Rescale around (x_n, t_n, ρ_n) :

$$v_n(y, s) := \rho_n u_n(x_n + \rho_n y, t_n + \rho_n^2 s), \quad q_n(y, s) := \rho_n^2 p_n(x_n + \rho_n y, t_n + \rho_n^2 s).$$

Using the critical Jacobian identity,

$$\int_{Q_1} |v_n|^3 dy ds + \int_{Q_1} |q_n - (q_n)_{B_1}(s)|^{3/2} dy ds = G_n(x_n, t_n, \rho_n) \geq \frac{1}{2}.$$

Hence this new frame carries positive local critical mass, i.e. a new local CKN concentration core, as required. $\square$

3.3. Terminal active-frame decoupling.

Definition 3.16 (Admissible terminal active family). Fix a compact cylinder $B_{2R} \times I$ and a selected active frame. A terminal active family is admissible on this cylinder if the following hold.

- (i) The active profiles satisfy the finite-mass and active-mass conditions of [Theorem 3.7](#).
- (ii) Every nonselected active profile is either separated from the selected physical point in the sense of [Theorem 3.13](#), or has the same physical point and a strictly outer scale in the sense of [Theorem 3.14](#), after comparable same-point scales have been grouped.
- (iii) The rescaled remainder r_n satisfies

$$\|r_n\|_{L^\infty L^3(B_{2R})} \rightarrow 0,$$

and there exist fields (π_n^r, f_n) such that on $B_{2R} \times I$,

$$\partial_t r_n - \Delta r_n + a_n(r_n + y \cdot \nabla r_n) + b_n \cdot \nabla r_n + \nabla \pi_n^r = f_n, \quad \nabla \cdot r_n = 0,$$

with

$$\|f_n\|_{L_I^1 H^{-1}(B_R)} \rightarrow 0,$$

$$\|\nabla \pi_n^r\|_{L_I^1 H^{-1}(B_R)} \rightarrow 0.$$

- (iv) The selected velocity satisfies

$$\sup_n \|U_n\|_{L_I^\infty L^3(B_{2R})} < \infty.$$

- (v) The selected modulation coefficients satisfy

$$\sup_n (\|a_n\|_{L^\infty(I)} + \|b_n\|_{L^\infty(I)}) < \infty.$$

- (vi) Local pressure decomposition is stable under the splitting: if tensor coefficients tend to zero in $L_I^1 L^{3/2}(B_R)$, then the corresponding pressure gradients tend to zero in $L_I^1 H^{-1}(B_R)$, after subtracting spatial means.
- (vii) Cutoff commutators generated by localizing to B_R are controlled by the $L_I^\infty L^3$, $L_I^1 L^{3/2}$, and pressure-gradient norms appearing above. In particular, for any compact $\chi_R \in C_c^\infty(B_{2R})$ with $\chi_R \equiv 1$ on B_R , the commutators satisfy

$$\|[\Delta, \chi_R]u\|_{H^{-1}(B_R)} + \|[\nabla \chi_R, \nabla \cdot](u \otimes u)\|_{H^{-1}(B_R)} \leq C_R \|u\|_{L^3(B_{2R})}^2.$$

uniformly for $u \in L^3(B_{2R})$ and the same type of estimate for S_n -quadratic commutator terms.

Theorem 3.17 (Terminal profile-evolution decoupling). *Consider an admissible terminal active family on $B_{2R} \times I$. Let U_n be the selected-frame velocity and let S_n be the sum, in that frame, of all nonselected active profiles together with the rescaled remainder. Then on $B_R \times I$:*

- (i) $\nabla \cdot S_n = 0$ distributionally;
- (ii) $S_n \rightarrow 0$ in $L_I^\infty L^3(B_{2R})$;
- (iii) $U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n \rightarrow 0$ in $L_I^1 L^{3/2}(B_R)$;

(iv)

$$\begin{aligned} \exists \pi_n, F_n : \quad & \partial_t S_n - \Delta S_n + a_n(y \cdot \nabla S_n + S_n) + b_n \cdot \nabla S_n + \nabla \pi_n = F_n, \\ \|F_n\|_{L_I^1 H^{-1}(B_R)} & \rightarrow 0, \quad \nabla \cdot S_n = 0 \quad \text{in } \mathcal{D}'(B_R \times I). \\ \|\nabla \pi_n\|_{L_I^1 H^{-1}(B_R)} & \rightarrow 0. \end{aligned}$$

- (v) every pressure source containing at least one factor of S_n has vanishing pressure gradient in $L_I^1 H^{-1}(B_R)$;
- (vi) localization errors introduced by compact cutoffs vanish in $L_I^1 H^{-1}(B_R)$;
- (vii) the effective modulation parameters of the selected frame remain bounded;
- (viii) after discarding S_n , the selected frame remains a positive finite-mass local Type II branch with pressure reconstruction and the compact-cylinder bounds required by the single-core analysis.

Proof. Each profile in the family is divergence-free, and the rescaled remainder is divergence-free. Since divergence commutes with the Navier–Stokes critical rescalings, finite sums remain divergence-free in distributions. This establishes (i).

We prove (ii). By admissibility, every nonselected active profile is either separated from the selected physical point or is a same-point strictly outer scale. The separated profiles vanish locally by [Theorem 3.13](#), and the same-point outer profiles vanish locally by [Theorem 3.14](#). The remainder tends to zero in $L_I^\infty L^3(B_{2R})$ by [Theorem 3.16](#). The number of active profiles is finite by [Theorem 3.9](#). Since every summand in S_n tends to zero in $L_I^\infty L^3(B_{2R})$, the triangle inequality gives

$$\|S_n\|_{L_I^\infty L^3(B_{2R})} \leq \sum_{j \neq j_*} \|S_n^j\|_{L_I^\infty L^3(B_{2R})} + \|r_n\|_{L_I^\infty L^3(B_{2R})} \rightarrow 0.$$

This establishes (ii).

For (iii), admissibility gives

$$\sup_n \|U_n\|_{L_I^\infty L^3(B_{2R})} < \infty.$$

Hence, by Hölder's inequality,

$$\|U_n \otimes S_n\|_{L_I^1 L^{3/2}(B_R)} \leq |I| \|U_n\|_{L_I^\infty L^3(B_R)} \|S_n\|_{L_I^\infty L^3(B_R)} \rightarrow 0,$$

and the same estimate applies to $S_n \otimes U_n$. Finally,

$$\|S_n \otimes S_n\|_{L_I^1 L^{3/2}(B_R)} \leq |I| \|S_n\|_{L_I^\infty L^3(B_R)}^2 \rightarrow 0.$$

For (iv), write the represented Navier–Stokes equation for the full profile sum $U_n + S_n$ and subtract the represented equation for the selected component U_n . The remaining equation for S_n has the displayed linear left-hand side

$$\partial_t S_n - \Delta S_n + a_n(y \cdot \nabla S_n + S_n) + b_n \cdot \nabla S_n.$$

Collect all right-hand terms into

$$F_n := -\nabla \cdot (U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n) + f_n + E_n^{\text{cut}} + E_n^{\text{mod}},$$

where f_n is the remainder forcing from item (iii) of admissibility, and $E_n^{\text{cut}}, E_n^{\text{mod}}$ are localization and modulation errors. The nonlinear stresses tend to zero in $L_I^1 L^{3/2}(B_R)$, hence in $L_I^1 H^{-1}(B_R)$, because $L^{3/2}(B_R) \hookrightarrow H^{-1}(B_R)$ on a bounded three-dimensional ball. The residual term vanishes by [Theorem 3.16](#). The cutoff terms vanish by the localization part of admissibility, recorded explicitly in (vi). This gives the displayed convergence in $L_I^1 H^{-1}(B_R)$.

For (v), decompose the pressure source of $U_n + S_n$ into the pure selected source, the pure exterior source, and the two mixed sources. The mixed sources are

$$\partial_i \partial_j (U_{n,i} S_{n,j} + S_{n,i} U_{n,j}) \quad \text{and} \quad \partial_i \partial_j (S_{n,i} S_{n,j}).$$

By (iii), their tensor coefficients tend to zero in $L_I^1 L^{3/2}(B_R)$. The pressure-stability part of admissibility, equivalently local Calderon–Zygmund estimates plus harmonic pressure control modulo spatial means, gives convergence of the corresponding pressure gradients to zero in $L_I^1 H^{-1}(B_R)$. Pure exterior sources are generated by profiles that are locally invisible in the selected frame; applying the same pressure stability after [Theorems 3.13](#) and [3.14](#) gives their vanishing pressure gradients.

For (vi), let χ_R be a cutoff equal to one on B_R and supported in B_{2R} . The commutators are finite sums of terms of the following types:

$$(\nabla \chi_R) \cdot \nabla S_n, \quad (\Delta \chi_R) S_n, \quad (\nabla \chi_R) \cdot (U_n \otimes S_n + S_n \otimes U_n + S_n \otimes S_n),$$

plus analogous pressure cutoff terms. Their supports lie in the fixed annulus $B_{2R} \setminus B_R$. By the localization part of admissibility, these commutators are controlled by the norms already shown to vanish in (ii), (iii), and (v). Hence all localization errors vanish in $L_I^1 H^{-1}(B_R)$.

For (vii), the only exterior contribution that can remain nonzero at first order in a compact selected frame is a constant ambient velocity. This has already been absorbed into the translation modulation. The remaining exterior profiles vanish locally, so their contribution to the modulation equations is small in the same local L^3 topology used above. The selected modulation coefficients are bounded by admissibility; hence the effective parameters remain bounded.

For (viii), the positive CKN mass of the selected profile is unchanged by subtracting a term that tends to zero in local L^3 and whose mixed pressure sources vanish by (v). The pressure reconstruction is stable under the same decomposition by the pressure-stability condition in [Theorem 3.16](#). The compact-cylinder bounds pass to the selected component because the full family is bounded and the discarded terms vanish on compact cylinders. Thus the selected frame remains a positive finite-mass local Type II branch with the analytic inputs needed for the single-core analysis. $\square$

3.4. Same-point and separated-point reductions.

Theorem 3.18 (Same-point cascade reduction). *Every same-point multibubble or strict same-point cascade reduces, in the selected innermost or compound frame, to one of the following: the single-core criterion, scale-rigidity, or a perturbative terminal branch covered by [Theorem 3.17](#), provided the terminal family generated in the selected innermost frame is admissible in the sense of [Theorem 3.16](#).*

Proof. Step 1: scale-comparability classes. Split active same-point profiles into equivalence classes by

$$i \sim j \iff \exists c, C \in (0, \infty) : c \leq \frac{\lambda_n^i}{\lambda_n^j} \leq C \quad \text{for all } n$$

after passing to a subsequence. Thus within each class there are constants c_{ij}, C_{ij} with $c_{ij} \leq \lambda_n^i / \lambda_n^j \leq C_{ij}$ for all n , so one may define a single common rescaling sequence (choose $\lambda_n^{(k)} = \lambda_n^{j_k}$) and write all class- k profiles in the same coordinates. Within each class, finite sums of divergence-free profiles remain divergence-free.

Step 2: compound profile reduction inside each class. For each class set

$$\Phi_n^k := \sum_{j \in J_k} \lambda_n^j \phi^j(x_n^j + \lambda_n^j y).$$

If the L^3 -size of this rescaled sum satisfies

$$\left\| \sum_{j \in J_k} \phi^j \right\|_{L^3(\mathbb{R}^3)} < \varepsilon_*,$$

with ε_* the perturbative small-data threshold, then standard local small-data theory for Navier–Stokes in the critical space eliminates this class from the active alternatives.

If the class is above threshold, it is retained as a single compound active core; at most one active core from that class is needed for local selection.

Step 3: local alternatives. If the retained compound core is unique and its scale-centering Jacobian is nondegenerate, we obtain the single selected core alternative. If this Jacobian is degenerate, the localized choice of active center or scale is not unique, which is the gauge-degenerate alternative in [Theorem 3.6](#).

Step 4: strict same-point scale separation. Assume now the branch is not reduced to comparable scales and choose the innermost active scale λ_n^1 in that physical point. By [Theorem 3.14](#), every profile with larger scale vanishes in the λ_n^1 -frame in local L^3 , up to constant drift terms. These constant drifts are absorbed into translation modulation by admissibility. Let S_n be the sum of all such outer profiles plus the rescaled remainder r_n . Then, by Step 2 and [Theorem 3.14](#),

$$S_n \rightarrow 0 \quad \text{in } L_I^\infty L^3(B_{2R}).$$

Subtracting the equation for the innermost selected profile from the full-frame equation gives a perturbation equation for the selected profile whose forcing belongs to $L_I^1 H^{-1}(B_R)$ and vanishes. Thus [Theorem 3.17](#) applies, and one of three outcomes holds:

- (a) **Single-core branch:** after discarding S_n , the branch is governed by one active component, so it reduces to the compact single-core branch of [Theorem 3.20](#);
- (b) **Scale-rigid branch:** the relative Jacobian between comparable scales becomes rigid in the sense of [Theorem 3.21](#), so [Theorem 3.21](#) excludes it;
- (c) **Purely perturbative branch:** $S_n \rightarrow 0$ and the selected profile satisfies a vanishing forcing perturbation in $L_I^1 H^{-1}(B_R)$. No independent same-point cascade mechanism remains; after discarding S_n , the selected component is reduced to the single-core or scale-rigidity analysis.

These are the strict same-point scale-separation alternatives, so the same-point reduction is exhausted. $\square$

Theorem 3.19 (Separated-point decoupling). *Separated active physical points are locally invisible in each other's active rescaled frames. Consequently every separated-point multibubble reduces in each selected active frame to a single-core, scale-rigid, or perturbative terminal branch, provided the terminal family in that selected active frame is admissible in the sense of [Theorem 3.16](#).*

Proof. Fix an active point p with frame (x_n^p, λ_n^p) . Let $q \neq p$ be another active point. By separation,

$$\frac{|x_n^q - x_n^p|}{\lambda_n^p} \rightarrow \infty.$$

Thus [Theorem 3.13](#) gives local L^3 convergence to zero of the q -profile in the p -frame on every compact ball. The same conclusion holds for each profile centered at any physical point different from p . Since only finitely many active profiles remain after the critical mass threshold is imposed, the sum of all profiles outside the p -frame tends to zero locally in L^3 .

The pure pressure generated by a separated profile is locally invisible by the same change of variables whenever its profile pressure belongs to $L_{\text{loc}}^{3/2}$ with the standard Calderon–Zygmund control. Mixed pressure terms contain at least one exterior velocity factor, hence vanish in $L^1 H_{\text{loc}}^{-1}$ by the pressure part of [Theorem 3.17](#). Localization and modulation commutators are localized to $B_{2R} \setminus B_R$, so their H^{-1} -norm is controlled by the L^3 -smallness already proved in [Theorems 3.13](#) and [3.14](#) and by admissibility item (vii).

Therefore, in the selected p -frame, all other physical points contribute only a perturbative forcing tending to zero in $L^1 H^{-1}$ on compact cylinders. The selected frame remains a positive finite-mass branch with the same compact-cylinder and pressure reconstruction properties. Hence the branch reduces to the single-core case, the scale-rigid case, or the perturbative terminal branch. Since p was arbitrary, the same reduction holds in every active frame. $\square$

3.5. Multibubble and cascade exclusion. The following assumptions isolate the analytic inputs needed for the multibubble reduction: the single-core exclusion, the scale-rigidity reduction, and terminal active-frame admissibility. They are stated explicitly so that the following theorem uses no unstated compactness, pressure, or profile completeness conclusion.

Assumption 3.20 (Local single-core exclusion). Every compact single-core branch with positive local CKN concentration, nondegenerate scale-center selection, pressure reconstruction, bounded compact-cylinder suitable estimates, bounded modulation, and vanishing localized dissipation is excluded from the local Type II regime.

Assumption 3.21 (Local scale-rigidity exclusion). Every scale-rigid local branch produced by the same-point or separated-point reductions, including a branch with degenerate relative scale-center Jacobian, either has a nonzero bounded selected-window limit or reduces to the single-core branch in [Theorem 3.20](#). Consequently no such branch is compatible with the local Type II condition.

Assumption 3.22 (Terminal admissibility in active frames). Every terminal family produced by the same-point and separated-point reductions is admissible on each compact selected cylinder in the sense of [Theorem 3.16](#).

Theorem 3.23 (Local multibubble and cascade exclusion). *Assume the local single-core and scale-rigidity exclusions [Theorems 3.20](#) and [3.21](#), and assume terminal admissibility in active frames, [Theorem 3.22](#). Let a positive local Type II branch admit an active family satisfying the hypotheses of [Theorem 3.11](#). Then no local multibubble, cascade, or gauge-degenerate Type II case can occur in this class.*

Proof. Let a local Type II branch be in the multibubble/cascade class of [Theorem 3.6](#). Apply [Theorem 3.11](#). The positive local concentration is carried by an active frame or compound profile, and one of the alternatives listed there occurs.

Case 1: compactness-failure reduction. Compactness failure yields, after changing radius by a fixed factor and rescaling as in [Theorem 3.15](#), a bounded-cylinder concentration branch. Relabel this branch as an active core. After passing to a subsequence, the new core and every previously retained active core have one of three relations: comparable scale at the same physical point, strictly separated scale at the same physical point, or separated physical center. Comparable same-point cores are grouped into compound profiles. A perturbative compound profile is removed by small-data stability; an active compound profile is treated as one selected core in its active frame. If the scale-center selection inside such a compound core is degenerate, the branch is gauge-degenerate and belongs to the non-uniqueness alternative in [Theorem 3.6](#), rather than to a separate core dynamics.

Case 1a: gauge degeneracy. If the scale-center selection of the selected compound or terminal frame is degenerate, then the branch is in the gauge-degenerate alternative of [Theorem 3.6](#). By the repaired-gauge representation from [Section 2](#), nondegeneracy is the hypothesis needed to enter the single-core branch. Persistent degeneracy therefore belongs to the scale-rigidity alternative in [Theorem 3.21](#); if the degeneracy disappears after passing to a subsequence, the branch enters the single-core or terminal-frame alternatives below.

Case 1b: same-point comparable scales. Same-point comparable-scale active frames are grouped into the compound profile of [Theorem 3.8](#). If the compound profile is below the critical small-data threshold, it is part of the perturbative remainder. If it is above threshold and its scale-center selection is nondegenerate, it is one selected active core and enters the single-core branch. If the corresponding scale-center selection is degenerate, it is covered by Case 1a. Thus comparable same-point profiles do not produce an additional multibubble mechanism.

Case 2: same-point cascading. Strict same-point scale cascades are treated by [Theorem 3.18](#). In the innermost frame, all outer scales are locally invisible after constant drifts are absorbed. The branch therefore reduces to one of the three terminal alternatives

(A) single-core, (B) scale-rigid, (C) terminal perturbative.

Case 3: separated-point interactions. For separated physical points, [Theorem 3.19](#) gives the same three terminal alternatives as in Case 2:

(A) single-core, (B) scale-rigid, (C) terminal perturbative.

The single-core alternative is excluded by [Theorem 3.20](#). The scale-rigid alternative is excluded by [Theorem 3.21](#), which includes the scale-collapse and autonomous-limit exclusions in [Section 5](#) in this setting. The terminal perturbative alternative gives no independent multibubble obstruction: [Theorem 3.17](#) removes all exterior profiles and remainders from the selected active equation in $L^1 H_{\text{loc}}^{-1}$, preserves the positive finite-mass selected branch, and leaves only the single-core or scale-rigid possibilities already excluded. Consequently every alternative in [Theorem 3.6](#) is incompatible with the local Type II regime. $\square$

Corollary 3.24 (Removal of the multibubble alternative). *Under the hypotheses of [Theorem 3.23](#), every failure of the single nondegenerate compact core alternative is excluded.*

Proof. By [Theorem 3.6](#), failure of the single nondegenerate compact core alternative means that at least one of the following occurs: the compact-cylinder suitable estimates are unbounded; positive local CKN mass splits between active cores; a strict same-point scale cascade occurs; separated physical points remain active in distinct local frames; or the localized scale or center selection is degenerate. The compactness-failure case produces an additional core by [Theorem 3.15](#). Same-point and separated cases are reduced by [Theorems 3.18](#) and [3.19](#). [Theorem 3.23](#) excludes the remaining cases. Hence no failure of the single nondegenerate compact core alternative remains. $\square$

4. LOCAL CACCIOPOLI AND ROUGH-CORE EXCLUSION

4.1. Analytic inputs and notation. The renormalized local energy inequality from the preceding renormalized-chart analysis is used as an analytic input. The renormalized suitable structure and pressure reconstruction are not rederived. All estimates are conditional on the following local hypotheses on the compact renormalized cylinders under consideration.

Assumption 4.1 (Renormalized local energy inequality). Let (V, P, a, b) be defined on a compact renormalized cylinder $Q_{2R,J} := B_{2R} \times J$, where $J \subset \mathbb{R}$ is a bounded interval. Assume

$$(6) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla P - \nu \Delta V = a(\tau)V + a(\tau)(y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0,$$

with $\nu > 0$, $\nabla \cdot V = 0$, and $a, b \in L^\infty(J)$. For every nonnegative $\phi \in C_c^\infty(Q_{2R,J})$ and every $\tau_1 < \tau_2$ in J ,

$$(7) \quad \begin{aligned} & \int_{B_{2R}} \frac{|V(\cdot, \tau_2)|^2}{2} \phi(\cdot, \tau_2) - \int_{B_{2R}} \frac{|V(\cdot, \tau_1)|^2}{2} \phi(\cdot, \tau_1) \\ & + \nu \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \phi |\nabla V|^2 \leq \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \frac{|V|^2}{2} (\partial_\tau \phi + \nu \Delta \phi) \\ & + \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \frac{|V|^2}{2} V \cdot \nabla \phi + \int_{\tau_1}^{\tau_2} \int_{B_{2R}} P V \cdot \nabla \phi \\ & + \int_{\tau_1}^{\tau_2} \int_{B_{2R}} a \frac{|V|^2}{2} (-\phi - y \cdot \nabla \phi) \\ & - \int_{\tau_1}^{\tau_2} \int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla \phi. \end{aligned}$$

The pressure is understood modulo functions of time, and the mean-normalized pressure on compact balls belongs to the local class used below.

Throughout, for a space-time set $E \subset \mathbb{R}^3 \times \mathbb{R}$, write

$$\int_E f := \int_E f(y, \tau) dy d\tau.$$

For $r > 0$ and a bounded interval $J \subset \mathbb{R}$, define

$$Q_{r,J} := B_r \times J, \quad |Q_{r,J}| := |B_r| |J|.$$

For a ball $B \subset \mathbb{R}^3$, denote the spatial mean of pressure by

$$(P)_B(\tau) := \frac{1}{|B|} \int_B P(y, \tau) dy.$$

The local quantities are

$$\begin{aligned} \mathcal{A}_J(r) &:= \operatorname{ess\,sup}_{\tau \in J} \int_{B_r} |V(y, \tau)|^2 dy, \\ \mathcal{H}_J(r) &:= \int_J \int_{B_r} |\nabla V|^2 dy d\tau, \\ \mathcal{C}_J(r) &:= \int_J \int_{B_r} |V|^3 dy d\tau, \quad \mathcal{D}_J(r) := \int_J \int_{B_r} |P - (P)_{B_r}(\tau)|^{3/2} dy d\tau. \end{aligned}$$

Lemma 4.2 (Finite-measure conversion). *For every bounded interval J and every $r > 0$,*

$$\int_J \int_{B_r} |V|^2 \leq |Q_{r,J}|^{1/3} \mathcal{C}_J(r)^{2/3} \leq |Q_{r,J}|^{1/3} (1 + \mathcal{C}_J(r)).$$

Proof. Hölder's inequality on $Q_{r,J}$ gives

$$\int_J \int_{B_r} |V|^2 \leq |Q_{r,J}|^{1/3} \left(\int_J \int_{B_r} |V|^3 \right)^{2/3}.$$

The second inequality follows from $s^{2/3} \leq 1 + s$ for $s \geq 0$. $\square$

Lemma 4.3 (Comparison of pressure normalizations). *Let $1 \leq p < \infty$ and $0 < r \leq R$. For almost every τ such that $P(\tau) \in L^p(B_R)$,*

$$\|P(\tau) - (P)_{B_r}(\tau)\|_{L^p(B_r)} \leq 2\|P(\tau) - (P)_{B_R}(\tau)\|_{L^p(B_R)}.$$

Consequently,

$$\mathcal{D}_J(r) \leq 2^{3/2} \mathcal{D}_J(R)$$

whenever the right-hand side is finite.

Proof. Set $c = (P)_{B_R}(\tau)$. Then

$$\|P - (P)_{B_r}\|_{L^p(B_r)} \leq \|P - c\|_{L^p(B_r)} + |B_r|^{1/p} |(P)_{B_r} - c|.$$

Since

$$|(P)_{B_r} - c| \leq |B_r|^{-1} \int_{B_r} |P - c| \leq |B_r|^{-1/p} \|P - c\|_{L^p(B_r)},$$

the displayed pointwise estimate follows. Raising to the power $3/2$ and integrating in time gives the last assertion. $\square$

4.2. Compact-cylinder Caccioppoli estimate.

Theorem 4.4 (Compact-cylinder Caccioppoli estimate). *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume [Theorem 4.1](#) on $Q_{2R,J}$ and assume*

$$\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty.$$

Then there is a constant

$$C = C(R, |J|, \nu, \|a\|_{L^\infty(J)}, \|b\|_{L^\infty(J)}, \text{dist}(J', \partial J))$$

such that

$$\mathcal{A}_{J'}(R) + \mathcal{H}_{J'}(R) \leq C(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R)).$$

The estimate is outer-to-inner: all right-hand quantities are measured on $B_{2R} \times J$, while the conclusion is on $B_R \times J'$.

Proof. Step 1: localized cut-off and the two energy inequalities. Set

$$\delta := \text{dist}(J', \partial J) > 0.$$

Choose $\eta \in C_c^\infty(J)$ and $\zeta \in C_c^\infty(B_{2R})$ with

$$0 \leq \eta \leq 1, \quad 0 \leq \zeta \leq 1, \quad \eta \equiv 1 \text{ on } J', \quad \zeta \equiv 1 \text{ on } B_R,$$

and

$$\|\eta'\|_{L^\infty(J)} \leq C_\eta \delta^{-1}, \quad \|\nabla \zeta\|_{L^\infty(B_{2R})} \leq C_\zeta R^{-1}, \quad \|\Delta \zeta\|_{L^\infty(B_{2R})} \leq C_\zeta R^{-2}.$$

Set

$$\phi(y, \tau) := \eta(\tau) \zeta(y)^2.$$

Then $\phi \in C_c^\infty(Q_{2R,J})$, $0 \leq \phi \leq 1$, and

$$\phi \equiv 1 \text{ on } B_R \times J'.$$

Choose $\tau_- < \inf J'$ and $\tau_+ > \sup J'$ in J with $\eta(\tau_-) = \eta(\tau_+) = 0$. Applying (7) on (τ_-, τ_+) gives

$$\nu \int_{\tau_-}^{\tau_+} \int_{B_{2R}} \phi |\nabla V|^2 \leq |R_\tau| + |R_\Delta| + |R_{\text{conv}}| + |R_{\text{pres}}| + |R_{\text{mod}}|,$$

where

$$\begin{aligned} R_\tau &:= \int_J \int_{B_{2R}} \frac{|V|^2}{2} \partial_\tau \phi, & R_\Delta &:= \nu \int_J \int_{B_{2R}} \frac{|V|^2}{2} \Delta \phi, \\ R_{\text{conv}} &:= \int_J \int_{B_{2R}} \left(\frac{|V|^2}{2} V \right) \cdot \nabla \phi, & R_{\text{pres}} &:= \int_J \int_{B_{2R}} (P - (P)_{B_{2R}}(\tau)) V \cdot \nabla \phi, \\ R_{\text{mod}} &:= \int_J \int_{B_{2R}} a \frac{|V|^2}{2} (-\phi - y \cdot \nabla \phi) - \int_J \int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla \phi. \end{aligned}$$

For almost every $\sigma \in J'$, applying (7) on (τ_-, σ) gives the same right-hand estimates and leaves the terminal term $\frac{1}{2} \int_{B_R} |V(\sigma)|^2$ on the left. Therefore the bounds below control both $\mathcal{H}_{J'}(R)$ and $\mathcal{A}_{J'}(R)$. Define

$$L := \int_J \int_{B_{2R}} |V|^2, \quad C_3 := C_J(2R), \quad D_{3/2} := \mathcal{D}_J(2R).$$

By Theorem 4.2,

$$L \leq |Q_{2R,J}|^{1/3} C_3^{2/3} \leq C(R, |J|)(1 + C_3).$$

Since $\phi(y, \tau) = \eta(\tau) \zeta(y)^2$,

$$\partial_\tau \phi = \eta'(\tau) \zeta(y)^2, \quad \nabla \phi = 2\eta(\tau) \zeta(y) \nabla \zeta(y), \quad \Delta \phi = \eta(\tau) \Delta(\zeta(y)^2),$$

hence

$$\|\partial_\tau \phi\|_{L^\infty(Q_{2R,J})} \leq C_\eta \delta^{-1}, \quad \|\nabla \phi\|_{L^\infty(Q_{2R,J})} \leq 2C_\zeta R^{-1}, \quad \|\Delta \phi\|_{L^\infty(Q_{2R,J})} \leq C_\zeta R^{-2}.$$

Step 2: derivative terms from the cut-off.

$$|R_\tau| \leq \frac{1}{2} \|\partial_\tau \phi\|_{L^\infty(Q_{2R,J})} L \leq C \delta^{-1} (1 + C_3),$$

$$|R_\Delta| \leq \frac{\nu}{2} \|\Delta \phi\|_{L^\infty(Q_{2R,J})} L \leq C \nu R^{-2} (1 + C_3).$$

Step 3: convection term. Using $|\nabla \phi| \leq C R^{-1}$,

$$|R_{\text{conv}}| \leq \frac{1}{2} \|\nabla \phi\|_{L^\infty(Q_{2R,J})} \int_J \int_{B_{2R}} |V|^3 \leq C R^{-1} C_3.$$

Step 4: pressure term. For each time τ ,

$$\int_{B_{2R}} (P - (P)_{B_{2R}}(\tau)) V \cdot \nabla \phi = \int_{B_{2R}} P V \cdot \nabla \phi,$$

with

$$\int_{B_{2R}} (P)_{B_{2R}}(\tau) V \cdot \nabla \phi = (P)_{B_{2R}}(\tau) \int_{B_{2R}} V \cdot \nabla \phi = -(P)_{B_{2R}}(\tau) \int_{B_{2R}} \phi \nabla \cdot V = 0$$

(using $\phi \in C_c^\infty(B_{2R})$ and $\nabla \cdot V = 0$). Thus subtracting any time-dependent pressure constant is legitimate in the pressure flux term. Hence

$$|R_{\text{pres}}| \leq \|\nabla \phi\|_{L^\infty(Q_{2R,J})} \int_J \|P - (P)_{B_{2R}}(\tau)\|_{L_x^{3/2}(B_{2R})} \|V(\tau, \cdot)\|_{L_x^3(B_{2R})} d\tau \leq C R^{-1} D_{3/2}^{2/3} C_3^{1/3}.$$

Young's inequality gives

$$|R_{\text{pres}}| \leq C(R)(C_3 + D_{3/2}).$$

Step 5: modulation terms. The local energy inequality already contains the modulation terms after the spatial integrations by parts from the repaired-gauge equation:

$$\int_{B_{2R}} (y \cdot \nabla V) \cdot V \phi = \frac{1}{2} \int_{B_{2R}} y \cdot \nabla (|V|^2) \phi = -\frac{1}{2} \int_{B_{2R}} |V|^2 \nabla \cdot (y \phi),$$

and

$$\int_{B_{2R}} (b(\tau) \cdot \nabla V) \cdot V \phi = \frac{1}{2} \int_{B_{2R}} b(\tau) \cdot \nabla (|V|^2) \phi = -\frac{1}{2} \int_{B_{2R}} |V|^2 b(\tau) \cdot \nabla \phi.$$

Thus we estimate the additional terms directly. Split

$$R_{\text{mod}} = R_a + R_{ay} + R_b,$$

where

$$R_a := - \int_J \int_{B_{2R}} a \frac{|V|^2}{2} \phi, \quad R_{ay} := - \int_J \int_{B_{2R}} a \frac{|V|^2}{2} y \cdot \nabla \phi, \quad R_b := - \int_J \int_{B_{2R}} \frac{|V|^2}{2} b \cdot \nabla \phi.$$

$$M_a := \|a\|_{L^\infty(J)}, \quad M_b := \|b\|_{L^\infty(J)}.$$

For R_a ,

$$|R_a| \leq \frac{1}{2} M_a L \leq C M_a (1 + C_3).$$

For the scale-gradient part, $|y| \leq 2R$ on B_{2R} and $\|\nabla \phi\|_{L^\infty(Q_{2R,J})} \leq C R^{-1}$, hence

$$|y \cdot \nabla \phi| \leq C.$$

Therefore

$$|R_{ay}| \leq C M_a L \leq C M_a (1 + C_3).$$

For R_b ,

$$|R_b| \leq C M_b R^{-1} L \leq C M_b R^{-1} (1 + C_3).$$

Step 6: absorption and conclusion. Collecting estimates from Steps 2–5 yields

$$\nu \int_{\tau_-}^{\tau_+} \int_{B_{2R}} \phi |\nabla V|^2 \leq C(1 + C_3 + D_{3/2}),$$

where C depends only on

$$R, |J|, \nu, \|a\|_{L^\infty(J)}, \|b\|_{L^\infty(J)}, \delta.$$

Since $\phi \equiv 1$ on $B_R \times J'$,

$$\mathcal{H}_{J'}(R) \leq C(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R)).$$

Repeating the same estimates with terminal time $\sigma \in J'$ gives

$$\frac{1}{2} \int_{B_R} |V(y, \sigma)|^2 dy \leq C(1 + \mathcal{C}_J(2R) + \mathcal{D}_J(2R))$$

for almost every $\sigma \in J'$. Taking the essential supremum over $\sigma \in J'$ and renaming C proves the theorem. $\square$

Definition 4.5 (Local compact-cylinder control). Local Caffarelli–Kohn–Nirenberg control on $Q_{r,J}$ means

$$\mathcal{C}_J(r) + \mathcal{D}_J(r) < \infty.$$

Local windowed H^1 control on $Q_{r,J}$ means

$$\mathcal{H}_J(r) < \infty.$$

Corollary 4.6 (Outer control implies inner suitable control). *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume [Theorem 4.1](#) on $Q_{2R,J}$ and $\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty$. Then*

$$\mathcal{A}_{J'}(R) + \mathcal{H}_{J'}(R) < \infty.$$

More quantitatively, the bound is the one in [Theorem 4.4](#). If an argument uses pressure means on a smaller ball, [Theorem 4.3](#) converts those normalizations to the outer-ball normalization above.

Proof. The estimate follows from [Theorem 4.4](#). The comparison of pressure normalizations follows from [Theorem 4.3](#). $\square$

Corollary 4.7 (Rough-core exclusion on compact windows). *Let $R > 0$, let $J \subset \mathbb{R}$ be a bounded open interval, and let $J' \Subset J$ be a compact subinterval. Assume [Theorem 4.1](#) on $Q_{2R,J}$. If*

$$\mathcal{C}_J(2R) + \mathcal{D}_J(2R) < \infty,$$

then

$$\mathcal{H}_{J'}(R) < \infty.$$

Equivalently, if $\mathcal{H}_{J'}(R) = +\infty$ for an inner cylinder $Q_{R,J'}$, then for every enclosing cylinder $Q_{2R,J}$ with $J' \Subset J$ on which the local energy inequality and bounded modulation hypotheses hold,

$$\mathcal{C}_J(2R) = +\infty \quad \text{or} \quad \mathcal{D}_J(2R) = +\infty.$$

Proof. The direct implication is the gradient part of [Theorem 4.4](#). The last assertion is its contrapositive with the same outer-to-inner geometry: finite outer $\mathcal{C} + \mathcal{D}$ on $Q_{2R,J}$ forces finite inner $\mathcal{H}$ on $Q_{R,J'}$. $\square$

Remark 4.8 (Meaning of rough core). Here a rough core means failure of local windowed H^1 control on compact inner renormalized cylinders while the modulation coefficients are bounded. [Theorem 4.7](#) shows that such a failure is not independent of compact-cylinder Caffarelli–Kohn–Nirenberg control: it can occur only if the velocity-pressure bounds $\mathcal{C} + \mathcal{D}$ fail on every relevant larger enclosing compact cylinder.

4.3. Consequences for the assembly. We shall use the following consequences in the final assembly.

- (1) On every compact renormalized window satisfying the local energy inequality and bounded modulation, finite $\mathcal{C} + \mathcal{D}$ on $Q_{2R,J}$ implies finite $\mathcal{A} + \mathcal{H}$ on $Q_{R,J'}$ whenever $J' \Subset J$.
- (2) The pressure normalization may be taken on any comparable ball, after using [Theorem 4.3](#).
- (3) Failure of local windowed H^1 control on an inner compact cylinder forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ bounds on each larger enclosing cylinder to which [Theorem 4.4](#) applies.

5. SCALE-COLLAPSE COST AND AUTONOMOUS LIMITS

Notation and functional conventions. For a bounded spatial region $K \subset \mathbb{R}^3$, we use

$$(f, g)_{L^2(K)} := \int_K f \cdot g \, dx, \quad \langle F, \phi \rangle_{H^{-1}(K), H_0^1(K)} := \int_0^T \langle F(\cdot, s), \phi(\cdot, s) \rangle_{H^{-1}(K), H_0^1(K)} \, ds$$

whenever $F \in L^1(0, T; H^{-1}(K))$ and $\phi \in L^\infty(0, T; H_0^1(K))$. Weak limits and weak convergences are taken in the distributional sense on C_c^∞ -test functions.

$$\|f\|_{L_t^p L_x^q} := \left(\int_0^T \|f(\cdot, s)\|_{L^q(\mathbb{R}^3)}^p \, ds \right)^{1/p},$$

for any fixed $T > 0$, with the usual modification $p = \infty$.

5.1. Renormalized scale-collapse setting. *Position of the result.* The scale-collapse analysis has two components:

- (1) Sections 5.2–5.4 contain the cost exclusion arguments, which do not use compactness.
- (2) Section 5.5 contains the autonomous compactness statement under explicitly stated compactness hypotheses.

Throughout the scale-collapse analysis, the renormalized representation established earlier is assumed: on the time interval under consideration, the fields (V, P) satisfy (8) and its distributional weak formulation. In particular, the solenoidal formulation is obtained by restricting the test fields below to divergence-free fields.

Let (u, p) be a divergence-free suitable weak solution of

$$\partial_t u + (u \cdot \nabla) u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0,$$

in $\mathbb{R}^3$, with viscosity $\nu > 0$. Let $\lambda(\tau) > 0$, $X(\tau) \in \mathbb{R}^3$, and a measurable gauge function $c(t)$ be given, and define the renormalized fields by

$$u(x, t) = \lambda(\tau)^{-1} V \left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau \right), \quad p(x, t) = \lambda(\tau)^{-2} P \left(\frac{x - X(\tau)}{\lambda(\tau)}, \tau \right) + c(t).$$

Pressure is defined only up to an arbitrary function of time; $c(t)$ has no effect on ∇p , so it is irrelevant for all subsequent estimates. Assume that (V, P) satisfies on $(0, \infty) \times \mathbb{R}^3$

$$(8) \quad \partial_\tau V + (V \cdot \nabla) V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0.$$

For every $\psi \in C_c^\infty((0, \infty) \times \mathbb{R}^3; \mathbb{R}^3)$ with $\nabla \cdot \psi = 0$, the pair (V, P) satisfies the weak formulation

$$\begin{aligned} 0 &= \int_0^\infty \int_{\mathbb{R}^3} (-V \cdot \partial_\tau \psi - (V \otimes V) : \nabla \psi + \nu \nabla V : \nabla \psi) \, dy \, d\tau \\ &\quad + \int_0^\infty \int_{\mathbb{R}^3} (-a(\tau)(V + y \cdot \nabla V) \cdot \psi - (b(\tau) \cdot \nabla V) \cdot \psi + P \nabla \cdot \psi) \, dy \, d\tau. \end{aligned}$$

For every $0 < T < \infty$, restricting the τ -integration to $(0, T)$ gives the same identity on $(0, T) \times \mathbb{R}^3$. Equivalently, for every $\psi \in C_c^\infty((0, \infty) \times \mathbb{R}^3; \mathbb{R}^3)$, not necessarily divergence free,

$$\begin{aligned} 0 &= \int_0^\infty \int_{\mathbb{R}^3} (-V \cdot \partial_\tau \psi - (V \otimes V) : \nabla \psi + \nu \nabla V : \nabla \psi - P \nabla \cdot \psi) \, dy \, d\tau \\ &\quad + \int_0^\infty \int_{\mathbb{R}^3} (-a(\tau)(V + y \cdot \nabla V) \cdot \psi - (b(\tau) \cdot \nabla V) \cdot \psi) \, dy \, d\tau. \end{aligned}$$

Define positive and negative parts of the scalar field a by

$$a_+(\tau) := \max\{a(\tau), 0\}, \quad a_-(\tau) := \max\{-a(\tau), 0\}.$$

Then for a.e. τ , $a = a_+ - a_-$, $a_+, a_- \geq 0$, and $a_+ a_- = 0$. Assume

$$a \in L^1_{\text{loc}}([\tau_0, \infty)), \quad b \in L^1_{\text{loc}}([\tau_0, \infty); \mathbb{R}^3),$$

and the scale convention

$$(9) \quad \frac{d}{d\tau} \log \lambda(\tau) = a(\tau)$$

for a.e. $\tau \geq \tau_0$, with $\log \lambda$ absolutely continuous on compact intervals.

All integrals below are interpreted in $[0, \infty]$. When we write $\int_{\tau_0}^{\infty}$, partial integrals are taken over $[\tau_0, T]$ and monotone convergence is used as $T \rightarrow \infty$.

Fix $R > 0$. Choose $\Phi \in C_c^\infty(\mathbb{R}^3; [0, 1])$ satisfying

$$(10) \quad \Phi(y) = 1 \text{ if } |y| \leq R, \quad \text{supp}_y \Phi \subset B_{2R}.$$

Define the localized kinetic mass

$$M_R(\tau) := \int_{\mathbb{R}^3} |V(y, \tau)|^2 \Phi(y) dy.$$

Since Φ is fixed in τ , M_R is measurable in τ , and only this measurability is used below.

Definition 5.1 (Nonvanishing localized core). A renormalized branch has nonvanishing localized core if there exist constants $m_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$M_R(\tau) \geq m_0 \quad \text{for a.e. } \tau \in [\tau_1, \infty).$$

In applications, this lower bound is supplied by the selected-core construction; here it is stated as an explicit hypothesis.

Definition 5.2 (Scale-collapse and absolute scale-drift densities). Define

$$\mathcal{C}_{\text{sc}}(\tau) := a_-(\tau) M_R(\tau), \quad \mathcal{C}_{\text{abs}}(\tau) := \nu \int_{\mathbb{R}^3} |\nabla V(y, \tau)|^2 \Phi(y) dy + |a(\tau)| M_R(\tau).$$

$\mathcal{C}_{\text{sc}}$ is the negative drift weighted by localized core mass, while $\mathcal{C}_{\text{abs}}$ controls both collapse drift and local dissipation. Their total costs are extended real-valued by

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\text{sc}}(\tau) d\tau, \quad \int_{\tau_0}^{\infty} \mathcal{C}_{\text{abs}}(\tau) d\tau.$$

Definition 5.3 (Finite-cost branch). A branch has finite scale-collapsing cost if

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\text{sc}}(\tau) d\tau < \infty.$$

It has finite absolute cost if

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\text{abs}}(\tau) d\tau < \infty.$$

5.2. Elementary alternative.

Lemma 5.4 (Finite-or-infinite alternative). *Let $f \geq 0$ be measurable on $[\tau_0, \infty)$. Then exactly one of the following holds:*

$$\int_{\tau_0}^{\infty} f(\tau) d\tau < \infty \quad \text{or} \quad \int_{\tau_0}^{\infty} f(\tau) d\tau = \infty.$$

Proof. (1) For $T > \tau_0$, define partial costs

$$F(T) := \int_{\tau_0}^T f(\tau) d\tau \in [0, \infty].$$

(2) If $T_2 > T_1$, then by nonnegativity of f ,

$$F(T_2) - F(T_1) = \int_{T_1}^{T_2} f(\tau) d\tau \geq 0,$$

so F is monotone nondecreasing.

(3) A monotone function on $[\tau_0, \infty)$ has an extended limit $L \in [0, \infty]$,

$$L := \lim_{T \rightarrow \infty} F(T).$$

By definition of improper integral, this limit equals $\int_{\tau_0}^{\infty} f(\tau) d\tau$.

(4) Therefore L is either finite or equal to $+\infty$, and these cases are mutually exclusive. Hence exactly one item in the lemma holds. $\square$

Lemma 5.5 (Logarithmic scale identity). *Under (9), for all τ_1, τ_2 with $\tau_0 \leq \tau_1 < \tau_2 < \infty$,*

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Proof. (1) Since $a \in L^1_{\text{loc}}$, the map $\log \lambda$ is absolutely continuous on $[\tau_1, \tau_2]$, hence has a representative with a.e. derivative equal to a .

(2) By the fundamental theorem of calculus for absolutely continuous functions, for every $\tau \in [\tau_1, \tau_2]$,

$$\log \lambda(\tau) = \log \lambda(\tau_1) + \int_{\tau_1}^{\tau} a(s) ds.$$

(3) Setting $\tau = \tau_2$ yields the identity. $\square$

5.3. Scale collapse versus localized core lower bounds.

Theorem 5.6 (Collapse with nonvanishing localized core forces infinite scale-collapsing cost).

Assume:

- (1) $\lambda(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$;
- (2) the branch has nonvanishing localized core.

Then

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\text{sc}}(\tau) d\tau = \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau = \infty.$$

Proof. (1) Choose $m_0 > 0$ and $\tau_1 \geq \tau_0$ from the nonvanishing core assumption so that

$$M_R(\tau) \geq m_0 \quad \text{for a.e. } \tau \in [\tau_1, \infty).$$

(2) By Lemma 5.5, for any $\tau_2 > \tau_1$,

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Taking $\tau_2 \rightarrow \infty$ and using $\lambda(\tau_2) \rightarrow 0$,

$$\lim_{\tau_2 \rightarrow \infty} \int_{\tau_1}^{\tau_2} a(\tau) d\tau = -\infty.$$

(3) Decompose $a = a_+ - a_-$. Then for each $\tau_2 > \tau_1$,

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \int_{\tau_1}^{\tau_2} a_+(\tau) d\tau - \int_{\tau_1}^{\tau_2} a_-(\tau) d\tau \geq - \int_{\tau_1}^{\tau_2} a_-(\tau) d\tau.$$

Hence

$$\int_{\tau_1}^{\tau_2} a_-(\tau) d\tau \geq - \int_{\tau_1}^{\tau_2} a(\tau) d\tau.$$

(4) Taking limits and using Step 2 yields

$$\int_{\tau_1}^{\infty} a_-(\tau) d\tau = \infty.$$

Indeed, by Step 2,

$$- \int_{\tau_1}^{\tau_2} a(\tau) d\tau \rightarrow +\infty.$$

Since the partial integrals of a_- dominate this quantity, they are unbounded. Because $a_- \geq 0$, monotone convergence gives $\int_{\tau_1}^{\infty} a_-(\tau) d\tau = \infty$.

(5) Split the cost into early and late times:

$$\int_{\tau_0}^{\infty} a_- M_R d\tau = \int_{\tau_0}^{\tau_1} a_- M_R d\tau + \int_{\tau_1}^{\infty} a_- M_R d\tau.$$

The first term is finite or infinite independently of the argument; the second term satisfies

$$\int_{\tau_1}^{\infty} a_-(\tau) M_R(\tau) d\tau \geq m_0 \int_{\tau_1}^{\infty} a_-(\tau) d\tau = \infty$$

by the bound in (1) and Step 4. Hence the total integral is infinite. $\square$

5.4. Finite-cost exclusions.

Theorem 5.7 (Scale-collapse cost exclusion). *No branch with finite scale-collapsing cost (Definition 5.3) can satisfy simultaneously $\lambda(\tau) \rightarrow 0$ and nonvanishing localized core.*

Proof. Assume, for contradiction, that such a branch exists. The two hypotheses are exactly those of Theorem 5.6; therefore

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\text{sc}}(\tau) d\tau = \infty.$$

This contradicts finite scale-collapsing cost. Therefore no such branch exists. $\square$

Lemma 5.8 (Absolute-cost domination). *For a.e. $\tau \in [\tau_0, \infty)$,*

$$\mathcal{C}_{\text{abs}}(\tau) \geq \mathcal{C}_{\text{sc}}(\tau).$$

Hence, pointwise and after integrating in τ ,

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\text{sc}}(\tau) d\tau \leq \int_{\tau_0}^{\infty} \mathcal{C}_{\text{abs}}(\tau) d\tau,$$

as extended real numbers. In particular, finite absolute cost implies finite scale-collapsing cost, and equivalently

$$\int_{\tau_0}^{\infty} \mathcal{C}_{\text{sc}}(\tau) d\tau = \infty \implies \int_{\tau_0}^{\infty} \mathcal{C}_{\text{abs}}(\tau) d\tau = \infty.$$

Proof. At each time τ ,

$$\mathcal{C}_{\text{abs}}(\tau) = \nu \int |\nabla V|^2 \Phi + (a_+ + a_-) M_R \geq (a_+ + a_-) M_R \geq a_- M_R = \mathcal{C}_{\text{sc}}(\tau),$$

since $a_+, a_- \geq 0$ and $\nu \int |\nabla V|^2 \Phi \geq 0$. Integrating over $[\tau_0, \infty)$ yields the claimed implication. $\square$

Corollary 5.9 (Absolute scale-cost exclusion). *No branch with finite absolute cost in the sense of Definition 5.3 can satisfy simultaneously $\lambda(\tau) \rightarrow 0$ and nonvanishing localized core.*

Proof. Assume instead such a branch exists. By Theorem 5.6, $\int \mathcal{C}_{\text{sc}} = \infty$. By Lemma 5.8, the absolute cost is pointwise above $\mathcal{C}_{\text{sc}}$, so its integral must also be infinite, contradicting finite absolute cost. $\square$

Remark 5.10. The exclusions above are purely integral and use neither compactness nor any autonomous-limit argument.

5.5. Autonomous reduced-limit theorem from the selected-window estimates. The preceding exclusions are integral statements and do not rely on compactness. On a fixed finite autonomous window, compactness requires not only the cost bound but also the selected-window estimates supplied by the preceding renormalized representation, pressure reconstruction, and local Caccioppoli estimates.

Analytic inputs. The preceding sections supply the following ingredients on selected compact renormalized windows:

- (1) the repaired-gauge renormalized equation and weak formulation for (V, P) ;
- (2) pressure reconstruction modulo spatial means, with local $L^{3/2}$ bounds on compact cylinders;
- (3) local windowed bounds $V \in L_t^\infty L_x^2 \cap L_t^2 H_x^1$ on compact cylinders;
- (4) a retained compact core lower bound preventing the selected profile from vanishing.

Together with the thick-window convergence of the modulation coefficients, these estimates imply the compactness and limit passage below.

Definition 5.11 (Thick autonomous windows). A branch has *thick autonomous windows* if there exist constants $T > 0$, $c > 0$, $M_1 \in (0, \infty)$, a sequence (τ_n) with $\tau_n \rightarrow \infty$, and limits $a_\infty \in \mathbb{R}$, $b_\infty \in \mathbb{R}^3$, such that for every n :

(i)

$$\frac{1}{T} \int_{\tau_n}^{\tau_n+T} a_-(\tau) M_R(\tau) d\tau \geq c;$$

(ii)

$$0 \leq M_R(\tau) \leq M_1 \quad \text{for a.e. } \tau \in [\tau_n, \tau_n + T];$$

(iii)

$$a(\tau_n + \cdot) \rightarrow a_\infty \text{ in } L^1(0, T), \quad b(\tau_n + \cdot) \rightarrow b_\infty \text{ in } L^1(0, T; \mathbb{R}^3).$$

Remark 5.12. This notion isolates a limiting drift coefficient $a_\infty < 0$. The compactness to extract a profile limit is obtained from the selected-window compactness estimates, not from the scalar cost inequalities of Sections 5.2–5.4.

Lemma 5.13 (Thick drift implies negative autonomous limit parameter). *On thick autonomous windows, the limiting autonomous coefficient is negative:*

$$a_\infty \leq -\frac{c}{M_1} < 0.$$

Proof. (1) From (ii), for a.e. $\tau \in [\tau_n, \tau_n + T]$, $0 \leq M_R(\tau) \leq M_1$. Therefore

$$\frac{a_-(\tau) M_R(\tau)}{M_1} \leq a_-(\tau).$$

Therefore

$$\frac{1}{T} \int_{\tau_n}^{\tau_n+T} a_-(\tau) d\tau \geq \frac{1}{M_1 T} \int_{\tau_n}^{\tau_n+T} a_-(\tau) M_R(\tau) d\tau \geq \frac{c}{M_1}.$$

- (2) Set $a_n(s) := a(\tau_n + s)$. By (iii), $a_n \rightarrow a_\infty$ in $L^1(0, T)$. The map $r \mapsto r^-$ is 1-Lipschitz, so $a_n^- \rightarrow a_\infty^-$ in $L^1(0, T)$:

$$\|a_n^- - a_\infty^-\|_{L^1(0, T)} \leq \|a_n - a_\infty\|_{L^1(0, T)} \rightarrow 0.$$

Thus

$$\lim_{n \rightarrow \infty} \frac{1}{T} \int_{\tau_n}^{\tau_n + T} a_-(\tau) d\tau = \frac{1}{T} \int_0^T (a_\infty)^- ds = (a_\infty)^-.$$

- (3) Combine Step 1 and Step 2: $(a_\infty)^- \geq c/M_1$, i.e. $a_\infty \leq -c/M_1 < 0$.

□

Assumption 5.14 (Selected-window suitable estimates). Let (τ_n) be a thick autonomous-window sequence and set

$$\begin{aligned} V_n(y, s) &:= V(y, \tau_n + s), & P_n(y, s) &:= P(y, \tau_n + s), \\ \alpha_n(s) &:= a(\tau_n + s), & \beta_n(s) &:= b(\tau_n + s), \end{aligned} \quad s \in (0, T).$$

For every bounded Lipschitz domain $K \Subset \mathbb{R}^3$, assume there is a constant $C_K < \infty$, independent of n , such that

$$\|V_n\|_{L^\infty(0, T; L^2(K))} + \|V_n\|_{L^2(0, T; H^1(K))} + \|P_n - (P_n)_K\|_{L^{3/2}((0, T) \times K)} \leq C_K,$$

where

$$(P_n)_K(s) := \frac{1}{|K|} \int_K P_n(y, s) dy.$$

Assume also that each V_n is divergence free and that there exist $\chi \in C_c^\infty(B_R)$, $\chi \geq 0$, $\eta > 0$, and $N \in \mathbb{N}$ such that for all $n \geq N$,

$$\int_{\mathbb{R}^3} |V_n(y, s)|^2 \chi(y) dy \geq \eta \quad \text{for a.e. } s \in (0, T).$$

Proposition 5.15 (Derivation of the selected-window estimates from the preceding sections). *Let a repaired-gauge branch satisfy the renormalized representation and pressure reconstruction of the compact-window representation theorem, Theorem 2.35, on the windows $[\tau_n, \tau_n + T]$. Assume that on each compact cylinder $(0, T) \times K$ the local compact-cylinder Caffarelli–Kohn–Nirenberg quantities are finite uniformly in n , and that the selected core has the localized lower bound stated in Assumption 5.14. Then Assumption 5.14 holds.*

Proof. The renormalized representation theorem supplies the shifted equation, divergence-free condition, and pressure reconstruction modulo functions of time. The pressure reconstruction gives, after subtracting spatial means,

$$\sup_n \|P_n - (P_n)_K\|_{L^{3/2}((0, T) \times K)} < \infty$$

on every compact K . The local Caccioppoli estimate applied on a slightly larger compact cylinder gives the inner bounds

$$\sup_n \|V_n\|_{L^\infty(0, T; L^2(K))} + \sup_n \|V_n\|_{L^2(0, T; H^1(K))} < \infty.$$

These are precisely the local velocity and pressure estimates in Assumption 5.14. The retained selected-core lower bound is the last condition in that assumption. Hence all items in Assumption 5.14 follow from the representation, pressure, and local Caccioppoli results. □

Lemma 5.16 (Local time regularity on autonomous windows). *Assume thick autonomous windows and Assumption 5.14. Let $K_0 \Subset K_1 \Subset \mathbb{R}^3$ be bounded Lipschitz domains, and choose an integer m so large that $H_0^m(K_1) \hookrightarrow W_0^{1, \infty}(K_1)$. Then*

$$\{\partial_s V_n\}_{n \geq 1} \quad \text{is bounded in } L^1(0, T; H^{-m}(K_1)).$$

Proof. Fix $\zeta \in C_c^\infty(K_1)$ with $\zeta \equiv 1$ on a neighborhood of K_0 . Let $\phi \in H_0^m(K_1; \mathbb{R}^3)$. Since $H_0^m(K_1) \hookrightarrow W_0^{1,\infty}(K_1)$, there is C_m such that

$$\|\phi\|_{L^\infty(K_1)} + \|\nabla \phi\|_{L^\infty(K_1)} \leq C_m \|\phi\|_{H^m(K_1)}.$$

The shifted equation is

$$\partial_s V_n = -(V_n \cdot \nabla) V_n - \nabla P_n + \nu \Delta V_n + \alpha_n (V_n + y \cdot \nabla V_n) + \beta_n \cdot \nabla V_n$$

in distributions. We estimate each term against ϕ .

For the convection term, integration by parts gives

$$|\langle (V_n \cdot \nabla) V_n, \phi \rangle| = \left| \int_{K_1} V_{n,i} V_{n,j} \partial_j \phi_i dy \right| \leq C_m \|V_n(s)\|_{L^2(K_1)}^2 \|\phi\|_{H^m(K_1)}.$$

For the viscous term,

$$|\langle \nu \Delta V_n, \phi \rangle| = \nu \left| \int_{K_1} \nabla V_n : \nabla \phi dy \right| \leq \nu C_m \|\nabla V_n(s)\|_{L^2(K_1)} |K_1|^{1/2} \|\phi\|_{H^m(K_1)}.$$

For the pressure term, subtracting the spatial mean does not change the gradient:

$$|\langle \nabla P_n, \phi \rangle| = \left| \int_{K_1} (P_n - (P_n)_{K_1}) \nabla \cdot \phi dy \right| \leq C_m |K_1|^{1/3} \|P_n - (P_n)_{K_1}\|_{L^{3/2}(K_1)} \|\phi\|_{H^m(K_1)}.$$

For the V_n -part of the scaling drift,

$$|\alpha_n(s)| \left| \int_{K_1} V_n \cdot \phi dy \right| \leq C_m |\alpha_n(s)| |K_1|^{1/2} \|V_n(s)\|_{L^2(K_1)} \|\phi\|_{H^m(K_1)}.$$

For the $y \cdot \nabla V_n$ -part, integrate by parts:

$$\int_{K_1} (y \cdot \nabla V_n) \cdot \phi dy = - \int_{K_1} V_n \cdot (y \cdot \nabla \phi + 3\phi) dy,$$

because ϕ has zero trace. Hence

$$|\alpha_n(s)| \left| \int_{K_1} (y \cdot \nabla V_n) \cdot \phi dy \right| \leq C_{K_1,m} |\alpha_n(s)| \|V_n(s)\|_{L^2(K_1)} \|\phi\|_{H^m(K_1)}.$$

Similarly,

$$\left| \int_{K_1} (\beta_n \cdot \nabla V_n) \cdot \phi dy \right| = \left| \int_{K_1} V_n \cdot (\beta_n \cdot \nabla \phi) dy \right| \leq C_m |\beta_n(s)| |K_1|^{1/2} \|V_n(s)\|_{L^2(K_1)} \|\phi\|_{H^m(K_1)}.$$

Combining the estimates and taking the supremum over $\|\phi\|_{H^m(K_1)} \leq 1$, we obtain

$$\begin{aligned} \|\partial_s V_n(s)\|_{H^{-m}(K_1)} &\leq C_{K_1,m} (\|V_n(s)\|_{L^2(K_1)}^2 + \|\nabla V_n(s)\|_{L^2(K_1)} \\ &\quad + \|P_n(s) - (P_n)_{K_1}(s)\|_{L^{3/2}(K_1)} + (|\alpha_n(s)| + |\beta_n(s)|) \|V_n(s)\|_{L^2(K_1)}). \end{aligned}$$

Integrating in $s \in (0, T)$, using Assumption 5.14, and using $\alpha_n \rightarrow a_\infty$, $\beta_n \rightarrow b_\infty$ in L^1 , hence uniform L^1 -bounds for α_n, β_n , proves the claimed $L^1(0, T; H^{-m})$ bound. $\square$

Lemma 5.17 (Local compactness on autonomous windows). *Assume thick autonomous windows and Assumption 5.14. Then after passing to a subsequence there exist*

$$U \in L^\infty(0, T; L_{\text{loc}}^2(\mathbb{R}^3)) \cap L^2(0, T; H_{\text{loc}}^1(\mathbb{R}^3)), \quad \Pi \in L_{\text{loc}}^{3/2}((0, T) \times \mathbb{R}^3),$$

such that, for every compact $K \Subset \mathbb{R}^3$,

(a)

$$V_n \rightarrow U \quad \text{strongly in } L^2(0, T; L^2(K));$$

(b)

$$V_n \rightharpoonup U \quad \text{weakly in } L^2(0, T; H^1(K));$$

(c)

$$V_n \rightharpoonup^* U \quad \text{weakly-}^* \text{ in } L^\infty(0, T; L^2(K));$$

(d)

$$P_n - (P_n)_K \rightharpoonup \Pi_K \quad \text{weakly in } L^{3/2}((0, T) \times K),$$

where the local distributions $\nabla \Pi_K$ agree on overlaps and define a pressure gradient $\nabla \Pi$ modulo functions of time.

Proof. Fix $K_0 \Subset K_1 \Subset \mathbb{R}^3$. Assumption 5.14 gives boundedness of V_n in $L^2(0, T; H^1(K_1))$, and Lemma 5.16 gives boundedness of $\partial_s V_n$ in $L^1(0, T; H^{-m}(K_1))$ for m large. Since

$$H^1(K_1) \Subset L^2(K_0) \hookrightarrow H^{-m}(K_1),$$

the Aubin–Lions–Simon compactness theorem implies, after extracting a subsequence,

$$V_n \rightarrow U \quad \text{strongly in } L^2(0, T; L^2(K_0)).$$

The local $L^\infty L^2$ and $L^2 H^1$ bounds also give, after extraction,

$$V_n \rightharpoonup^* U \quad \text{in } L^\infty(0, T; L^2(K_0)), \quad V_n \rightharpoonup U \quad \text{in } L^2(0, T; H^1(K_0)).$$

The pressure bound in Assumption 5.14 gives weak compactness of $P_n - (P_n)_{K_0}$ in $L^{3/2}((0, T) \times K_0)$. A diagonal extraction over balls B_j gives a single subsequence for every compact K . On overlaps, two pressure limits can differ only by a function of s , because both represent the same distributional pressure gradient in the limit equation. Thus the gradients assemble into $\nabla \Pi$, with Π defined modulo functions of time. $\square$

Lemma 5.18 (Compatibility of local pressure limits). *Let $K_1, K_2 \Subset \mathbb{R}^3$ be bounded Lipschitz domains with nonempty connected overlap $K_{12} := K_1 \cap K_2$. Suppose that, after passing to a common subsequence,*

$$P_n - (P_n)_{K_i} \rightharpoonup \Pi_i \quad \text{in } L^{3/2}((0, T) \times K_i), \quad i = 1, 2.$$

Then there exists $c_{12} \in L^{3/2}(0, T)$ such that

$$\Pi_1(y, s) - \Pi_2(y, s) = c_{12}(s) \quad \text{for a.e. } (y, s) \in K_{12} \times (0, T).$$

Consequently the pressure gradient $\nabla \Pi$ obtained from the local limits is globally well-defined on $(0, T) \times \mathbb{R}^3$, with the pressure itself defined modulo functions of time.

Proof. For each n ,

$$(P_n - (P_n)_{K_1}) - (P_n - (P_n)_{K_2}) = (P_n)_{K_2}(s) - (P_n)_{K_1}(s)$$

on K_{12} . Hence its spatial gradient is zero in $\mathcal{D}'((0, T) \times K_{12})$. Passing to the weak limit gives

$$\nabla_y(\Pi_1 - \Pi_2) = 0 \quad \text{in } \mathcal{D}'((0, T) \times K_{12}).$$

Since K_{12} is connected, the standard distributional characterization of functions with zero spatial gradient implies that $\Pi_1 - \Pi_2 = c_{12}(s)$ for some $c_{12} \in L^{3/2}(0, T)$. The last assertion follows because adding a time-dependent scalar to pressure does not change $\nabla \Pi$. $\square$

Theorem 5.19 (Autonomous reduced-limit theorem). *Assume that a branch has thick autonomous windows and satisfies Assumption 5.14 on the same window sequence. Let (U, Π) be the limit extracted in Lemma 5.17. Then*

$$\nabla \cdot U = 0,$$

$$U \not\equiv 0$$

and (U, Π) satisfies

$$(11) \quad \partial_s U + (U \cdot \nabla)U + \nabla \Pi = \nu \Delta U + a_\infty(U + y \cdot \nabla U) + b_\infty \cdot \nabla U, \quad \nabla \cdot U = 0,$$

in $\mathcal{D}'((0, T) \times \mathbb{R}^3)$. Equivalently, for every $\varphi \in C_c^\infty((0, T) \times \mathbb{R}^3; \mathbb{R}^3)$,

$$\int_0^T \int (-U \cdot \partial_s \varphi - (U \otimes U) : \nabla \varphi + \nu \nabla U : \nabla \varphi - \Pi \nabla \cdot \varphi - a_\infty(U + y \cdot \nabla U) \cdot \varphi - (b_\infty \cdot \nabla U) \cdot \varphi) dy ds = 0.$$

Remark 5.20. The theorem asserts only the compact finite-window autonomous limit from the selected-window estimates. A rigidity contradiction requires a separate Liouville theorem for the class containing (U, Π) .

Proof. Step 1: divergence-free property of the limit.

Each V_n is divergence free by Assumption 5.14, so for any compactly supported scalar $\psi \in C_c^\infty((0, T) \times \mathbb{R}^3)$,

$$\int_{(0, T) \times \mathbb{R}^3} V_n \cdot \nabla \psi dy ds = 0.$$

Hence, for every such ψ , since $\nabla \psi \in C_c^\infty((0, T) \times \mathbb{R}^3)$ and $V_n \rightarrow U$ in $L^2(0, T; L_{\text{loc}}^2)$,

$$\int_{(0, T) \times \mathbb{R}^3} (V_n - U) \cdot \nabla \psi dy ds \rightarrow 0,$$

we deduce

$$\int_{(0, T) \times \mathbb{R}^3} U \cdot \nabla \psi dy ds = 0.$$

Hence $\nabla \cdot U = 0$ in $\mathcal{D}'((0, T) \times \mathbb{R}^3)$.

Step 2: nontriviality of the limit profile. Define for $s \in (0, T)$

$$f_n(s) := \int_{\mathbb{R}^3} |V_n(y, s)|^2 \chi(y) dy, \quad f(s) := \int_{\mathbb{R}^3} |U(y, s)|^2 \chi(y) dy.$$

Here $K_\chi := \text{supp } \chi$, so $\chi^{1/2} V_n, \chi^{1/2} U \in L^2(0, T; L^2(K_\chi))$. Then

$$f_n - f = \int_{\mathbb{R}^3} (V_n - U) \cdot (V_n + U) \chi dy.$$

Applying Cauchy–Schwarz in space,

$$|f_n(s) - f(s)| \leq \|\chi^{1/2}(V_n(s) - U(s))\|_{L^2(\mathbb{R}^3)} \|\chi^{1/2}(V_n(s) + U(s))\|_{L^2(\mathbb{R}^3)}.$$

Integrating in s and applying Cauchy–Schwarz in time gives

$$\|f_n - f\|_{L^1(0, T)} \leq \|\chi^{1/2}(V_n - U)\|_{L^2(0, T; L^2)} \|\chi^{1/2}(V_n + U)\|_{L^2(0, T; L^2)}.$$

Since K_χ is compact, Assumption 5.14 and the membership $U \in L^2(0, T; L^2(K_\chi))$ give

$$\sup_n \|\chi^{1/2}(V_n + U)\|_{L^2(0, T; L^2)} \leq C_\chi \left(\sup_n \|V_n\|_{L^2(0, T; L^2(K_\chi))} + \|U\|_{L^2(0, T; L^2(K_\chi))} \right) < \infty.$$

Moreover

$$\|\chi^{1/2}(V_n - U)\|_{L^2(0, T; L^2)} \leq \|\chi\|_{L^\infty}^{1/2} \|V_n - U\|_{L^2(0, T; L^2(K_\chi))} \rightarrow 0.$$

Therefore

$$\|f_n - f\|_{L^1(0, T)} \rightarrow 0.$$

Hence $f_n \rightarrow f$ in $L^1(0, T)$.

Assumption 5.14 gives for large n , $f_n(s) \geq \eta$ for a.e.s. Therefore

$$\int_0^T f(s) ds = \lim_{n \rightarrow \infty} \int_0^T f_n(s) ds \geq \eta T > 0.$$

Hence $f \not\equiv 0$, so $U \not\equiv 0$.

Step 3: shifted weak formulation for V_n . Take any test field $\varphi \in C_c^\infty((0, T) \times \mathbb{R}^3; \mathbb{R}^3)$, and set

$$K_\varphi := \overline{\{x \in \mathbb{R}^3 : \exists s \in (0, T), \varphi(x, s) \neq 0\}} \subset \mathbb{R}^3, \quad K_\varphi \text{ bounded.}$$

Since (V, P) satisfies (8) in distributions, the change of variables $\tau = \tau_n + s$ gives the shifted weak formulation

$$(12) \quad \begin{aligned} 0 = & \int_0^T \int_{\mathbb{R}^3} \left(-V_n \cdot \partial_s \varphi - (V_n \otimes V_n) : \nabla \varphi + \nu \nabla V_n : \nabla \varphi - P_n \nabla \cdot \varphi \right) dy ds \\ & + \int_0^T \int_{\mathbb{R}^3} \left(-a(\tau_n + s)(V_n + y \cdot \nabla V_n) \cdot \varphi - (b(\tau_n + s) \cdot \nabla V_n) \cdot \varphi \right) dy ds. \end{aligned}$$

Step 4: term-by-term passage to the limit. Write $\alpha_n(s) := a(\tau_n + s)$, $\beta_n(s) := b(\tau_n + s)$.

(i) *Time derivative.*

By strong convergence in $L^2(0, T; L^2(K_\varphi))$,

$$\left| \int_0^T \int (V_n - U) \cdot \partial_s \varphi dy ds \right| \leq \|V_n - U\|_{L^2(0, T; L^2(K_\varphi))} \|\partial_s \varphi\|_{L^2(0, T; L^2(K_\varphi))} \rightarrow 0.$$

Hence

$$\int V_n \cdot \partial_s \varphi \rightarrow \int U \cdot \partial_s \varphi.$$

(ii) *Convection term.* Decompose

$$V_n \otimes V_n - U \otimes U = (V_n - U) \otimes V_n + U \otimes (V_n - U).$$

Then

$$\begin{aligned} & \left| \int_0^T \int ((V_n \otimes V_n - U \otimes U) : \nabla \varphi) dy ds \right| \\ & \leq \|\nabla \varphi\|_{L_{x,t}^\infty} \|V_n - U\|_{L^2(0, T; L^2(K_\varphi))} (\|V_n\|_{L^2(0, T; L^2(K_\varphi))} + \|U\|_{L^2(0, T; L^2(K_\varphi))}) \rightarrow 0, \end{aligned}$$

using boundedness of V_n in $L^2(0, T; L^2(K_\varphi))$. Therefore $\int (V_n \otimes V_n) : \nabla \varphi \rightarrow \int (U \otimes U) : \nabla \varphi$.

(iii) *Viscous term.* From weak convergence of V_n in $L^2(0, T; H_{\text{loc}}^1)$,

$$\int_0^T \int \nabla V_n : \nabla \varphi dy ds \rightarrow \int_0^T \int \nabla U : \nabla \varphi dy ds.$$

(iv) *Pressure term.* Let K be a bounded Lipschitz domain with $K_\varphi \Subset K$. Since φ is compactly supported in K ,

$$\int_K (P_n)_K(s) \nabla \cdot \varphi(y, s) dy = (P_n)_K(s) \int_K \nabla \cdot \varphi(y, s) dy = 0.$$

Therefore

$$\int_0^T \int_{\mathbb{R}^3} P_n \nabla \cdot \varphi dy ds = \int_0^T \int_K (P_n - (P_n)_K) \nabla \cdot \varphi dy ds.$$

By Lemma 5.17,

$$P_n - (P_n)_K \rightharpoonup \Pi_K \quad \text{in } L^{3/2}((0, T) \times K),$$

and $\nabla \cdot \varphi \in L^3((0, T) \times K)$. The compatibility Lemma 5.18 permits us to write the resulting pressure distribution as Π , modulo functions of time. Hence

$$\int_0^T \int_{\mathbb{R}^3} P_n \nabla \cdot \varphi dy ds \rightarrow \int_0^T \int_K \Pi_K \nabla \cdot \varphi dy ds.$$

The value of the last integral is independent of the chosen K , because replacing Π_K by $\Pi_K + c(s)$ changes the integral by $\int_0^T c(s) \int_K \nabla \cdot \varphi dy ds = 0$. The limit is therefore the pressure term $\int \Pi \nabla \cdot \varphi$.

(v) *Drift term with a : part involving V_n .* Define

$$Y_n^0(s) := \int_{\mathbb{R}^3} V_n(y, s) \cdot \varphi(y, s) dy, \quad Y^0(s) := \int_{\mathbb{R}^3} U(y, s) \cdot \varphi(y, s) dy.$$

For each s , Cauchy–Schwarz gives

$$|Y_n^0(s)| \leq \|V_n(s)\|_{L^2(K_\varphi)} \|\varphi(s)\|_{L^2(K_\varphi)}.$$

By Assumption 5.14, there is $C_0 < \infty$ such that $\sup_n \|Y_n^0\|_{L^\infty(0,T)} \leq C_0$. Moreover,

$$Y_n^0 \rightarrow Y^0 \quad \text{in } L^2(0, T),$$

because $V_n \rightarrow U$ in $L^2(0, T; L^2(K_\varphi))$. Now

$$\int_0^T \alpha_n(s) Y_n^0(s) ds = \int_0^T (\alpha_n(s) - a_\infty) Y_n^0(s) ds + a_\infty \int_0^T Y_n^0(s) ds.$$

For the first term,

$$\left| \int_0^T (\alpha_n - a_\infty) Y_n^0 ds \right| \leq \sup_{s \in (0, T)} |Y_n^0(s)| \|\alpha_n - a_\infty\|_{L^1(0, T)} \rightarrow 0.$$

For the second term, strong convergence in $L^2(0, T)$ implies $\int_0^T Y_n^0 \rightarrow \int_0^T Y^0$, so

$$\int_0^T \alpha_n(s) \int_{\mathbb{R}^3} V_n \cdot \varphi dy ds \rightarrow a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot \varphi dy ds.$$

Thus the signed contribution in (12) satisfies

$$- \int_0^T \alpha_n(s) \int_{\mathbb{R}^3} V_n \cdot \varphi dy ds \rightarrow -a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot \varphi dy ds.$$

(vi) *Drift term with a : part involving $y \cdot \nabla V_n$.* Since $\varphi \in C_c^\infty$, integrating by parts in y gives

$$\int_{\mathbb{R}^3} (y \cdot \nabla V_n) \cdot \varphi = - \int_{\mathbb{R}^3} V_n \cdot (y \cdot \nabla \varphi + 3\varphi) dy.$$

Define

$$Y_n^1(s) := \int_{\mathbb{R}^3} V_n \cdot (y \cdot \nabla \varphi + 3\varphi) dy, \quad Y^1(s) := \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla \varphi + 3\varphi) dy.$$

For each s , Cauchy–Schwarz gives

$$|Y_n^1(s)| \leq \|V_n(s)\|_{L^2(K_\varphi)} \|y \cdot \nabla \varphi(s) + 3\varphi(s)\|_{L^2(K_\varphi)}.$$

The $L^\infty(0, T; L^2(K_\varphi))$ bound in Assumption 5.14 gives

$$\sup_n \|Y_n^1\|_{L^\infty(0, T)} \leq C_1 < \infty.$$

Also,

$$\|Y_n^1 - Y^1\|_{L^2(0, T)} \leq \|V_n - U\|_{L^2(0, T; L^2(K_\varphi))} \|y \cdot \nabla \varphi + 3\varphi\|_{L^\infty(0, T; L^2(K_\varphi))} \rightarrow 0.$$

Therefore

$$- \int_0^T \alpha_n(s) \int_{\mathbb{R}^3} (y \cdot \nabla V_n) \cdot \varphi dy ds \rightarrow a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla \varphi + 3\varphi) dy ds.$$

Integrating by parts once more in the limit gives

$$a_\infty \int_0^T \int_{\mathbb{R}^3} U \cdot (y \cdot \nabla \varphi + 3\varphi) dy ds = -a_\infty \int_0^T \int_{\mathbb{R}^3} (y \cdot \nabla U) \cdot \varphi dy ds.$$

(vii) *Term with b .* Since $\beta_n(s)$ is independent of y ,

$$\int_{\mathbb{R}^3} (\beta_n \cdot \nabla V_n) \cdot \varphi = - \int_{\mathbb{R}^3} V_n \cdot (\beta_n \cdot \nabla \varphi) dy,$$

so

$$- \int_0^T \int_{\mathbb{R}^3} (\beta_n \cdot \nabla V_n) \cdot \varphi dy ds = \int_0^T \int_{\mathbb{R}^3} V_n \cdot (\beta_n \cdot \nabla \varphi) dy ds.$$

Let

$$\begin{aligned} Z_n(s) &:= \left(\int_{\mathbb{R}^3} V_n \cdot \partial_1 \varphi dy, \int_{\mathbb{R}^3} V_n \cdot \partial_2 \varphi dy, \int_{\mathbb{R}^3} V_n \cdot \partial_3 \varphi dy \right), \\ Z(s) &:= \left(\int_{\mathbb{R}^3} U \cdot \partial_1 \varphi dy, \int_{\mathbb{R}^3} U \cdot \partial_2 \varphi dy, \int_{\mathbb{R}^3} U \cdot \partial_3 \varphi dy \right). \end{aligned}$$

Then

$$\int_{\mathbb{R}^3} V_n \cdot (\beta_n \cdot \nabla \varphi) dy = \beta_n(s) \cdot Z_n(s), \quad \int_{\mathbb{R}^3} U \cdot (b_\infty \cdot \nabla \varphi) dy = b_\infty \cdot Z(s).$$

Consequently

$$\left| \int_0^T (\beta_n - b_\infty) \cdot Z_n ds \right| \leq \|\beta_n - b_\infty\|_{L^1(0,T)} \sup_{s \in (0,T)} |Z_n(s)| \rightarrow 0,$$

because

$$|Z_n(s)| \leq \|V_n(s)\|_{L^2(K_\varphi)} \left(\sum_{j=1}^3 \|\partial_j \varphi(s)\|_{L^2(K_\varphi)}^2 \right)^{1/2}$$

and Assumption 5.14 bounds the right-hand side uniformly in n and s . Furthermore,

$$\|Z_n - Z\|_{L^2(0,T;\mathbb{R}^3)} \leq \|V_n - U\|_{L^2(0,T;L^2(K_\varphi))} \left(\sum_{j=1}^3 \|\partial_j \varphi\|_{L^\infty(0,T;L^2(K_\varphi))}^2 \right)^{1/2} \rightarrow 0.$$

Therefore

$$\int_0^T b_\infty \cdot Z_n(s) ds \rightarrow \int_0^T b_\infty \cdot Z(s) ds.$$

Hence

$$- \int_0^T \int_{\mathbb{R}^3} (\beta_n \cdot \nabla V_n) \cdot \varphi dy ds \rightarrow \int_0^T \int_{\mathbb{R}^3} U \cdot (b_\infty \cdot \nabla \varphi) dy ds.$$

Since b_∞ is constant in y , integration by parts yields

$$\int_0^T \int_{\mathbb{R}^3} U \cdot (b_\infty \cdot \nabla \varphi) dy ds = - \int_0^T \int_{\mathbb{R}^3} (b_\infty \cdot \nabla U) \cdot \varphi dy ds.$$

Step 5: assemble the limit. Passing to the limit in (12) in each term (Steps 4(i)–4(vii)) gives

$$\int_0^T \int_{\mathbb{R}^3} (-U \cdot \partial_s \varphi - (U \otimes U) : \nabla \varphi + \nu \nabla U : \nabla \varphi - \Pi \nabla \cdot \varphi - a_\infty (U + y \cdot \nabla U) \cdot \varphi - (b_\infty \cdot \nabla U) \cdot \varphi) dy ds = 0.$$

Since this holds for every compactly supported vector field φ , equation (11) holds in distributions on $(0, T) \times \mathbb{R}^3$. $\square$

5.6. Decomposition of alternatives of the scale-collapse branch. The cost argument in Sections 5.2 and 5.4 excludes the finite-cost scale-collapse branch. The remaining scale-collapse obstruction is treated by a decomposition of alternatives, following the local estimate and autonomous-reduction argument of the preceding sections. The conclusion separates the excluded branches from the residual alternatives; it does not assert a Liouville theorem for every autonomous negative-drift equation.

Theorem 5.21 (Selected core retention). *Let a represented Type II branch be obtained from the selected-core construction of the preceding sections. Then there exist $R > 0$, a cutoff $\chi \in C_c^\infty(B_R)$, $\chi \geq 0$, a constant $\eta > 0$, and a terminal time τ_1 such that, along every selected window used below,*

$$\int_{\mathbb{R}^3} |V(y, \tau)|^2 \chi(y) dy \geq \eta \quad \text{for a.e. } \tau \geq \tau_1.$$

In particular, every translated sequence $V_n(y, s) = V(y, \tau_n + s)$ satisfies the localized nontriviality condition in Assumption 5.14, after discarding finitely many n .

Proof. The selected-core construction chooses a compact renormalized core carrying a fixed positive amount of localized kinetic mass. Since the cutoff χ is supported strictly inside that core and equals a positive spatial weight there, the selected-core lower bound gives the stated estimate. Translating $\tau = \tau_n + s$ gives the final assertion on every finite selected window. $\square$

Theorem 5.22 (Autonomous modulation selection). *Let a represented scale-collapse branch satisfy the modulation bounds obtained in the repaired gauge construction. Suppose that a sequence of windows $[\tau_n, \tau_n + T]$ is selected in the autonomous regime. Then, after passing to a subsequence, there exist constants $a_\infty \in \mathbb{R}$ and $b_\infty \in \mathbb{R}^3$ such that*

$$a(\tau_n + \cdot) \rightarrow a_\infty \quad \text{in } L^1(0, T), \quad b(\tau_n + \cdot) \rightarrow b_\infty \quad \text{in } L^1(0, T; \mathbb{R}^3).$$

Proof. The autonomous-window selection in the repaired-gauge modulation analysis gives convergence of the modulation functions to constant autonomous parameters on the fixed window length T . Passing to the corresponding subsequence gives the displayed L^1 -convergences. $\square$

Theorem 5.23 (Scale-collapse decomposition of alternatives). *Let a represented Type II branch have genuine scale collapse and nonvanishing selected core. Then exactly one of the following alternatives occurs after passing to selected terminal windows:*

- (i) *the branch has finite scale-collapsing cost, which is excluded by Theorem 5.7;*
- (ii) *the branch has finite absolute cost, which is excluded by Corollary 5.9;*
- (iii) *the branch admits thick autonomous windows in the sense of Definition 5.11, satisfies the selected-window estimates of Assumption 5.14, and has autonomous modulation convergence;*
- (iv) *the branch has a thin-drift alternative: no selected fixed-length window carries a uniform positive lower bound for the averaged weighted negative drift;*
- (v) *the branch has an autonomous-modulation alternative: thick windows exist, but no subsequence has a constant-coefficient L^1 modulation limit in the sense needed for Theorem 5.19;*
- (vi) *the branch has a compactness alternative: the local estimates needed for Assumption 5.14 fail on the selected windows.*

Proof. The scale-collapse selection theorem decomposes terminal scale-collapse windows according to the averaged negative-drift mass

$$\frac{1}{T} \int_{\tau_n}^{\tau_n + T} a_-(\tau) M_R(\tau) d\tau.$$

If the accumulated cost is finite, alternatives (i) or (ii) apply according to the cost functional under consideration. If the averaged negative-drift mass stays bounded below on a subsequence, one obtains thick windows; otherwise the branch lies in the thin-drift alternative. On thick windows, either the local estimates and core retention hold, in which case Proposition 5.15 and Theorem 5.21 give Assumption 5.14, or the branch lies in the compactness alternative. If the estimates hold but no subsequence has the required constant-coefficient modulation limit, the branch lies in the autonomous-modulation alternative. The remaining case is the thick autonomous alternative. $\square$

Definition 5.24 (Stationary omega-limit class). Let (U, Π) be a nonzero autonomous limit produced by Theorem 5.19. We say that it has a stationary omega-limit in the admissible class if there exists a nonzero pair

$$(W, Q), \quad W \in L^3(\mathbb{R}^3),$$

with the local energy and pressure reconstruction properties inherited from the selected windows, such that

$$(W \cdot \nabla)W + \nabla Q = \nu \Delta W + a_\infty(W + y \cdot \nabla W) + b_\infty \cdot \nabla W, \quad \nabla \cdot W = 0,$$

in $\mathcal{D}'(\mathbb{R}^3)$.

Definition 5.25 (Rigidity-covered stationary class). For a parameter pair (a_∞, b_∞) , the stationary class is rigidity-covered if every admissible stationary solution in Definition 5.24 is either zero or is not a terminal Type II branch. In the classical backward self-similar normalization this is the role of the Nečas–Růžička–Šverák rigidity theorem, after the harmless translation drift is removed [NRŠ96]. Other parameter regimes require their own stated Liouville or classification theorem.

Theorem 5.26 (Stationary-rigidity exclusion). *Assume a thick autonomous branch produces a nonzero autonomous limit by Theorem 5.19. If that autonomous limit has a stationary omega-limit in the admissible class and the corresponding stationary class is rigidity-covered, then the branch cannot occur as a terminal Type II scale-collapse branch.*

Proof. The stationary omega-limit gives a nonzero stationary profile (W, Q) in the admissible class. The rigidity-covered hypothesis says that no such nonzero profile can represent a terminal Type II branch. This contradicts membership in the terminal scale-collapse Type II class. $\square$

Theorem 5.27 (Scale-collapse alternatives). *Let a represented Type II branch have genuine scale collapse and a nonvanishing selected core. Then one of the following mutually exclusive outcomes occurs:*

- (i) *the finite scale-collapsing cost exclusion applies;*
- (ii) *the finite absolute cost exclusion applies;*
- (iii) *the branch has a thin-drift alternative;*
- (iv) *the branch has an autonomous-modulation alternative;*
- (v) *the branch has a compactness alternative;*
- (vi) *the branch has a nonzero autonomous reduced limit, and either stationary rigidity excludes it or the remaining obstruction is the corresponding stationary/generalized self-similar Liouville problem.*

Proof. Apply Theorem 5.23. The finite-cost alternatives are items (i) and (ii). The thin-drift, modulation, and compactness alternatives are items (iii)–(v). In the remaining thick autonomous case, Lemma 5.13 gives $a_\infty < 0$, and Theorem 5.19 produces a nonzero autonomous reduced limit. If Definitions 5.24 and 5.25 apply, then Theorem 5.26 excludes the branch. If not, the remaining obstruction is the stationary or generalized self-similar Liouville problem for the parameter pair (a_∞, b_∞) . $\square$

5.7. Conclusion. The exclusions in Sections 5.2–5.4 are proved unconditionally within the renormalized setting: no branch with finite scale-collapsing (hence finite absolute) cost can have $\lambda(\tau) \rightarrow 0$ together with a nonvanishing localized core.

Lemma 5.13 shows that any thick autonomous window admits a negative autonomous drift limit a_∞ . Lemmas 5.16 and 5.17, together with Theorem 5.19, assemble the selected-window estimates from the preceding sections into a nontrivial autonomous finite-window limit (U, Π) satisfying the full distributional autonomous equation.

Section 5.6 gives the corresponding decomposition of alternatives for the scale-collapse branch. The autonomous reduction gives a contradiction only in parameter regimes where a stationary omega-limit exists and the resulting stationary class is covered by a Liouville or rigidity theorem, such as the classical Nečas–Růžička–Šverák self-similar class. Outside those regimes the conclusion is the thin-drift, modulation, compactness, or Liouville alternative, rather than an unconditional exclusion.

6. CORRECTED MONOTONICITY AND SCALE-DRIFT BARRIERS

This section records the remaining conditional analytic inputs needed to use a barrier argument on represented Type II branches. The statements are written entirely in the variables of the repaired gauge:

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0.$$

All pressure terms are understood modulo functions of time.

6.1. Application Data for a Conditional Type II Barrier.

Definition 6.1 (Barrier-application data). A represented Type II branch is said to have barrier-application data if the following analytic facts hold.

- (i) the branch is a scale-supercritical local Type II branch in the sense of the local concentration route;
- (ii) the physical Navier–Stokes energy inequality gives bounded energy along the branch;
- (iii) the branch is represented by the repaired-gauge variables (V, P, a, b) above, including the scale, center, pressure normalization, and modulation data;
- (iv) the active localized barrier cost is identified with

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y, \tau)|^2 \phi_{R_0}(y) dy + a_+(\tau) \int_{\mathbb{R}^3} |V(y, \tau)|^2 \phi_{R_0}(y) dy,$$

or with an explicitly stated tail-comparable nonnegative replacement;

- (v) a localized energy identity or suitable-solution local energy inequality is available for the cutoff used in the cost;
- (vi) a corrected localized monotonicity inequality, in the sense of [Theorem 6.5](#), is available;
- (vii) the conditional Type II barrier principle used in the final assembly has the same conclusion as the local Type II exclusion target: once the branch is scale-supercritical and its admissible nonnegative barrier cost is divergent, the branch cannot remain in the represented Type II class.

Lemma 6.2 (Scale route and energy). *If a branch is routed through the local Type II concentration alternative and satisfies the physical Navier–Stokes energy inequality, then it satisfies [Theorem 6.1](#)(i) and (ii).*

Proof. The local Type II route asserts precisely that the selected branch is scale-supercritical and carries a concentration profile. The physical energy inequality gives the bounded-energy part. These are precisely items (i) and (ii). $\square$

Lemma 6.3 (Cost identification). *Assume the Type II barrier cost is the localized Navier–Stokes cost $\tilde{\mathfrak{D}}_{R_0}$. Then divergence of*

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau) d\tau$$

is the barrier-cost divergence used by the conditional Type II barrier principle.

Proof. The active cost is assumed to be equal to $\tilde{\mathfrak{D}}_{R_0}$. Thus the cost appearing in the local Navier–Stokes estimates and the cost consumed by the barrier principle are the same nonnegative functional. The divergence condition is therefore identical in the two formulations. $\square$

Lemma 6.4 (Parabolic representation class). *If the repaired-gauge representation theorem applies to a Type II branch, then the branch lies in the parabolic transport-diffusion class required by the conditional Type II barrier principle.*

Proof. The repaired-gauge theorem supplies the renormalized orbit (V, P, a, b) , the scale and center, the pressure reconstruction modulo functions of time, and the modulation coefficients. The displayed equation is a parabolic Navier–Stokes equation with transport, dilation, and translation drift terms. Consequently the branch has the parabolic represented structure required by the barrier principle, with the same scale-supercritical and barrier-cost interpretations. $\square$

Definition 6.5 (Corrected localized monotonicity input). A represented branch has corrected localized monotonicity if there are a localized renormalized energy $\mathcal{E}_{R_0}$, a nonnegative barrier cost $\mathcal{D}_{R_0}$, a constant $c_0 > 0$, and a nonnegative remainder $\mathcal{R}_{R_0}$ such that, on the tail of the branch,

$$\frac{d}{d\tau} \mathcal{E}_{R_0}(\tau) + c_0 \mathcal{D}_{R_0}(\tau) + \mathcal{R}_{R_0}(\tau) \leq 0$$

in distributions. If the localized energy identity has an integrable error B_{R_0} , the corrected energy is

$$\mathcal{E}_{R_0}^{\text{corr}}(\tau) = \mathcal{E}_{R_0}(\tau) + \int_{\tau}^{\infty} B_{R_0}(s) ds,$$

and the same inequality is required for $\mathcal{E}_{R_0}^{\text{corr}}$. Multiplication of a nonnegative barrier cost by $c_0 > 0$ does not change whether its tail integral diverges.

Lemma 6.6 (Energy and Caccioppoli bounds alone are not monotonicity). *The global physical energy inequality and compact-cylinder Caccioppoli estimates do not, by themselves, imply the corrected localized monotonicity input of Theorem 6.5.*

Proof. The global physical energy inequality controls total physical energy and physical dissipation, but it does not identify a localized renormalized energy at the active Type II scale whose derivative is monotone after pressure flux, cutoff flux, transport, and modulation terms are included. The Caccioppoli estimate gives local spacetime H^1 control from the local energy inequality. It is an a priori estimate, not a corrected monotonicity formula. Therefore an additional absorption or tail-correction argument is needed. $\square$

Theorem 6.7 (Conditional barrier application). *Let a represented Type II branch have the data of Theorem 6.1. If its admissible nonnegative barrier cost has divergent tail integral, then the conditional Type II barrier principle applies to the branch, and the branch is excluded from the represented Type II class covered by that principle.*

Proof. Theorem 6.2 gives the scale-supercritical and bounded-energy inputs. Theorem 6.3 identifies the localized Navier–Stokes cost with the cost used in the barrier argument. Theorem 6.4 gives the represented parabolic class. The corrected monotonicity input is an explicit hypothesis of Theorem 6.1. These are precisely the analytic inputs listed in that definition, and the last item of the definition is the conditional Type II barrier implication with the same local conclusion. $\square$

6.2. Fixed-Cutoff Monotonicity. Let $\phi = \phi_{R_0} \in C_c^\infty(\mathbb{R}^3)$, $0 \leq \phi \leq 1$, be the cutoff used in the localized cost. Define

$$\mathcal{E}_\phi(\tau) = \frac{1}{2} \int_{\mathbb{R}^3} |V(y, \tau)|^2 \phi(y) dy,$$

$$A_\phi(\tau) = \int_{\mathbb{R}^3} |\nabla V(y, \tau)|^2 \phi(y) dy, \quad M_\phi(\tau) = \int_{\mathbb{R}^3} |V(y, \tau)|^2 \phi(y) dy,$$

and

$$\tilde{\mathcal{D}}_{R_0}(\tau) = \nu A_\phi(\tau) + a_+(\tau) M_\phi(\tau), \quad a_+(\tau) = \max\{a(\tau), 0\}.$$

The localized energy identity used below is

$$\frac{d}{d\tau} \mathcal{E}_\phi = -\nu A_\phi + \frac{\nu}{2} \int |V|^2 \Delta \phi + \frac{1}{2} \int |V|^2 V \cdot \nabla \phi + \int P V \cdot \nabla \phi$$

$$(13) \quad -\frac{a}{2}M_\phi - \frac{a}{2} \int |V|^2 y \cdot \nabla \phi - \frac{1}{2}b \cdot \int |V|^2 \nabla \phi.$$

Definition 6.8 (Fixed-cutoff error density). Set $a_-(\tau) = \max\{-a(\tau), 0\}$ and define

$$B_{R_0}(\tau) := \frac{\nu}{2} \left| \int |V|^2 \Delta \phi \right| + \frac{1}{2} \left| \int |V|^2 V \cdot \nabla \phi \right| + \left| \int P V \cdot \nabla \phi \right| \\ + \frac{1}{2} |b(\tau)| \left| \int |V|^2 \nabla \phi \right| + \frac{1}{2} a_-(\tau) M_\phi(\tau) + \frac{1}{2} |a(\tau)| \left| \int |V|^2 y \cdot \nabla \phi \right|.$$

The fixed-cutoff finite-error condition is

$$\int_{\tau_0}^{\infty} B_{R_0}(\tau) d\tau < \infty.$$

Remark 6.9 (Pressure normalization). The pressure term in B_{R_0} is independent of the representative of the pressure modulo functions of time. Replacing P by $P + c(\tau)$ changes the pressure flux by

$$c(\tau) \int V \cdot \nabla \phi = -c(\tau) \int \phi \nabla \cdot V = 0.$$

Theorem 6.10 (Fixed-cutoff corrected monotonicity). Assume (13) holds on (τ_0, ∞) and that $B_{R_0} \in L^1(\tau_0, \infty)$. Define

$$\mathcal{E}_\phi^{\text{corr}}(\tau) = \mathcal{E}_\phi(\tau) + \int_{\tau}^{\infty} B_{R_0}(s) ds.$$

Then, in distributions on (τ_0, ∞) ,

$$\frac{d}{d\tau} \mathcal{E}_\phi^{\text{corr}}(\tau) + \frac{1}{2} \tilde{\mathfrak{D}}_{R_0}(\tau) \leq 0.$$

Proof. Add

$$\frac{1}{2} \tilde{\mathfrak{D}}_{R_0} = \frac{\nu}{2} A_\phi + \frac{1}{2} a_+ M_\phi$$

to both sides of (13). The gradient term becomes

$$-\nu A_\phi + \frac{\nu}{2} A_\phi = -\frac{\nu}{2} A_\phi \leq 0.$$

The core scale term satisfies

$$-\frac{a}{2} M_\phi + \frac{1}{2} a_+ M_\phi = \frac{1}{2} a_- M_\phi \leq B_{R_0}(\tau).$$

Every remaining term in (13) is bounded above by its corresponding absolute-value contribution in B_{R_0} . Therefore

$$\frac{d}{d\tau} \mathcal{E}_\phi(\tau) + \frac{1}{2} \tilde{\mathfrak{D}}_{R_0}(\tau) \leq B_{R_0}(\tau)$$

in distributions. Since $B_{R_0} \in L^1(\tau_0, \infty)$, the correction tail is finite for every $\tau \geq \tau_0$, absolutely continuous on compact subintervals, and

$$\frac{d}{d\tau} \int_{\tau}^{\infty} B_{R_0}(s) ds = -B_{R_0}(\tau).$$

Adding this identity gives the claimed inequality. $\square$

Corollary 6.11 (Fixed-cutoff monotonicity input). *If the repaired-gauge representation, the cost identification, the localized energy identity, and the fixed-cutoff finite-error condition all hold, then Theorem 6.5 holds with $\mathcal{D}_{R_0} = \tilde{\mathfrak{D}}_{R_0}$ and $c_0 = 1/2$.*

Proof. The repaired-gauge representation supplies (V, P, a, b) . The cost identification supplies $\tilde{\mathfrak{D}}_{R_0}$. The localized energy identity and finite-error condition allow [Theorem 6.10](#) to be applied. $\square$

Remark 6.12 (Finite-tail subconditions). The fixed-cutoff finite-error condition is equivalent to the conjunction of tail integrability for the viscous cutoff term, the convective flux, the pressure flux, the center-drift cutoff term, the negative-scale term, and the scale-cutoff term:

$$\begin{aligned} & \left| \int |V|^2 \Delta \phi \right|, \quad \left| \int |V|^2 V \cdot \nabla \phi \right|, \quad \left| \int P V \cdot \nabla \phi \right|, \\ & |b| \left| \int |V|^2 \nabla \phi \right|, \quad a_- M_\phi, \quad |a| \left| \int |V|^2 y \cdot \nabla \phi \right|. \end{aligned}$$

Uniform compact-window bounds do not by themselves give this global tail integrability on $[\tau_0, \infty)$.

Definition 6.13 (Direct fixed-cutoff application data). A represented Type II branch has direct fixed-cutoff application data if it has the scale-supercritical concentration route, bounded physical energy, repaired-gauge representation, cost identification with $\tilde{\mathfrak{D}}_{R_0}$, divergent admissible barrier cost, localized energy identity, and the finite-tail bound $\int_{\tau_0}^\infty B_{R_0} < \infty$.

Theorem 6.14 (Direct fixed-cutoff application). *Every branch with direct fixed-cutoff application data is covered by [Theorem 6.7](#).*

Proof. By [Theorem 6.11](#), the finite-tail condition supplies the corrected monotonicity input. The remaining assumptions are precisely the other items in [Theorem 6.1](#), together with cost divergence. $\square$

Definition 6.15 (Compact-barrier replacement). A compact-barrier replacement is available when the compact Type II hypotheses of the preceding sections imply

$$\int_{\tau_0}^\infty \tilde{\mathfrak{D}}_{R_0}(\tau) d\tau = \infty,$$

and this divergence is the admissible barrier-cost divergence used in [Theorem 6.7](#). In particular, it may be supplied by the combination of classification of local alternatives, uniform critical tightness, local windowed H^1 control, and the cost identification above whenever those are the active route to the compact Type II theorem.

Theorem 6.16 (Expanded compact-barrier application). *If a represented branch has all direct fixed-cutoff application data except that cost divergence is supplied through the compact-barrier replacement of [Theorem 6.15](#), then the conditional Type II barrier principle applies.*

Proof. The compact-barrier replacement supplies the missing divergent admissible cost. Substituting this into the direct fixed-cutoff data gives the hypotheses of [Theorem 6.14](#). $\square$

6.3. Moving-Cutoff Monotonicity. Let $\phi \in C_c^\infty(\mathbb{R}^3)$ be radial, $0 \leq \phi \leq 1$, $\phi = 1$ on B_1 , and $\phi = 0$ outside B_2 . For $R(\tau) \geq R_0$, locally absolutely continuous and nondecreasing, set

$$\Phi(\tau, y) = \phi(y/R(\tau)), \quad A_{R(\tau)} = \{R(\tau) \leq |y| \leq 2R(\tau)\},$$

$$\mathcal{E}_R(\tau) = \frac{1}{2} \int |V(y, \tau)|^2 \Phi(\tau, y) dy,$$

and

$$\tilde{\mathfrak{D}}_{R(\tau)}(\tau) = \nu \int |\nabla V|^2 \Phi + a_+(\tau) \int |V|^2 \Phi.$$

Definition 6.17 (Moving-cutoff error density). Define

$$B_{\text{mov}}(\tau) := \frac{1}{2} \left| \int |V|^2 \partial_\tau \Phi \right| + \frac{\nu}{2} \left| \int |V|^2 \Delta \Phi \right| + \frac{1}{2} \left| \int |V|^2 V \cdot \nabla \Phi \right| + \left| \int P V \cdot \nabla \Phi \right| \\ + \frac{1}{2} |b(\tau)| \left| \int |V|^2 \nabla \Phi \right| + \frac{1}{2} a_-(\tau) \int |V|^2 \Phi + \frac{1}{2} |a(\tau)| \left| \int |V|^2 y \cdot \nabla \Phi \right|.$$

The moving finite-error condition is

$$\int_{\tau_0}^{\infty} B_{\text{mov}}(\tau) d\tau < \infty.$$

Theorem 6.18 (Moving-cutoff corrected monotonicity). *Assume the localized energy identity is valid with the time-dependent cutoff Φ , and assume $B_{\text{mov}} \in L^1(\tau_0, \infty)$. Then*

$$\mathcal{E}_R^{\text{corr}}(\tau) = \mathcal{E}_R(\tau) + \int_{\tau}^{\infty} B_{\text{mov}}(s) ds$$

satisfies

$$\frac{d}{d\tau} \mathcal{E}_R^{\text{corr}}(\tau) + \frac{1}{2} \tilde{\mathfrak{D}}_{R(\tau)}(\tau) \leq 0$$

in distributions.

Proof. Use the localized energy identity with the time-dependent test function $\Phi(\tau, y)$. Compared with the fixed-cutoff identity, the only new term is

$$\frac{1}{2} \int |V|^2 \partial_\tau \Phi,$$

which comes from differentiating the cutoff inside $\mathcal{E}_R$. All other terms are the fixed-cutoff terms with ϕ replaced by Φ . Add $\frac{1}{2} \tilde{\mathfrak{D}}_{R(\tau)}$ to both sides. The gradient term leaves

$$-\frac{\nu}{2} \int |\nabla V|^2 \Phi \leq 0,$$

and the core scale term contributes

$$\frac{1}{2} a_- \int |V|^2 \Phi.$$

Each remaining term is bounded above by the corresponding absolute-value term in B_{mov} . Hence

$$\frac{d}{d\tau} \mathcal{E}_R(\tau) + \frac{1}{2} \tilde{\mathfrak{D}}_{R(\tau)}(\tau) \leq B_{\text{mov}}(\tau).$$

The moving finite-error condition makes the correction tail finite and absolutely continuous on compact subintervals. Differentiating that tail subtracts B_{mov} , giving the stated inequality. $\square$

Definition 6.19 (Summable moving-annulus errors). The moving-annulus summability condition is the existence of a nondecreasing locally absolutely continuous $R(\tau) \rightarrow \infty$ such that each of the seven quantities

$$\left| \int |V|^2 \partial_\tau \Phi \right|, \quad \left| \int |V|^2 \Delta \Phi \right|, \quad \left| \int |V|^2 V \cdot \nabla \Phi \right|, \quad \left| \int P V \cdot \nabla \Phi \right|, \\ |b| \left| \int |V|^2 \nabla \Phi \right|, \quad a_- \int |V|^2 \Phi, \quad |a| \left| \int |V|^2 y \cdot \nabla \Phi \right|$$

is integrable on $[\tau_0, \infty)$.

Lemma 6.20 (Summable annuli imply moving finite error). *If [Theorem 6.19](#) holds, then $B_{\text{mov}} \in L^1(\tau_0, \infty)$.*

Proof. The density B_{mov} is a finite positive linear combination of the seven quantities listed in [Theorem 6.19](#). If each is integrable, then their linear combination is integrable. $\square$

Definition 6.21 (Summable tightness schedule). A summable tightness schedule is a nondecreasing sequence $R_n \rightarrow \infty$ such that, on unit windows $I_n = [\tau_0 + n, \tau_0 + n + 1]$,

$$\sum_{n=0}^{\infty} \sup_{\tau \in I_n} \|V(\tau)\|_{L^3(A_{R_n})}^3 < \infty,$$

and the same schedule is compatible with the pressure-tail normalization used for P . This is stronger than uniform critical tightness, which permits large radii on each window but does not by itself give a monotone radius schedule with a summable series.

Definition 6.22 (Moving-cost admissibility). The moving localized cost is admissible if

$$\tilde{\mathfrak{D}}_{R(\tau)}(\tau) = \nu \int |\nabla V|^2 \phi_{R(\tau)} + a_+(\tau) \int |V|^2 \phi_{R(\tau)}$$

is accepted as the active Type II barrier cost, or is compared to the fixed cost by a positive constant on the tail so that cost divergence is unchanged.

Theorem 6.23 (Moving-cutoff replacement). *Assume the repaired-gauge representation, localized energy identity, moving-cost admissibility, and summable moving-annulus errors. Then the moving cutoff supplies the corrected monotonicity input needed in [Theorem 6.7](#).*

Proof. [Theorem 6.20](#) gives $B_{\text{mov}} \in L^1(\tau_0, \infty)$. Applying [Theorem 6.18](#) gives corrected monotonicity with cost $\tilde{\mathfrak{D}}_{R(\tau)}$. The moving-cost admissibility assumption identifies this cost, or a tail-comparable version of it, with the active barrier cost. $\square$

6.4. Negative Scale Drift. Set

$$M_R(\tau) = \int |V(y, \tau)|^2 \Phi(\tau, y) dy.$$

The negative-scale contribution in the moving cutoff argument is

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau.$$

Definition 6.24 (Negative scale-drift alternatives). We say that the branch has finite negative drift if

$$\int_{\tau_0}^{\infty} a_-(\tau) d\tau < \infty.$$

It has bounded moving L^2 core mass if

$$\sup_{\tau \geq \tau_0} M_R(\tau) < \infty.$$

It has a moving L^2 core floor if there are $m_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$M_R(\tau) \geq m_0 \quad \text{for a.e. } \tau \geq \tau_1.$$

The scale-collapse drift obstruction is the divergent alternative

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau = \infty.$$

Lemma 6.25 (Finite negative drift controls the scale term). *If the branch has bounded moving L^2 core mass and finite negative drift, then*

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau < \infty.$$

Proof. Let $M_* = \sup_{\tau \geq \tau_0} M_R(\tau) < \infty$. Since $a_- \geq 0$,

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau \leq M_* \int_{\tau_0}^{\infty} a_-(\tau) d\tau < \infty.$$

□

Theorem 6.26 (Finite-or-infinite scale-negative alternative). *For every represented orbit for which a_- and M_R are measurable and nonnegative, precisely one of the following alternatives holds:*

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau < \infty, \quad \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau = \infty.$$

Proof. The integral

$$I_R = \int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau$$

is an extended nonnegative number. Hence either $I_R < \infty$ or $I_R = \infty$, and the two cases are mutually exclusive. □

Lemma 6.27 (Logarithmic scale identity). *For the repaired-gauge convention*

$$\tau_t = \lambda(t)^{-2}, \quad a(\tau) = \lambda(t) \lambda_t(t),$$

one has

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1)$$

whenever $\tau_0 \leq \tau_1 < \tau_2 < \infty$.

Proof. The chain rule gives

$$\frac{d}{d\tau} \log \lambda = \frac{\lambda_t}{\lambda} \frac{dt}{d\tau} = \frac{\lambda_t}{\lambda} \lambda^2 = \lambda \lambda_t = a.$$

Integrating over $[\tau_1, \tau_2]$ proves the identity. □

Theorem 6.28 (Collapse with core floor forces scale-drift obstruction). *Assume*

$$\lambda(\tau) \rightarrow 0 \quad \text{as } \tau \rightarrow \infty,$$

and assume the moving L^2 core floor of [Theorem 6.24](#). Then

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau = \infty.$$

Proof. By [Theorem 6.27](#),

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

Since $\lambda(\tau) \rightarrow 0$, the right-hand side tends to $-\infty$ as $\tau_2 \rightarrow \infty$. If $\int_{\tau_1}^{\infty} a_-(\tau) d\tau < \infty$, then

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \int_{\tau_1}^{\tau_2} a_+(\tau) d\tau - \int_{\tau_1}^{\tau_2} a_-(\tau) d\tau \geq - \int_{\tau_1}^{\infty} a_-(\tau) d\tau > -\infty,$$

which contradicts the preceding limit. Thus $\int_{\tau_1}^{\infty} a_-(\tau) d\tau = \infty$.

By the moving L^2 core floor, $M_R(\tau) \geq m_0$ for a.e. $\tau \geq \tau_1$. Therefore

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau \geq m_0 \int_{\tau_1}^{\infty} a_-(\tau) d\tau = \infty.$$

□

Corollary 6.29 (Scale-drift routing). *In the moving-cutoff setup, the negative-scale term has the following exhaustive routing. If bounded moving L^2 core mass and finite negative drift hold, then the negative-scale term is integrable. If the weighted negative-drift integral is infinite, the branch lies in the scale-collapse drift obstruction. If genuine collapse and a moving L^2 core floor also hold, then the obstruction is forced.*

Proof. The first case is [Theorem 6.25](#). The exhaustive split is [Theorem 6.26](#). The final assertion is [Theorem 6.28](#). $\square$

6.5. Scale-Drift Barrier Costs.

Definition 6.30 (Scale-collapse drift cost). For a represented branch define

$$\mathfrak{C}_{\text{sc}}(\tau) = a_-(\tau)M_R(\tau), \quad M_R(\tau) = \int |V(y, \tau)|^2 \phi_{R(\tau)}(y) dy,$$

and

$$\mathcal{C}_{\text{sc}}[\tau_0, \infty) = \int_{\tau_0}^{\infty} \mathfrak{C}_{\text{sc}}(\tau) d\tau.$$

Thus the scale-collapse drift obstruction is precisely $\mathcal{C}_{\text{sc}}[\tau_0, \infty) = \infty$. The density is nonnegative and local in the renormalized core.

Definition 6.31 (Scale-collapse cost admissibility). The scale-collapse drift cost is admissible when:

- (i) $\mathfrak{C}_{\text{sc}}$ is measurable, nonnegative, and locally integrable on finite τ -intervals for represented branches;
- (ii) divergence of $\mathcal{C}_{\text{sc}}[\tau_0, \infty)$ is an accepted Type II barrier condition;
- (iii) the resulting barrier conclusion is the same local Type II exclusion conclusion used for the fixed localized cost.

Theorem 6.32 (Scale-collapse drift is a barrier when admissible). *If the scale-collapse drift obstruction holds and the cost $\mathfrak{C}_{\text{sc}}$ is admissible, then the divergent-cost hypothesis of [Theorem 6.7](#) is satisfied for the scale-collapse branch.*

Proof. By [Theorem 6.30](#), the obstruction is precisely

$$\mathcal{C}_{\text{sc}}[\tau_0, \infty) = \infty.$$

By [Theorem 6.31](#), this divergence is an accepted nonnegative Type II barrier condition with the same local conclusion. $\square$

Corollary 6.33 (Scale-collapse suppression under the conditional principle). *Assume a scale-supercritical represented Type II branch has scale-collapse drift obstruction, admissible scale-collapse drift cost, and all other application data required by the conditional Type II barrier principle. Then the branch is excluded from the represented Type II class covered by that principle.*

Proof. [Theorem 6.32](#) supplies the divergent admissible barrier cost. Applying [Theorem 6.7](#) gives the exclusion. $\square$

Definition 6.34 (Absolute scale-drift cost). Define

$$\mathfrak{C}_{\text{abs}}(\tau) = \nu \int |\nabla V|^2 \phi_{R(\tau)} + |a(\tau)|M_R(\tau).$$

The absolute scale-drift cost is admissible if divergence of $\int_{\tau_0}^{\infty} \mathfrak{C}_{\text{abs}}(\tau) d\tau$ is accepted as a Type II barrier condition with the same local conclusion.

Lemma 6.35 (Scale-collapse drift forces absolute-cost divergence). *If the scale-collapse drift obstruction holds, then*

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\text{abs}}(\tau) d\tau = \infty.$$

Proof. Since

$$\mathfrak{C}_{\text{abs}}(\tau) = \nu \int |\nabla V|^2 \phi_{R(\tau)} + |a(\tau)| M_R(\tau) \geq a_-(\tau) M_R(\tau),$$

divergence of $\int a_- M_R$ forces divergence of $\int \mathfrak{C}_{\text{abs}}$. $\square$

Corollary 6.36 (Absolute scale-cost barrier). *If the scale-collapse drift obstruction holds and the absolute scale-drift cost is admissible, then the divergent-cost hypothesis of Theorem 6.7 is satisfied.*

Proof. Theorem 6.35 gives divergence of the absolute scale-drift cost, and admissibility makes this divergence an accepted Type II barrier condition. $\square$

Theorem 6.37 (Moving route with scale-collapse fallback). *Assume the moving-cutoff setup and a scale-supercritical represented Type II branch.*

- (i) *If the negative-scale term is integrable and the remaining moving-annulus errors are summable, then the moving-cutoff monotonicity route is available.*
- (ii) *If the scale-collapse drift obstruction holds and either the scale-collapse drift cost or the absolute scale-drift cost is admissible, then the divergent-cost hypothesis of the conditional barrier principle is satisfied directly.*
- (iii) *If, in addition, the remaining application data of Theorem 6.1 hold, then the represented Type II branch is excluded.*

Proof. The first branch is Theorem 6.23. The second branch is Theorem 6.32 or Theorem 6.36, depending on which admissible cost is used. The third branch is Theorem 6.7. $\square$

6.6. Terminal Conditional Assembly.

Definition 6.38 (Abstract final Type II data). The abstract final Type II data consist of:

- (i) classification of represented Type II branches into the local alternatives used in the final assembly;
- (ii) exclusion of the radiative or noncompact alternative, equivalently the uniform critical tightness conclusion on represented branches;
- (iii) exclusion of the rough-core alternative, equivalently uniform local windowed H^1 control on represented branches;
- (iv) the conditional Type II barrier principle of Theorem 6.7.

Theorem 6.39 (Abstract final conditional Type II theorem). *If the abstract final Type II data hold, then every represented Type II branch in the covered class is excluded.*

Proof. The classification data place every represented Type II branch in one of the local alternatives. The radiative and rough-core alternatives are excluded by items (ii) and (iii) of Theorem 6.38. Any branch carrying a divergent admissible Type II barrier cost is excluded by item (iv). The remaining compact, multibubble, cascade, and scale-rigid alternatives are the ones treated in the preceding local sections and in the final assembly below. $\square$

Definition 6.40 (Expanded terminal data). Expanded terminal data are the following analytic inputs:

- (i) classification of represented Type II branches into the ordered local alternatives;

- (ii) the aggregate technical Type II exclusions used for compact, multibubble, scale-collapse, and rough-core branches, excluding the conditional barrier principle itself;
- (iii) the no-radiation route yielding uniform critical tightness;
- (iv) an exhaustive terminal state-space partition and a local finite critical-mass package for retained active terminal strata;
- (v) small-data stability in L^3 and a critical Navier–Stokes profile decomposition with nonlinear stability;
- (vi) removal of the scattering branch and discard of physically exterior regular profiles;
- (vii) the repaired-gauge representation theorem;
- (viii) the Caccioppoli/windowed regularity input;
- (ix) the scale-collapse rigidity input used by the multibubble and scale-collapse reductions;
- (x) the conditional Type II barrier principle.

Lemma 6.41 (Terminal data give the radiative exclusion). *The expanded terminal data imply exclusion of the radiative or noncompact alternative.*

Proof. The repaired-gauge representation turns represented orbits into terminal sequences. The terminal state-space partition, the finite critical-mass input, small-data L^3 stability, and the critical profile decomposition give the profile completeness needed for the terminal profile route. The scattering branch and exterior regular profiles are removed by hypothesis. Together with the aggregate technical Type II exclusions, the Caccioppoli input, and the scale-collapse rigidity input, the multibubble alternatives are removed. The remaining non-bubble noncompact branch is the radiative alternative, and the no-radiation route excludes it, equivalently giving uniform critical tightness. $\square$

Lemma 6.42 (Terminal data give the rough-core exclusion). *The expanded terminal data imply exclusion of the rough-core alternative.*

Proof. The rough-core composition requires the represented branch, local pressure and modulation control, compact-cylinder critical bounds, and Caccioppoli/windowed regularity. These are included in the expanded terminal data through the aggregate technical exclusions, the repaired-gauge representation, and the Caccioppoli input. Hence the universal rough-core exclusion is available. $\square$

Lemma 6.43 (Expanded terminal data give abstract final data). *The expanded terminal data imply the abstract final Type II data.*

Proof. The expanded terminal data include classification of represented Type II branches and the conditional Type II barrier principle. By [Theorem 6.41](#) they give the radiative exclusion, and by [Theorem 6.42](#) they give the rough-core exclusion. These are precisely the four entries in [Theorem 6.38](#). $\square$

Theorem 6.44 (Expanded terminal Type II theorem). *If the expanded terminal data hold, then every represented Type II branch in the covered class is excluded.*

Proof. [Theorem 6.43](#) gives the abstract final Type II data. Applying [Theorem 6.39](#) excludes every represented Type II branch in the covered class. $\square$

Remark 6.45 (Direct branchwise reading). Equivalently, compact single-core branches are removed by the compact Type II barrier; scale-collapse branches are removed by the scale-collapse rigidity input, or by [Theorem 6.32](#) when the scale-collapse drift cost is admissible; multibubble and radiative branches are removed by the terminal profile and no-radiation inputs; rough-core branches are removed by the rough-core estimate; and every divergent admissible barrier cost is converted to the same local Type II exclusion conclusion by the conditional barrier principle.

7. ASSEMBLY OF THE LOCAL TYPE II EXCLUSION

7.1. Local Entry. Let (u, p) be a suitable weak solution of the three-dimensional Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0,$$

on $\mathbb{R}^3 \times (T - \delta, T)$. For $z_0 = (x_0, T)$ and $r > 0$, set

$$Q_r(z_0) = B_r(x_0) \times (T - r^2, T),$$

$$C(z_0, r) = r^{-2} \int_{Q_r(z_0)} |u|^3 dx dt,$$

and

$$D(z_0, r) = r^{-2} \int_{T-r^2}^T \int_{B_r(x_0)} |p(x, t) - (p)_{B_r(x_0)}(t)|^{3/2} dx dt.$$

The pressure is always taken modulo functions of time, and the spatial mean normalization is the one used in the Caffarelli–Kohn–Nirenberg criterion.

Definition 7.1 (Singular set and local Type II point). Let $\Sigma(T)$ be the set of points x_0 for which u is not locally bounded in any backward cylinder $Q_r(x_0, T)$. A point $x_0 \in \Sigma(T)$ is in the local Type II alternative if it carries a positive local Caffarelli–Kohn–Nirenberg concentration sequence and no local Type I scale-invariant bound holds at (x_0, T) .

Definition 7.2 (Positive local Type II concentration sequence). A sequence $r_n \downarrow 0$ is a positive local Type II concentration sequence at (x_0, T) if there exists $\eta > 0$ such that

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n,$$

and (x_0, T) is in the local Type II alternative. The associated parabolic rescalings are

$$u_n(y, s) = r_n u(x_0 + r_n y, T + r_n^2 s), \quad p_n(y, s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Theorem 7.3 (Local concentration and Type I/Type II dichotomy). *If $\Sigma(T) \neq \emptyset$, then there are $x_0 \in \Sigma(T)$, $\eta > 0$, and $r_n \downarrow 0$ such that*

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

At the concentration point exactly one of the following alternatives holds: there is a local Type I scale-invariant bound, or (x_0, T) is in the local Type II alternative.

Proof. The theorem is the standing local concentration dichotomy assumed in the assembly. $\square$

7.2. Local Theorems Used in the Assembly. The local results needed for the final assembly are restated in the following form.

Theorem 7.4 (Repaired-gauge representation). *Let a positive local Type II concentration sequence contain a nondegenerate selected core. Then, after passing to selected windows, there are local scale-center variables $\lambda(\tau) > 0$, $x_c(\tau)$, renormalized fields*

$$V(y, \tau) = \lambda(\tau) u(x_c(\tau) + \lambda(\tau) y, t(\tau)), \quad P(y, \tau) = \lambda(\tau)^2 p(x_c(\tau) + \lambda(\tau) y, t(\tau)) + \pi(\tau),$$

and modulation coefficients a, b such that

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0.$$

On each compact renormalized window one has the local suitable energy inequality, local pressure reconstruction modulo functions of time, stability under subsequences, and the compact-window bounds proved in [Section 2](#). Failure of the nondegenerate scale-center selection is part of the multibubble, cascade, or scale-rigidity alternatives.

Proof. [Theorem 2.35](#) is precisely the compact-window representation theorem. $\square$

Theorem 7.5 (Compact single-core exclusion). *No compact single-core vanishing-dissipation branch can be a local Type II sequence. More explicitly, let a represented selected-window sequence have:*

- (i) *positive local CKN concentration on a fixed compact cylinder;*
- (ii) *uniformly bounded compact-cylinder suitable estimates $A + E + C + D$;*
- (iii) *nondegenerate local scale-center representation;*
- (iv) *bounded modulation coefficients;*
- (v) *local pressure reconstruction and pressure stability modulo spatial means;*
- (vi) *vanishing localized dissipation on some retained compact cylinder.*

Then the sequence is incompatible with the local Type II condition.

Proof. [Theorem 1.18](#) gives the stated exclusion. $\square$

Theorem 7.6 (Multibubble and cascade exclusion). *Assume the terminal admissibility and scale-rigidity hypotheses stated in the multibubble/cascade analysis. Then no local multibubble, cascade, or gauge-degenerate Type II branch can occur in the class produced by the local active-core extraction. Equivalently, every failure of the single nondegenerate compact-core alternative is excluded by the local multibubble/cascade analysis under those stated hypotheses.*

Proof. [Theorem 3.23](#), together with its removal corollary [Theorem 3.24](#), gives the stated exclusion. $\square$

Theorem 7.7 (Rough-core reduction). *On any compact renormalized window satisfying the local energy inequality and bounded modulation hypotheses, finite outer velocity-pressure CKN control $C + D$ implies finite inner windowed H^1 control. Hence failure of local windowed H^1 control on a retained inner cylinder forces failure of the compact-cylinder $C + D$ bounds on every relevant enclosing cylinder.*

Proof. [Theorem 4.7](#) gives the stated reduction. $\square$

Theorem 7.8 (Finite-cost scale-collapse exclusion). *No represented branch with finite scale-collapsing cost, and hence no branch with finite absolute scale cost, can have genuine scale collapse together with a nonvanishing selected core.*

Proof. [Theorems 5.7](#) and [5.9](#) give the stated exclusion. $\square$

Theorem 7.9 (Scale-collapse decomposition of alternatives). *Let a represented Type II branch have genuine scale collapse and a nonvanishing selected core. After passing to selected terminal windows, exactly one of the alternatives isolated in [Section 5](#) occurs:*

- (i) *finite scale-collapsing cost;*
- (ii) *finite absolute scale cost;*
- (iii) *thin-drift alternative;*
- (iv) *autonomous-modulation alternative;*
- (v) *compactness alternative;*
- (vi) *a nonzero autonomous reduced limit in the scale-rigid state.*

In the local Type II class of alternatives used by the multibubble/cascade and scale-collapse analyses, alternatives (iii)–(vi) do not constitute additional compact single-core mechanisms: the thin-drift, modulation, and compactness alternatives lie outside the compact finite-cost scale-collapse branch, and the scale-rigid state is covered by the scale-rigidity hypothesis in the multibubble/cascade theorem.

Proof. The assertion follows by combining the scale-collapse decomposition of alternatives with the scale-rigidity alternative used in the multibubble/cascade reduction. $\square$

Definition 7.10 (Virial-dissipative terminal state). Let $G_\nu(y) = e^{-|y|^2/(4\nu)}$. A scale-rigid terminal state is called velocity-virial dissipative if its selected terminal representative (V, P) satisfies, after the usual pressure normalization, the autonomous self-similar equation

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \nabla \cdot V = 0,$$

and the Gaussian identity below is justified by cutoff approximation on the selected renormalized windows. It is called swirl-virial dissipative if, in addition, the representative is axisymmetric, $\Gamma = RV_\theta$ is the scale-invariant circulation, and the corresponding weighted circulation identity below is justified.

Theorem 7.11 (Virial exclusion for dissipative terminal states). *Let a scale-rigid terminal state satisfy the hypotheses of Theorem 7.10 and be bounded for all negative renormalized times in the sense needed for the Gaussian integrals below.*

(a) *Suppose the velocity virial defect is absorbable: for some $\kappa > 0$ and a.e. τ ,*

$$-\frac{1}{2\nu} \int_{\mathbb{R}^3} |V|^2 (V \cdot y) G_\nu dy - \frac{1}{\nu} \int_{\mathbb{R}^3} P (V \cdot y) G_\nu dy \leq (1-\kappa) \int_{\mathbb{R}^3} |V|^2 G_\nu dy + 2\nu(1-\kappa) \int_{\mathbb{R}^3} |\nabla V|^2 G_\nu dy.$$

Then

$$\int_{\mathbb{R}^3} |V(y, \tau)|^2 G_\nu(y) dy = 0 \quad \text{for every } \tau.$$

Consequently no retained terminal core with positive Gaussian velocity mass can belong to this branch.

(b) *Suppose the terminal state is axisymmetric and the circulation defect is absorbable: for some $\kappa > 0$ and a.e. τ ,*

$$-\frac{1}{2\nu} \int_{\mathbb{R}^3} \Gamma^2 (V \cdot y) G_\nu dy \leq (1-\kappa) \int_{\mathbb{R}^3} \Gamma^2 G_\nu dy + 2\nu(1-\kappa) \int_{\mathbb{R}^3} |\nabla \Gamma|^2 G_\nu dy.$$

Then

$$\int_{\mathbb{R}^3} \Gamma(y, \tau)^2 G_\nu(y) dy = 0 \quad \text{for every } \tau.$$

Thus every retained axisymmetric terminal core whose remaining obstruction is a positive Gaussian circulation mass is excluded by the weighted circulation identity.

Proof. For the velocity statement set

$$\mathcal{E}(\tau) = \int_{\mathbb{R}^3} |V(y, \tau)|^2 G_\nu(y) dy.$$

The Gaussian virial identity gives

$$\partial_\tau \mathcal{E} = -2\nu \int_{\mathbb{R}^3} |\nabla V|^2 G_\nu dy - \int_{\mathbb{R}^3} |V|^2 G_\nu dy - \frac{1}{2\nu} \int_{\mathbb{R}^3} |V|^2 (V \cdot y) G_\nu dy - \frac{1}{\nu} \int_{\mathbb{R}^3} P (V \cdot y) G_\nu dy.$$

The absorbability hypothesis therefore implies

$$\partial_\tau \mathcal{E} + \kappa \mathcal{E} + 2\nu \kappa \int_{\mathbb{R}^3} |\nabla V|^2 G_\nu dy \leq 0.$$

In particular $\partial_\tau \mathcal{E} + \kappa \mathcal{E} \leq 0$. Integrating from $s < \tau$ to τ gives

$$\mathcal{E}(\tau) \leq e^{-\kappa(\tau-s)} \mathcal{E}(s).$$

The ancient boundedness of the terminal state makes $\sup_{s < \tau} \mathcal{E}(s) < \infty$. Sending $s \rightarrow -\infty$ yields $\mathcal{E}(\tau) = 0$.

For the axisymmetric statement set

$$\mathcal{G}(\tau) = \int_{\mathbb{R}^3} \Gamma(y, \tau)^2 G_\nu(y) dy.$$

The weighted circulation identity is

$$\partial_\tau \mathcal{G} + \mathcal{G} + 2\nu \int_{\mathbb{R}^3} |\nabla \Gamma|^2 G_\nu dy = -\frac{1}{2\nu} \int_{\mathbb{R}^3} \Gamma^2 (V \cdot y) G_\nu dy.$$

The assumed absorption gives

$$\partial_\tau \mathcal{G} + \kappa \mathcal{G} + 2\nu \kappa \int_{\mathbb{R}^3} |\nabla \Gamma|^2 G_\nu dy \leq 0.$$

The same backward Gronwall argument gives $\mathcal{G}(\tau) = 0$ for every τ . $\square$

Remark 7.12 (Controlled swirl alone). If the axisymmetric representative satisfies $\Gamma \in L^\infty$, the same weighted circulation identity gives the a priori estimate

$$\mathcal{G}(\tau)^{1/2} \lesssim \|V\|_{L^\infty} \|\Gamma\|_{L^\infty}$$

for bounded ancient terminal states. This estimate enters the local decomposition of alternatives but does not by itself exclude a terminal state. The exclusion requires the stated absorption, vanishing conclusion, or an equivalent condition which makes the virial inequality dissipative.

7.3. Assembly of Alternatives.

Theorem 7.13 (Local Type II decomposition into local alternatives). *Every positive local Type II concentration sequence in Theorem 7.2, after passing to subsequences and selected windows, enters one of the following alternatives:*

- (i) a compact single-core vanishing-dissipation branch;
- (ii) a multibubble, cascade, or gauge-degenerate branch;
- (iii) a rough-core branch, meaning loss of local windowed H^1 control on a retained compact cylinder;
- (iv) a finite-cost scale-collapse branch with nonvanishing selected core;
- (v) a scale-rigid state covered by the scale-rigidity hypothesis in Theorem 7.6.

In the last alternative, any terminal subbranch satisfying Theorem 7.11 is eliminated before the remaining scale-rigid state is passed to the scale-rigidity hypothesis.

Proof. This decomposition is obtained by combining Theorems 7.4, 7.6, 7.7 and 7.9. The representation theorem gives the nondegenerate single-core chart whenever a unique retained core is selected. Failure of that selection, splitting of positive CKN mass, compactness failure, or scale cascade is the multibubble/cascade side. On a represented retained core, rough-core loss is equivalent by Theorem 7.7 to failure of compact CKN control on an enclosing cylinder, hence returns to the multibubble/cascade side. Genuine scale collapse is divided by Theorem 7.9 into finite-cost scale collapse or one of the scale-rigid terminal states used in Theorem 7.6. These cases exhaust the local alternatives. $\square$

Proposition 7.14 (Elimination of local Type II states). *No positive local Type II concentration sequence can lie in any of the alternatives (i)–(v) of Theorem 7.13.*

Proof. Alternative (i) is excluded by the compact single-core criterion, Theorem 7.5.

Alternative (ii) is excluded by the local multibubble and cascade theorem, Theorem 7.6, under its terminal admissibility and scale-rigidity hypotheses.

Alternative (iii) is not an independent terminal state. By Theorem 7.7, rough-core loss of local H^1 control forces failure of the compact-cylinder $\mathcal{C} + \mathcal{D}$ bounds on enclosing compact cylinders.

Such failure yields an active compact-cylinder concentration branch and therefore belongs to the multibubble/cascade alternative, already excluded by [Theorem 7.6](#).

Alternative (iv) is excluded by the finite-cost scale-collapse theorem, [Theorem 7.8](#).

Alternative (v) is the scale-rigid terminal state in [Theorem 7.6](#). If this terminal state is velocity-virial dissipative, [Theorem 7.11](#) forces its Gaussian velocity mass to vanish, which contradicts retention of a positive terminal core. If it is axisymmetric and swirl-virial dissipative with positive retained circulation mass, the same theorem forces $\Gamma \equiv 0$, contradicting that positive circulation mass. The remaining scale-rigid terminal states are covered by the scale-rigidity hypothesis in [Theorem 7.6](#). Under that hypothesis, the state either has a nonzero bounded selected-window limit, which is incompatible with the local Type II condition, or it reduces to the compact single-core branch, which is excluded by [Theorem 7.5](#). Hence it cannot be a local Type II branch. $\square$

7.4. Assembly Theorem.

Theorem 7.15 (Local Type II exclusion). *Let (u, p) be a suitable weak solution on $\mathbb{R}^3 \times (T - \delta, T)$. Then no point of $\Sigma(T)$ lies in the local Type II alternative, within the class of local alternatives covered by [Theorem 7.13](#).*

Proof. Assume, for contradiction, that $x_0 \in \Sigma(T)$ lies in the local Type II alternative. By [Theorem 7.3](#), there is a positive local Type II concentration sequence $r_n \downarrow 0$ at (x_0, T) . By the decomposition into local alternatives, [Theorem 7.13](#), this sequence lies, after passing to selected windows, in one of the five alternatives listed there. Each alternative is impossible by [Theorem 7.14](#). This contradiction proves that no such x_0 exists in the covered local class of alternatives. $\square$

Corollary 7.16 (Final local dichotomy after Type I separation). *Every singular point at time T in the covered class of local alternatives belongs to the local Type I alternative. Equivalently, once the Type I case is covered by the Type I part of the theory, the local Type II class of alternatives contains no remaining singular branch.*

Proof. Let $x_0 \in \Sigma(T)$. By [Theorem 7.3](#), a concentration point is either in the local Type I alternative or in the local Type II alternative. [Theorem 7.15](#) excludes the latter. $\square$

7.5. Conclusion. The proof uses only local PDE statements. The local concentration theorem gives positive local CKN concentration and the Type I/Type II dichotomy. The repaired-gauge theorem gives the local renormalized representation for nondegenerate selected cores. The single-core theorem excludes compact single-core vanishing-dissipation branches. The multibubble/cascade theorem excludes splitting, cascades, gauge degeneracy, and the scale-rigid terminal state under its stated local hypotheses. The virial identities eliminate the velocity-virial dissipative terminal states and the axisymmetric swirl-virial dissipative terminal states before the remaining scale-rigid alternatives are covered by that hypothesis. The rough-core estimate reduces loss of windowed H^1 control to failure of compact CKN control. The scale-collapse theorem excludes finite-cost scale collapse and decomposes the remaining scale-collapse alternatives locally. Thus the local alternatives are exhausted without adding any further classification theorem.

8. PDE TRANSLATION OF THE C1–C5 CLASSIFICATION INPUTS

This section records, in PDE notation, the Type II classification inputs encoded in the C1–C5 cluster notes. The statements are conditional classification statements for declared Type II branches. They do not assert the existence of such branches and do not add any regularity theorem beyond the analytic hypotheses explicitly stated below.

8.1. Declared Type II branches and exhaustion.

Definition 8.1 (Declared local Type II class). Let $\mathcal{U}_{\text{II}}^{NS}$ be the class of candidate terminal branches (u, p, T^*) such that:

- (i) (u, p) is a three-dimensional incompressible Navier–Stokes solution on a terminal interval $[0, T^*)$ in the analytic class under consideration;
- (ii) $T^* < \infty$ is the terminal time of the branch;
- (iii) the branch is not in the local Type I class associated with the scale-invariant Type I criterion;
- (iv) the branch is a Type II candidate rather than a continuation, Type I, boundary, or non-Navier–Stokes-domain alternative.

Membership in $\mathcal{U}_{\text{II}}^{NS}$ is only a declaration of the classification problem; no nonemptiness assertion is made.

Definition 8.2 (Type II route data). A branch $\omega = (u, p, T^*) \in \mathcal{U}_{\text{II}}^{NS}$ has the Type II route data if the following three analytic outputs are present:

- (i) a concentration profile or blow-up germ modulo the Navier–Stokes symmetry group;
- (ii) a supercritical scale route, meaning that the branch enters the Type II scale alternative rather than the Type I scale alternative;
- (iii) a complete local profile output in the Navier–Stokes profile class used for the subsequent compactness and representation arguments.

The route is exhaustive if every declared Type II branch either has these three outputs or falls into one of the ordered failures in [Theorem 8.3](#).

Definition 8.3 (Ordered exhaustion failures). For a declared Type II branch, the ordered failures of the Type II route are:

- (i) concentration extraction fails;
- (ii) concentration extraction succeeds but the supercritical scale route fails;
- (iii) concentration extraction and scale routing succeed but the local profile output is not complete;
- (iv) the declared Type II branch lies outside the certified concentration/scale/profile route.

The first applicable failure in this order is the emitted exhaustion alternative.

Theorem 8.4 (Type II branch exhaustion). *Assume the Type II route data of [Theorem 8.2](#) are exhaustive on $\mathcal{U}_{\text{II}}^{NS}$. Then every $\omega \in \mathcal{U}_{\text{II}}^{NS}$ has the concentration output, the supercritical scale route, and the complete local profile output.*

Proof. Fix $\omega = (u, p, T^*) \in \mathcal{U}_{\text{II}}^{NS}$. By the concentration-extraction component of the route data, the branch has a concentration profile or blow-up germ. By the scale-routing component, the same branch lies in the supercritical Type II scale route. By profile completeness, the extracted profile belongs to the local Navier–Stokes profile class used in the subsequent arguments. Since ω was arbitrary, the three outputs hold for every declared Type II branch. $\square$

Corollary 8.5 (Ordered C1 classification). *For each $\omega \in \mathcal{U}_{\text{II}}^{NS}$, exactly one ordered output occurs: the Type II route data hold, or one of the four failures in [Theorem 8.3](#) occurs. The Type II route is exhaustive on $\mathcal{U}_{\text{II}}^{NS}$ if and only if the first output holds for every $\omega \in \mathcal{U}_{\text{II}}^{NS}$.*

Proof. Test the four route requirements in the stated order. If concentration extraction, scale routing, and profile completeness all hold, the branch has the Type II route data. If a test fails, the first failed test gives the corresponding ordered failure. The alternatives are disjoint by the first failure convention and exhaustive by construction. The final equivalence is the definition of exhaustion on $\mathcal{U}_{\text{II}}^{NS}$. $\square$

Theorem 8.6 (Entry into the downstream Type II analysis). *Assume the Type II route is exhaustive. Then each declared Type II branch enters the downstream analysis in exactly one of the following ways:*

- (i) *it reaches the compact Type II barrier hypotheses after the representation, critical-mass, tightness, and windowed- H^1 tests;*
- (ii) *it falls into an ordered representation failure, critical-mass failure, loss of critical tightness, or loss of local windowed H^1 control.*

Proof. By Theorem 8.4, every declared Type II branch has the concentration, scale, and profile route data. The representation step then either gives a repaired-gauge orbit or one of the ordered representation failures in Theorem 8.12. If a repaired-gauge orbit exists, the critical L^3 test gives either the positive finite annulus or one of the ordered critical-mass alternatives in Theorem 8.22. If the positive finite annulus holds, the remaining compact-barrier inputs are critical tightness and local windowed H^1 control. Their failures are exactly the radiative/noncompact and rough-core alternatives of Theorem 8.30. If both hold, the compact Type II barrier applies. $\square$

8.2. Repaired-gauge representation from a Type II route.

Definition 8.7 (Raw Type II chart). A raw Type II chart consists of

$$(u, p, T^*, x_c, \lambda, \tau, V, P, \mathcal{I}), \quad \mathcal{I} = (t_0, T^*),$$

where u, p solve

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0,$$

$\lambda(t) > 0$, $x_c(t) \in \mathbb{R}^3$, $x_c, \lambda \in AC_{\text{loc}}$, $\lambda(t) \rightarrow 0$ as $t \uparrow T^*$, and

$$\frac{d\tau}{dt} = \lambda(t)^{-2}, \quad \tau(t) \rightarrow \infty \quad \text{as } t \uparrow T^*.$$

The renormalized variables are

$$y = \frac{x - x_c(t)}{\lambda(t)}, \quad u(x, t) = \lambda(t)^{-1}V(y, \tau(t)), \quad p(x, t) = \lambda(t)^{-2}P(y, \tau(t)),$$

with enough regularity to justify the distributional chain rule on compact y -sets.

Lemma 8.8 (Raw renormalized equation). *For a raw Type II chart, define*

$$a_{\text{raw}}(\tau(t)) = \lambda(t)\lambda_t(t), \quad b_{\text{raw}}(\tau(t)) = \lambda(t)x'_c(t).$$

Then V, P satisfy, distributionally on compact y -sets,

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a_{\text{raw}}(\tau)(V + y \cdot \nabla V) + b_{\text{raw}}(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0.$$

Proof. Write

$$u(x, t) = \lambda(t)^{-1}V(y, \tau(t)), \quad y = \frac{x - x_c(t)}{\lambda(t)}, \quad \tau_t = \lambda^{-2}.$$

Then

$$\nabla_x u = \lambda^{-2} \nabla_y V, \quad \Delta_x u = \lambda^{-3} \Delta_y V, \quad (u \cdot \nabla_x)u = \lambda^{-3}(V \cdot \nabla_y)V.$$

Since

$$y_t = -\frac{x'_c(t)}{\lambda(t)} - \frac{\lambda_t(t)}{\lambda(t)}y,$$

one has

$$\partial_t u = \lambda^{-3} \partial_\tau V - \lambda_t \lambda^{-2} (V + y \cdot \nabla V) - \lambda^{-2} x'_c(t) \cdot \nabla V.$$

Moreover $\nabla_x p = \lambda^{-3} \nabla_y P$. Substitution in Navier–Stokes and multiplication by λ^3 give

$$\partial_\tau V - \lambda \lambda_t (V + y \cdot \nabla V) - \lambda x'_c(t) \cdot \nabla V + (V \cdot \nabla) V + \nabla P = \nu \Delta V.$$

Moving the modulation terms to the right-hand side gives the equation with $a_{\text{raw}} = \lambda \lambda_t$ and $b_{\text{raw}} = \lambda x'_c$. Finally, $\nabla_x \cdot u = \lambda^{-2} \nabla_y \cdot V$, so incompressibility of u implies $\nabla_y \cdot V = 0$. $\square$

Definition 8.9 (Repaired-gauge realization). Let $0 < p < 3$, $\Theta_0 > 0$, and let ψ_R be a smooth compactly supported centering cutoff. A raw Type II chart admits a repaired-gauge realization if, after an admissible time-dependent scale and translation correction, the profile satisfies

$$G_{\text{sc}}(V) = \int_{\mathbb{R}^3} |y|^{-p} |V(y)|^3 dy - \Theta_0 = 0,$$

and

$$G_j(V) = \int_{\mathbb{R}^3} y_j |V(y)|^2 \psi_R(y) dy = 0, \quad j = 1, 2, 3,$$

for a.e. renormalized time. The corrected profile belongs to the admissible gauge class $\mathcal{A}_p$, has the regularity needed to differentiate these functionals, and all time-dependent symmetry corrections have been absorbed into final modulation coefficients a, b .

Definition 8.10 (Repaired-gauge representation). A Type II branch has a repaired-gauge representation if it has:

- (i) a raw Type II chart;
- (ii) a repaired-gauge realization;
- (iii) pressure reconstruction modulo functions of time;
- (iv) final modulation coefficients a, b appearing in the repaired-gauge equation.

The output is the tuple

$$(V, P, a, b, G_{\text{sc}}, G_1, G_2, G_3, \tau_0).$$

Theorem 8.11 (C2 representation implication). *Assume a Type II profile branch has a raw Type II chart, a repaired-gauge realization, pressure reconstruction modulo functions of time, and final modulation coefficients. Then the branch has a repaired-gauge representation in the sense of [Theorem 8.10](#). In particular, (V, P, a, b) satisfies the PDE class required by the compact Type II barrier.*

Proof. The raw chart gives V, P and, by [Theorem 8.8](#), the renormalized Navier–Stokes equation with raw modulation coefficients. The repaired-gauge realization places the profile on the scale and centering gauge surface and absorbs the time-dependent symmetry correction into updated coefficients. The pressure reconstruction identifies the pressure associated with V , modulo the usual functions of time. The modulation data identify the final coefficients a, b . These are the components of [Theorem 8.10](#). $\square$

Definition 8.12 (Ordered representation failures). If a repaired-gauge representation is not obtained, the ordered failures are:

- (i) no raw orbit is supplied by the concentration/profile output;
- (ii) the raw orbit exists but the repaired gauge cannot be realized;
- (iii) pressure reconstruction modulo functions of time is unavailable;
- (iv) the final modulation coefficients are unavailable.

The first applicable failure is the emitted representation alternative.

Corollary 8.13 (Ordered C2 representation classification). *Every Type II profile branch in the declared class has exactly one ordered output: a repaired-gauge representation, or one of the four failures in [Theorem 8.12](#).*

Proof. Verify the four representation inputs in order. If all are present, [Theorem 8.11](#) gives the repaired-gauge representation. If not, the first missing input gives the corresponding ordered failure. This is exhaustive and disjoint by first-failure ordering. $\square$

8.3. Minimal chart and gauge realization.

Definition 8.14 (Minimal representation data). The minimal data for a routed Type II branch consist of:

- (i) an absolutely continuous concentration chart

$$(u, p, T^*, x_c, \lambda, \rho, U, Q, \mathcal{I}),$$

where $\mathcal{I} = (t_0, T^*)$, $\lambda(t) > 0$, $\lambda(t) \rightarrow 0$, $\rho_t = \lambda^{-2}$, and

$$u(x, t) = \lambda(t)^{-1}U(y, \rho(t)), \quad p(x, t) = \lambda(t)^{-2}Q(y, \rho(t)) + c(t), \quad y = \frac{x - x_c(t)}{\lambda(t)};$$

- (ii) an absolutely continuous repaired-gauge correction consisting of $\rho = \rho(\tau)$, $\mu(\tau) > 0$, and $q(\tau) \in \mathbb{R}^3$, with

$$\rho_\tau = \mu(\tau)^2,$$

$$V(Y, \tau) = \mu(\tau)U(\mu(\tau)Y + q(\tau), \rho(\tau)), \quad P(Y, \tau) = \mu(\tau)^2Q(\mu(\tau)Y + q(\tau), \rho(\tau)),$$

and

$$G_{\text{sc}}(V(\tau)) = 0, \quad G_j(V(\tau)) = 0, \quad j = 1, 2, 3.$$

The condition $\rho_\tau = \mu^2$ preserves the viscosity coefficient after the time-dependent scale correction.

Lemma 8.15 (Chart data imply the raw orbit). *Under the chart data in [Theorem 8.14](#), U, Q satisfy*

$$\partial_\rho U + (U \cdot \nabla)U + \nabla Q = \nu \Delta U + A(\rho)(U + y \cdot \nabla U) + B(\rho) \cdot \nabla U, \quad \nabla \cdot U = 0,$$

where

$$A(\rho(t)) = \lambda(t)\lambda_t(t), \quad B(\rho(t)) = \lambda(t)x'_c(t).$$

Proof. The chart data are the raw orbit data with raw time ρ . The additive pressure function $c(t)$ has zero spatial gradient. The chain-rule computation is the one in [Theorem 8.8](#), with τ replaced by ρ . Thus spatial derivatives scale as $\nabla_x u = \lambda^{-2}\nabla_y U$, $\Delta_x u = \lambda^{-3}\Delta_y U$, the convection term scales as $\lambda^{-3}(U \cdot \nabla)U$, and $\rho_t = \lambda^{-2}$. The terms generated by λ_t and x'_c give $A = \lambda\lambda_t$ and $B = \lambda x'_c$. Incompressibility pulls back as $\nabla_x \cdot u = \lambda^{-2}\nabla_y \cdot U$. $\square$

Lemma 8.16 (Pressure pullback). *Under the chart data,*

$$-\Delta Q = \partial_i \partial_j (U_i U_j)$$

in distributions, modulo functions of ρ . After any admissible repaired-gauge correction, the transformed pressure satisfies

$$-\Delta P = \partial_i \partial_j (V_i V_j)$$

in distributions, modulo functions of τ .

Proof. Taking divergence of the physical Navier–Stokes equation gives

$$-\Delta p = \partial_i \partial_j (u_i u_j),$$

with pressure determined up to a function of time. Pulling this identity back through $p = \lambda^{-2}Q + c(t)$, $u = \lambda^{-1}U$, and $y = (x - x_c)/\lambda$ gives the pressure equation for Q . If

$$V(Y) = \mu U(\mu Y + q), \quad P(Y) = \mu^2 Q(\mu Y + q),$$

then

$$-\Delta_Y P = \mu^4 (-\Delta Q)(\mu Y + q) = \mu^4 \partial_i \partial_j (U_i U_j)(\mu Y + q) = \partial_i \partial_j (V_i V_j).$$

Adding a function of ρ or τ to the pressure does not affect the identity. $\square$

Lemma 8.17 (Forced modulation coefficients). *Assume the minimal data of [Theorem 8.14](#). Then (V, P) satisfies*

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + Y \cdot \nabla V) + b(\tau) \cdot \nabla V,$$

where

$$a(\tau) = \mu(\tau)^2 A(\rho(\tau)) + \frac{\mu_\tau(\tau)}{\mu(\tau)}$$

and

$$b(\tau) = \mu(\tau)B(\rho(\tau)) + \mu(\tau)A(\rho(\tau))q(\tau) + \frac{q_\tau(\tau)}{\mu(\tau)}.$$

Proof. Let

$$z = \mu(\tau)Y + q(\tau), \quad V(Y, \tau) = \mu(\tau)U(z, \rho(\tau)), \quad \rho_\tau = \mu^2.$$

The raw equation in (z, ρ) is

$$U_\rho + (U \cdot \nabla_z)U + \nabla_z Q = \nu \Delta_z U + A(U + z \cdot \nabla_z U) + B \cdot \nabla_z U.$$

The identities

$$\nabla_Y V = \mu^2 \nabla_z U, \quad \Delta_Y V = \mu^3 \Delta_z U, \quad (V \cdot \nabla_Y)V = \mu^3 (U \cdot \nabla_z)U, \quad \nabla_Y P = \mu^3 \nabla_z Q$$

together with differentiation in τ give

$$V_\tau + (V \cdot \nabla_Y)V + \nabla_Y P - \nu \Delta_Y V = \left(\frac{\mu_\tau}{\mu} + \mu^2 A \right) (V + Y \cdot \nabla_Y V) + \left(\mu B + \mu A q + \frac{q_\tau}{\mu} \right) \cdot \nabla_Y V.$$

This is the displayed equation. Since μ, q, ρ are absolutely continuous and A, B are the raw chart coefficients, a, b are measurable and locally integrable on every finite renormalized window. $\square$

Theorem 8.18 (Minimal representation theorem). *If a routed Type II branch has the minimal data of [Theorem 8.14](#), then it has a repaired-gauge representation in the sense of [Theorem 8.10](#).*

Proof. [Theorem 8.15](#) gives the raw orbit. The gauge correction places the profile on the repaired scale and centering surface, hence gives the repaired-gauge realization. [Theorem 8.16](#) gives pressure reconstruction modulo functions of time. [Theorem 8.17](#) gives the final modulation coefficients. These are the four inputs in [Theorem 8.10](#). $\square$

Definition 8.19 (Minimal representation failures). Outside the declared repaired-gauge representation class, the ordered failures are:

- (i) no absolutely continuous concentration chart is available;
- (ii) the chart exists but the repaired scale and centering gauges cannot be solved admissibly and absolutely continuously.

Pressure-pullback and modulation-coefficient failures are not independent alternatives when the chart and the absolutely continuous gauge solve exist.

Corollary 8.20 (Ordered minimal representation classification). *Every routed Type II branch has exactly one ordered output: a repaired-gauge representation, failure of the raw chart, or failure of the repaired-gauge solve.*

Proof. If the chart is absent, the first failure occurs. If the chart exists but the admissible absolutely continuous gauge solve is absent, the second failure occurs. If both are present, [Theorem 8.18](#) gives the repaired-gauge representation. The alternatives are disjoint by their order and exhaustive for the representation row. $\square$

8.4. Critical L^3 annulus.

Definition 8.21 (Critical mass and positive finite annulus). For a repaired-gauge renormalized Type II candidate (V, P, a, b) on $[\tau_0, \infty)$, define

$$N_3(\tau) = \|V(\tau)\|_{L^3(\mathbb{R}^3)}.$$

The orbit has positive finite critical mass if there are constants $0 < \eta \leq M < \infty$ such that

$$\eta \leq N_3(\tau) \leq M, \quad \tau \geq \tau_0.$$

This condition replaces the exact normalization $\|V(\tau)\|_{L^3} = 1$; no amplitude rescaling is performed.

Definition 8.22 (Critical-mass alternatives). For a represented orbit, the ordered critical-mass alternatives are:

- (i) N_3 is not a well-defined extended measurable critical norm on the represented branch;
- (ii) $N_3(\tau) = \infty$ for some τ , or $\sup_{\tau \geq \tau_0} N_3(\tau) = \infty$;
- (iii) $\sup_{\tau \geq \tau_0} N_3(\tau) < \infty$ and $\liminf_{\tau \rightarrow \infty} N_3(\tau) = 0$;
- (iv) after discarding at most a finite initial time interval, the positive finite critical annulus of [Theorem 8.21](#) holds.

The first applicable alternative is emitted.

Theorem 8.23 (Nonzero-mass compact Type II barrier). *Let (V, P, a, b) be a repaired-gauge renormalized Type II candidate. Assume:*

- (i) *the positive finite critical annulus holds:*

$$0 < \eta \leq \|V(\tau)\|_{L^3(\mathbb{R}^3)} \leq M < \infty;$$

- (ii) *uniform critical tightness holds:*

$$\forall \varepsilon > 0 \exists R_\varepsilon : \sup_{\tau \geq \tau_0} \int_{|y| > R_\varepsilon} |V(y, \tau)|^3 dy < \varepsilon;$$

- (iii) *local windowed H^1 control holds:*

$$\forall m \geq 1 : \sup_{n \in \mathbb{N}} \int_{\tau_0+n}^{\tau_0+n+1} \|V(\tau)\|_{H^1(B_m)}^2 d\tau < \infty.$$

Then finite total localized renormalization cost is impossible:

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau) d\tau = \infty.$$

Proof. The proof is the compact Type II barrier proof with the exact normalization $\|V(\tau)\|_{L^3} = 1$ replaced by the lower bound $N_3(\tau) \geq \eta$. Assume for contradiction that the total localized renormalization cost is finite. The good-window selection and compactness portion of the compact barrier gives times $\sigma_j \rightarrow \infty$ such that, after passing to a subsequence,

$$V(\sigma_j) \rightarrow V_\infty \quad \text{strongly in } L_{\text{loc}}^3,$$

the selected sequence has vanishing local dissipation on the exhaustion used by the compact barrier, and consequently

$$\nabla V_\infty = 0$$

distributionally on every fixed ball. These conclusions use only tightness, local windowed H^1 control, and finite total cost.

The low-dissipation limit is spatially constant on each fixed ball. The constants are compatible as the balls increase, so $V_\infty \equiv c$ on $\mathbb{R}^3$. The upper critical-mass bound gives, for every R ,

$$|c|^3 |B_R| = \lim_{j \rightarrow \infty} \int_{B_R} |V(y, \sigma_j)|^3 dy \leq M^3.$$

Letting $R \rightarrow \infty$ forces $c = 0$. Thus

$$V(\sigma_j) \rightarrow 0 \quad \text{strongly in } L^3_{\text{loc}}.$$

Choose R so that the critical L^3 -tail outside B_R is $< \eta^3/4$ uniformly in τ . Strong convergence on B_R gives

$$\int_{B_R} |V(y, \sigma_j)|^3 dy < \eta^3/4$$

for j large. Therefore

$$\|V(\sigma_j)\|_{L^3(\mathbb{R}^3)}^3 < \eta^3/2,$$

contradicting $N_3(\sigma_j) \geq \eta$. Hence the total localized renormalization cost is infinite. $\square$

Theorem 8.24 (Critical L^3 -annulus classification). *For a represented repaired-gauge Type II candidate, exactly one of the ordered alternatives in [Theorem 8.22](#) occurs. In the fourth case the compact Type II barrier remains valid with the positive finite critical annulus in place of exact unit L^3 -normalization.*

Proof. If N_3 is not a well-defined measurable extended norm, the first alternative holds. Otherwise N_3 is an extended nonnegative measurable function. If it is infinite at some time, or has infinite supremum on the tail, the second alternative holds. If the supremum is finite but $\liminf_{\tau \rightarrow \infty} N_3(\tau) = 0$, the third alternative holds. In the remaining case,

$$0 < \liminf_{\tau \rightarrow \infty} N_3(\tau) \quad \text{and} \quad \sup_{\tau \geq \tau_0} N_3(\tau) < \infty.$$

After discarding a finite initial time interval, there exist $0 < \eta \leq M < \infty$ such that

$$\eta \leq N_3(\tau) \leq M$$

for all remaining τ . Relabeling this tail initial time as τ_0 gives the fourth alternative. The barrier statement is [Theorem 8.23](#). $\square$

8.5. Localized cost and the Type II barrier condition.

Definition 8.25 (Localized Type II cost). For a repaired-gauge Type II candidate, let

$$\tilde{\mathfrak{D}}_{R_0}(\tau) = \nu \int_{\mathbb{R}^3} |\nabla V(y, \tau)|^2 \phi_{R_0}(y) dy + a_+(\tau) \int_{\mathbb{R}^3} |V(y, \tau)|^2 \phi_{R_0}(y) dy,$$

where $a_+ = \max(a, 0)$, ϕ_{R_0} is the compact Type II cutoff, and $\nu > 0$. A Type II barrier cost is a measurable nonnegative function $\mathfrak{C}_{\text{II}}$ that is locally integrable on finite τ -intervals and whose divergence on $[\tau_0, \infty)$ is accepted as the blocked Type II scale condition.

Definition 8.26 (Cost comparison). A cost comparison for a repaired-gauge Type II candidate consists of $R_0, \phi_{R_0}, \mathfrak{C}_{\text{II}}, c_0, \tau_1$, with $c_0 > 0$ and $\tau_1 \geq \tau_0$, such that

$$\mathfrak{C}_{\text{II}}(\tau) \geq c_0 \tilde{\mathfrak{D}}_{R_0}(\tau) \quad \text{for a.e. } \tau \geq \tau_1.$$

The canonical Navier–Stokes choice is

$$\mathfrak{C}_{\text{II}}(\tau) = \tilde{\mathfrak{D}}_{R_0}(\tau), \quad c_0 = 1, \quad \tau_1 = \tau_0.$$

Lemma 8.27 (Identity-cost bridge). *If $\tilde{\mathfrak{D}}_{R_0}$ is measurable, nonnegative, and locally integrable on finite τ -intervals, then the canonical choice $\mathfrak{C}_{\text{II}} = \tilde{\mathfrak{D}}_{R_0}$ defines a valid Type II barrier cost and satisfies the cost comparison with $c_0 = 1$ and $\tau_1 = \tau_0$.*

Proof. The assumed measurability, nonnegativity, and local finite-interval integrability give the cost properties in [Theorem 8.25](#). The identity

$$\mathfrak{C}_{\text{II}} = \tilde{\mathfrak{D}}_{R_0}$$

gives the comparison in [Theorem 8.26](#) with $c_0 = 1$ and $\tau_1 = \tau_0$. $\square$

Theorem 8.28 (Cost bridge). *Let (V, P, a, b) be a repaired-gauge Type II candidate. Assume:*

- (i) $\tilde{\mathfrak{D}}_{R_0}$ is measurable, nonnegative, and locally integrable on finite τ -intervals;
- (ii)

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau) d\tau = \infty;$$

- (iii) either the canonical cost $\mathfrak{C}_{\text{II}} = \tilde{\mathfrak{D}}_{R_0}$ is used, or a cost comparison as in [Theorem 8.26](#) is available.

Then the blocked Type II scale condition associated with $\mathfrak{C}_{\text{II}}$ holds.

Proof. In the canonical case, [Theorem 8.27](#) supplies the cost comparison. In the general case, the comparison is assumed. Hence there are $c_0 > 0$ and $\tau_1 \geq \tau_0$ such that

$$\mathfrak{C}_{\text{II}}(\tau) \geq c_0 \tilde{\mathfrak{D}}_{R_0}(\tau)$$

for a.e. $\tau \geq \tau_1$. Since $\tilde{\mathfrak{D}}_{R_0} \geq 0$ and is locally integrable on finite intervals, divergence on $[\tau_0, \infty)$ implies

$$\int_{\tau_1}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau) d\tau = \infty.$$

Therefore

$$\int_{\tau_1}^{\infty} \mathfrak{C}_{\text{II}}(\tau) d\tau \geq c_0 \int_{\tau_1}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau) d\tau = \infty.$$

Since $\mathfrak{C}_{\text{II}}$ is nonnegative and locally integrable on finite intervals,

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\text{II}}(\tau) d\tau = \int_{\tau_0}^{\tau_1} \mathfrak{C}_{\text{II}}(\tau) d\tau + \int_{\tau_1}^{\infty} \mathfrak{C}_{\text{II}}(\tau) d\tau = \infty.$$

By definition of the Type II barrier cost, this divergence is the blocked Type II scale condition. $\square$

Corollary 8.29 (Compact barrier implies blocked scale). *Assume the hypotheses of [Theorem 8.23](#) and use the canonical Navier–Stokes cost $\mathfrak{C}_{\text{II}} = \tilde{\mathfrak{D}}_{R_0}$. Then the blocked Type II scale condition holds. If, in addition, the supercritical scale route is present and the corresponding Navier–Stokes promotion hypothesis is available, then the promoted Type II suppression condition holds.*

Proof. [Theorem 8.23](#) gives

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau) d\tau = \infty.$$

By [Theorem 8.27](#), the canonical cost satisfies the cost comparison. [Theorem 8.28](#) converts the infinite PDE cost into the blocked Type II scale condition. Combining the blocked condition with the supercritical scale route gives the promoted suppression condition only under the stated Navier–Stokes promotion hypothesis. $\square$

8.6. Two-bucket finite-cost classification.

Definition 8.30 (Remaining compact-barrier tests). For a represented repaired-gauge Type II orbit, define the critical-tightness condition by

$$\forall \varepsilon > 0 \exists R_\varepsilon : \sup_{\tau \geq \tau_0} \int_{|y| > R_\varepsilon} |V(y, \tau)|^3 dy < \varepsilon.$$

Its failure is the radiative/noncompact alternative:

$$\exists \varepsilon_0 > 0 \forall R > 0 \exists \tau_R \geq \tau_0 : \int_{|y| > R} |V(y, \tau_R)|^3 dy \geq \varepsilon_0.$$

Define local windowed H^1 control by

$$\forall m \geq 1 : \sup_{n \in \mathbb{N}} \int_{\tau_0+n}^{\tau_0+n+1} \|V(\tau)\|_{H^1(B_m)}^2 d\tau < \infty.$$

Its failure is the rough-core alternative:

$$\exists m \geq 1 : \sup_{n \in \mathbb{N}} \int_{\tau_0+n}^{\tau_0+n+1} \|V(\tau)\|_{H^1(B_m)}^2 d\tau = \infty.$$

Definition 8.31 (Classification-ready compact route). A declared Type II branch reaches the classification-ready compact route if it has:

- (i) the Type II route data of [Theorem 8.2](#);
- (ii) the repaired-gauge representation of [Theorem 8.10](#);
- (iii) the positive finite critical annulus of [Theorem 8.21](#);
- (iv) the locally integrable nonnegative cost and cost comparison of [Theorems 8.25](#) and [8.26](#).

This condition does not include critical tightness or windowed H^1 control; those are the two remaining compact-barrier tests.

Lemma 8.32 (Compact positive branch is suppressed). *Assume a declared Type II candidate reaches the classification-ready compact route and satisfies both compact-barrier tests in [Theorem 8.30](#). If the Navier–Stokes promotion hypothesis for the blocked Type II scale condition is available, then the candidate satisfies the promoted Type II suppression condition.*

Proof. The classification-ready compact route gives the supercritical scale route, a repaired-gauge orbit (V, P, a, b) , the positive finite critical annulus, and the cost comparison. Together with critical tightness and local windowed H^1 control, the hypotheses of [Theorem 8.23](#) hold. Thus

$$\int_{\tau_0}^{\infty} \tilde{\mathfrak{D}}_{R_0}(\tau) d\tau = \infty.$$

[Theorem 8.28](#) converts this divergence into the blocked Type II scale condition. Combining the blocked condition with the supercritical scale route gives the promoted Type II suppression condition under the stated Navier–Stokes promotion hypothesis. $\square$

Theorem 8.33 (Suppressed-or-two-bucket classification). *Assume every declared Type II candidate reaches the classification-ready compact route and that the Navier–Stokes promotion hypothesis is available. Then each declared Type II candidate has exactly one ordered output:*

- (i) *the promoted compact Type II suppression condition;*
- (ii) *the radiative/noncompact alternative, i.e. failure of critical tightness;*
- (iii) *the rough-core alternative, i.e. failure of local windowed H^1 control after critical tightness has been verified.*

Proof. Fix a declared Type II candidate. Since it reaches the classification-ready compact route, only the two compact-barrier tests remain. First evaluate critical tightness. If tightness fails, the candidate is in the radiative/noncompact alternative. If tightness holds, evaluate local windowed H^1 control. If the windowed H^1 condition fails, the candidate is in the rough-core alternative. If it holds, then [Theorem 8.32](#) gives the promoted compact Type II suppression condition. The outputs are exhaustive and disjoint by this ordered evaluation. $\square$

Definition 8.34 (Finite-cost non-suppressed branch). A declared Type II branch is finite-cost if

$$\int_{\tau_0}^{\infty} \mathfrak{C}_{\text{II}}(\tau) d\tau < \infty.$$

In the canonical Navier–Stokes cost this is equivalent to finite localized PDE cost, because $\mathfrak{C}_{\text{II}} = \tilde{\mathfrak{D}}_{R_0}$. It is non-suppressed if the promoted compact Type II suppression condition is not emitted for that branch.

Corollary 8.35 (Two-bucket finite-cost classification). *Under the hypotheses of [Theorem 8.33](#), every finite-cost non-suppressed declared Type II branch is either radiative/noncompact or rough-core.*

Proof. Apply [Theorem 8.33](#). Since the branch is non-suppressed, the first output is unavailable. Therefore the ordered output is the radiative/noncompact alternative or the rough-core alternative. The finite-cost assumption records that the resulting branch remains finite-cost; the exclusion of the suppressed case is the explicit non-suppression hypothesis. $\square$

Remark 8.36 (Content of the C1–C5 translation). The preceding statements translate the C1–C5 cluster proof obligations into PDE form. Exhaustion failure, missing repaired-gauge representation, ambiguous critical L^3 normalization, and cost-comparison failure are all replaced by ordered alternatives. Once those alternatives are excluded or assumed absent, the only finite-cost non-suppressed Type II alternatives left by C5 are the radiative/noncompact alternative and the rough-core alternative.

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